



GOLDEN
TREE

INTEGRATED SCIENCE

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7
Class

FOCUS AREAS OF NEP 2020



Created with the new dimension of joyful learning based on **National Education Policy, 2020**.



Each chapter begins with **Put into Focus** includes concept



Key Words is a list of difficult words to help learners to understand the topics.



Cognitive Skills gives descriptive writing tasks to develop the written skills of the learners.



Critical Thinking is the ability to examine learner's thoughts and feelings to make decisions based on their life.



Observing Skills includes learning process whereby students learn with their own senses.





Communication Skills includes discussion which is generally a face-to-face interaction promotes confidence level.



Linking Chain
increases the ability
of students to excel
in two or more
different
subjects.



Activity includes varied activities in which learners construct and demonstrate understanding through an art form.



Computational Thinking provides learners problem solving technique.



Project work is the ability to do effective work together with others by doing some experiments.





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PREFACE

Science is an integral part of our lives. It is, therefore imperative that we help our children to learn the importance of science in such a way that it reflects through their attitude. This series '**Integrated Science**' aims to achieve the objective and to fulfil the guidelines of National Education Policy 2020. The series has been written in such a way as to kindle a genuine interest and motivation in a child to develop a scientific attitude and problem solving abilities from an early age.

This series focuses on the development of cognitive, effective and psychomotor skills through process based, activity-oriented learning, rather than through rote memorization of content-efforts have been made to help children in learning without burdening them, through simple text, child-friendly illustrations, pictures and ample activities. The aim is to encourage them to observe, interact, explore the world around them and learn by doing. Based on the experience gained through exploration, children are further guided and encouraged to **construct knowledge** in order to strengthen the foundation of their learning as well as develop language skills.

Wherever possible, the books of this series instil into children '**Values**' such as the importance of conserving nature, keeping health etc.

Respiration

1

- Need of Respiration
- Breathing
- Parts of Respiratory System
- Breathing in Other Animals
- Respiration in Plants

We see in our daily life that running makes a person breathe faster. To understand this phenomenon, we first have to understand why do we breathe. In fact, breathing is a part of respiration.

NEED OF RESPIRATION

All living things need energy to move and grow. They get energy by breaking down food substances. We need oxygen to break down the food in every cell of our body. Our lungs take oxygen from the air during respiration (breathing).

When we breathe, oxygen is transported to the lungs and gets mixed with blood. The oxygen mixed blood goes to all parts of the body and finally to all the cells. When oxygen combines with the food in cells, oxidation (burning) of food takes place. During this process, energy is released along with water and carbon dioxide as waste. The process of oxidation of food to release energy along with water and carbon dioxide as wastes in living cells is called **respiration** or **cellular respiration**.



Respiration

Breathing is the process that moves air in and out of the lungs. **Respiration** is the process by which living organisms use oxygen of air and food to produce energy. Carbon dioxide is also produced during this process. The parts of body used in the process of breathing form the **respiratory system**.

Respiration is also called slow combustion at body temperature. But the two differ widely.

Differences between Respiration and Combustion

Respiration	Combustion
1. Respiration is a slow process, that is, oxidation of food in the body takes place slowly.	1. Combustion is a fast process. 2. Combustion usually takes place at a much higher temperature.

2. Respiration takes place at a relatively lower temperature.	4. Combustion of fuels is usually not complete, so some carbon monoxide and unburnt carbon particles are also formed.
3. In respiration, complete combustion of food takes place.	5. Ordinary combustion takes place outside the living body.
4. Respiration takes place inside the living body.	6. Combustion is usually not a continuous process.
5. Respiration is a continuous process, and continuous throughout the life of the living body.	

BREATHING

Breathing means taking in air rich in oxygen and giving out air rich in carbon dioxide with the help of respiratory organs.

The taking in of air rich in oxygen into the body is called **inhalation** and giving out of air rich in carbon dioxide is called **exhalation**. It is a continuous process going on all the time and throughout the life of all organisms.

Breathing Rate

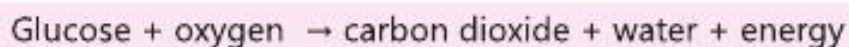
During breathing, inhalation and exhalation take place alternately. A breath means one inhalation plus one exhalation. The number of times a person breathes in a minute is termed as the **breathing rate**.

On an average an adult human being at rest breathes in and out 15-18 times in a minute. During heavy exercise, the breathing rate can increase upto 25 times per minute.

There are two parts (processes) of respiration.

Cellular Respiration or Internal Respiration

Cellular respiration takes place inside the cells. In this process, glucose obtained during the digestion of food is oxidised by the inhaled oxygen to carbon dioxide and water, and energy is released.



Differences between breathing and respiration	
Breathing	Respiration
1. It is a physical process because only the air moves from one place to another.	1. It is a chemical process because the food undergoes chemical changes.
2. Energy is not released.	2. Energy is released.
3. It takes place in breathing organs.	3. It takes place in living cells.

Types of Respiration

Respiration may be of two types: (a) Aerobic respiration and (b) Anaerobic respiration.

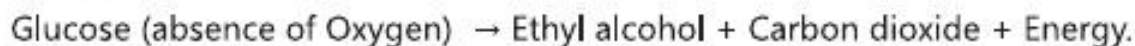
Aerobic Respiration

Most of the living organisms use oxygen to break up the food in order to get energy. So, the respiration that requires usage of oxygen is called **aerobic respiration**. It is represented by the equation below :



Anaerobic Respiration

Some microorganisms like yeast and bacteria obtain energy from food in the absence of oxygen. So, the respiration which takes place in the absence of free oxygen is called **anaerobic respiration**. Anaerobic respiration takes place in our skeletal muscles.



Bacteria and fungi can respire anaerobically, which is useful in converting sugar into alcohol. Alcohol, on one hand can be bad for the society. But on the other hand, it can be used as a fuel. Yeast is a one-celled fungus and respire anaerobically to produce alcohol. Therefore, it is used to make wine and beer.

Differences between Aerobic and Anaerobic Respiration

Aerobic Respiration

1. Aerobic respiration takes place in the presence of oxygen.
2. Food molecules (glucose) are completely broken down into carbon dioxide and water, releasing energy.
3. Energy released is more.
4. It occurs in most plants and animals.

Anaerobic Respiration

1. Anaerobic respiration takes place in the absence of oxygen.
2. Food molecules (glucose) are partially broken down into ethyl alcohol, carbon dioxide and energy (or lactic acid and energy.)
3. Energy released is much less.
4. It occurs in yeast, some bacteria and parasitic worms.

Conclusion : Whenever a person needs extra energy, he/she breathes faster. As a result more oxygen is supplied to body cells. It speeds up the breakdown of food and more energy is released.

Parts of Respiratory System

Our respiratory system consists of the nose and throat, the wind pipe (trachea), the breathing muscles and the lungs .

Nose and Throat

The air enters through our nose or mouth. Our nose has hair and mucus to clean, moisten and warm the air. Mucus is a sticky liquid. Dust particles and germs present in the air stick to the mucus.

The air enters the throat and passes through the larynx. Our vocal cords are present in the larynx to produce sound.

Trachea (Windpipe)

Air passes from the larynx into the trachea or windpipe. Our windpipe is made of C-shaped rings of cartilage. These rings keep our windpipe open. Mucus and tiny hair in the trachea also filter the air.

Bronchi and Lungs

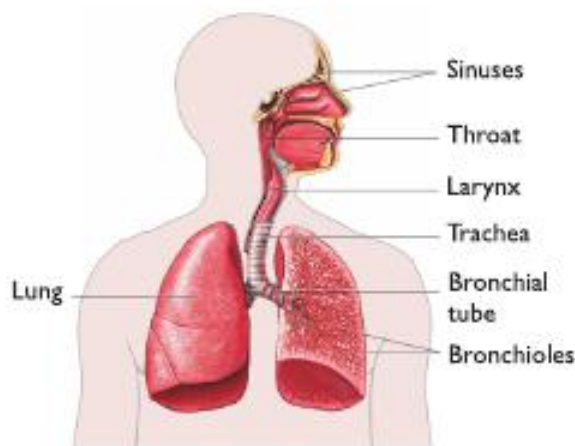
The trachea divides into two branches called bronchi (singular bronchus). Bronchi carry air into the lungs.

Our lungs are the most important organs of the respiratory system. In each lung, the bronchus divides into smaller tubes called **bronchioles**. At the end of each bronchiole, tiny air sacs called **alveoli** are present. Alveoli are surrounded by blood capillaries. When we breathe in, the air enters the lungs and reaches the alveoli. Oxygen of the air passes through the walls of alveoli into the capillaries. The red blood cells carry this oxygen to every cell of our body. Cells in our body use oxygen and food to produce energy and carbon dioxide. The blood brings carbon dioxide back to the lungs. Carbon dioxide leaves our body when we breathe out.

FACT

The air we breathe in contains about 0.04% carbon dioxide. The air we breathe out contains about 4% carbon dioxide.

In other words, the exhaled air contains about 100 times the concentration of carbon dioxide than inhaled air does.



Respiratory System

HOW DO WE BREATHE?

Our lungs have no muscles. Two types of muscles work during the breathing process; the intercostal muscles of ribs, and the dome-shaped diaphragm. The process of breathing is completed in two steps :

Inhaling

When the intercostal muscles pull our ribs outward and the diaphragm contracts, the air enters the lungs. It is called **inhaling**.

Exhaling

When the intercostal muscles and diaphragm muscles relax, the air moves out of the lungs. It is called **exhaling**.

As the breath goes downward, the chest expands, the rib cage elevates, diaphragm goes down and the belly comes out.

The breathing process functions briefly as follows. Consider the lungs to be two very flexible and strong bags which can be inflated or deflated as air flows in or out of them during inhalation and exhalation respectively. The lungs have the ability to expand greatly and also can be contracted into a small space.

The lungs are surrounded at the top and sides by the ribcage, and at the bottom by a very important flat muscle called the diaphragm. The diaphragm separates the abdomen from the lungs. The lungs are not fixed to the diaphragm or the ribcage but they nevertheless closely follow any change in shape of this enclosure.



Sneezing

The air around us has various types of unwanted particles, such as smoke, dust, pollens etc. When we inhale, the particles get trapped in the hair present in our nasal cavity. The brain instructs the nasal chamber to initiate sneezing. This is a method to remove the unwanted dust particles from the nostril. sneezing is a type of defense mechanism used by our body.

What do we breathe out?

Breathed out air contains carbon dioxide and water (along with nitrogen and unused oxygen).

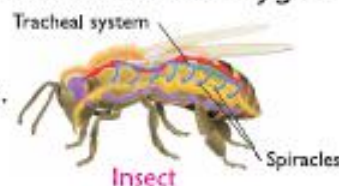
BREATHING IN OTHER ANIMALS

Like human beings, animals and plants also breathe and respire. The basic process of respiration is the same in all organisms. Let us study the structures of some animals and how they enable them to respire.

- (a) In unicellular and smaller multicellular animals, all the cells take up oxygen from the surrounding air or water and give out carbon dioxide by diffusion.

Eg. amoeba, paramecium.

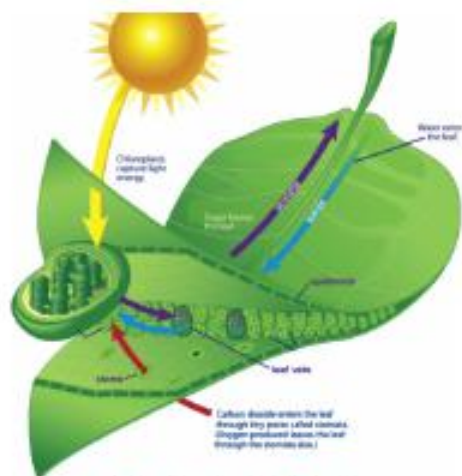
- (b) Animals like the earthworm and the leeches breathe through their moist skin. Gases can easily pass through the skin and the blood vessels under the skin.
- (c) In insects, there are several small openings called **spiracles** on the lateral side of their bodies. These spiracles lead to air tubes called **trachea**. Exchange of gases takes place through spiracles into trachea.
- (d) Fishes have special organs called gills, which are used to absorb the dissolved oxygen in water.
- (e) Animals like reptiles, birds and mammals have lungs for breathing.
- (f) Animals such as frogs respire through their skin and lungs.



Whales and dolphins live in water but they are not fish. They are mammals and have lungs for breathing. They often come up to the water surface to breathe (inhale and exhale). Moving upwards, they exhale. The pressure of exhaling gases move the water upwards in the form of a fountain of water sometimes.

RESPIRATION IN PLANTS

Like other living organisms, plants also respire to get energy from food. Generally, plants do not have any special organ for breathing. Respiration in plants takes place through different parts such as roots, stems and leaves. They do not show breathing movements like that of animals.



Respiration in Plants

Plants breathe through tiny pores in the leaves called **stomata**. Oxygen from the air diffuses into the leaves and the carbon dioxide from the leaves diffuses out through stomata. Stems have minute openings on their surfaces. These openings help in the exchange of gases. Roots also respire independently. Roots draw in air from the air spaces present between the soil particles. Thus, all parts of the plant like root, stem and leaves respire independently. Aquatic plants directly exchange gases with the water, that surrounds their leaves, roots and stems. The process of photosynthesis in plants takes place during the day.

During this process, carbon dioxide is used and oxygen is released.

A part of the oxygen released during photosynthesis is used by the plants for respiration and the rest is sent out through the stomata. Carbon dioxide released during respiration is used up. The carbon dioxide that is released as a result of respiration is sent out through the stomata into the atmosphere. The oxygen in the atmosphere is taken in and used for respiration.

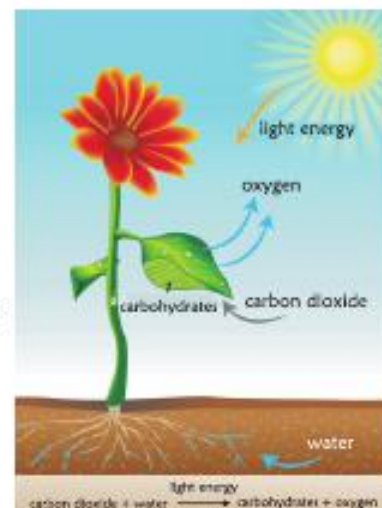
Differences between Respiration and Photosynthesis

Respiration

1. It takes place during day and night.
2. All living organisms respire.
3. Food is consumed.
4. During this process oxygen is taken in and carbon dioxide is given out.

Photosynthesis

1. It takes place only during day time.
2. Only green plants prepare food.
3. Food is synthesized.
4. During this process carbon dioxide is taken in and oxygen is given out.



Photosynthesis

Keep Your Lungs Healthy

1. Fruits and vegetables contain vitamins. Vitamins keep our lungs healthy.
2. Exercises like running, walking, swimming, jumping, cycling, etc. are good for our lungs.
3. Avoid smoking. Smoking is the main cause of lung cancer.
4. Leafy green vegetables contain such chemicals that are good for our lungs.



Healthy lungs



Diseased lungs

REWIND

- ▶ Our respiratory system helps to produce energy which we use in our activities.
- ▶ Breathing and burning are similar processes, but release of energy is very slow during breathing.
- ▶ The process of taking in oxygen into the cells, using it for producing energy and exhaling carbon dioxide and water is called respiration.
- ▶ Respiration in the presence of oxygen is called aerobic respiration while in the absence of oxygen anaerobic.
- ▶ Most plants, animals and humans do aerobic respiration. But in animals and humans, anaerobic is respiration takes place in muscles sometimes.
- ▶ The number of times a person breathes in one minute is called breathing rate.
- ▶ During inhalation our lungs expand and then come back to the original state as the air moves out during exhalation.
- ▶ In mammals, birds and snakes, the respiratory organs and the process of breathing are similar to those in humans.
- ▶ In earthworm, the exchange of gases occurs through the moist skin. In fishes it takes place through gills and in insects through the tracheae.
- ▶ In a plant, the roots take in air present in the soil. Leaves have tiny pores called stomata through which they exchange gases.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- Breathing is a

(i) Chemical process	☐	(ii) Physical process	☐
(iii) Biological process	☐	(iv) Biochemical process	☐
- In anaerobic respiration which chemical substance is produced?

(i) Lactic acid	☐	(ii) Formic acid	☐	(iii) Alcohol	☐	(iv) Citric acid	☐
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- Inhalation is the

(i) Intake of oxygen	☐	(ii) Release of oxygen	☐
(iii) Intake of carbon dioxide	☐	(iv) Release of carbon dioxide	☐
- More energy is released in

(i) Anaerobic respiration	☐	(ii) Aerobic respiration	☐
(iii) Fermentation	☐	(iv) Breathing	☐
- Plants breathe through their stomata located in their

(i) Leaves	☐	(ii) Stem	☐	(iii) Roots	☐	(iv) Flowers	☐
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- Gills are the respiratory organs in

(i) Fish	☐	(ii) Cockroach	☐	(iii) Amoeba	☐	(iv) Earthworm	☐
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2 Fill in the blanks :

- The leaves of plants have small pores called _____.
- Frogs have lungs and _____ as respiratory organ.
- _____ respiration takes place in the presence of oxygen.
- The oxygen carrying components in blood is _____.
- Muscle cell produce _____ during sudden activity.
- _____ is produced in anaerobic respiration.

3 Write T for true and F for false :

- Frogs breathe through their skins as well as their lungs.
- During heavy exercise the breathing rate of a person slows down.
- The size of the chest cavity increases during inhalation.
- Haemoglobin is present in white blood cell of mammals.
- Anaerobic respiration produces more energy than aerobic respiration.
- Gill is the respiratory organ in fish.

4 Answer these questions in short :

- What is the need of respiration?
- Why do we breathe?
- What is the role of oxygen in our body?

- d. State the role of diaphragm and rib muscles in breathing?
- e. Describe briefly the respiratory system in plants.
- f. What is a diaphragm?

5 Match the following :

- | | |
|--------------------------|-----------------------------------|
| a. Aerobic respiration | (i) Oxidation of food in the body |
| b. Anaerobic respiration | (ii) Breathing |
| c. Vitamins | (iii) Absence of oxygen |
| d. External respiration | (iv) Plant food |
| e. Internal respiration | (v) Energy and carbon dioxide gas |
| f. Photosynthesis | (vi) Lungs |

6 Give answers in detail :

- a. Compare between aerobic and anaerobic respiration.
- b. What is anaerobic respiration? When does it happen in humans and why?
- c. Describe the respiratory system of humans with a labelled diagram.
- d. Describe all different ways in which animals breathe?
- e. What are inhaling and exhaling?
- f. Differentiate between
 - (i) Respiration and Photosynthesis
 - (ii) Aerobic and Anaerobic Respiration

NEP QUERY



PROJECT WORK

- ♦ In the laboratory : Observe fish in an aquarium. You will find flap like structures on both sides of their heads. These are flaps which cover the gills. These flaps open and close alternately. On the basis of these observations, explain the process of respiration in the fish.

(Learning by Experiment)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- "Importance of artificial respiration."

(Discussion)

- Formation of Soil
- Properties of Soil
- Soil Erosion
- Horizons of Soil
- Soils and Crops
- Soil Pollution

The term 'Soil' has been derived from Latin word 'Solum' which means earthy material necessary for the growth of plants. Soil may be defined as a mixture of organic as well as weathered rock materials present in the uppermost part of earth crust.

The soil is formed through a number of physical, chemical and biological processes occurring slowly for long periods of time. It is an important abiotic factor in which the root grows, anchors the plant and supplies necessary water and nutrients to the plant. Thus soil is a store house of minerals, a reservoir of water, a conserver of soil fertility, a producer of vegetable crops, a home of wildlife and live stocks.



Soil

FORMATION OF SOIL

The transformation of rocks into soil may be termed as **soil formation**. It starts with the weathering of rocks. For millions of years, there was continuous breaking of rocks into small particles due to heat, rain and wind. These small particles were carried away by wind and water and ultimately got deposited over other rocks. In this way, thick layer of clay, sand and small pieces of rocks are formed. This is known as soil. The process of soil formation is known as **weathering**. Weathering is the process of disintegration of parent rocks into smaller particles.

Water enters into the cracks of rocks during rain. In cold climate, water gets frozen and increase in volume. Due to increase in volume, rocks get cracked. Once rocks are broken down, the particles are carried downhill under the force of gravity. These small particles were carried away by wind and water and ultimately got deposited over other rocks. In this way, thick layer of clay, sand and small pieces of rocks are formed. This is known as soil.

The rock pieces in decreasing order of their sizes are called stones, gravel, sand, silt and clay. Their touch varies from rough to smooth. As there are different types of rocks of different colours and minerals, the soil vary in colour and minerals. Their mixing in different proportions makes a variegated spectrum of colours as well as minerals. The size of the particles and their hardness make them of different textures.

HORIZONS OF SOIL

Soil is made up of distinct horizontal layers; these layers are called **horizons**. They range from rich, organic upper layers (humus and topsoil) to underlying rocky layers (subsoil, regolith and bedrock).

O Horizon

The top, organic layer of soil, mostly made up of leaf litter and humus (decomposed organic matter).

A Horizon

This layer called topsoil; it is found below the O horizon and above the E horizon. Seeds germinate and plant roots grow in this dark-coloured layer. It is made up of humus (decomposed organic matter) mixed with mineral particles.

E Horizon

This eluviation (leaching) layer is light in colour; this layer is beneath the A horizon and above the B horizon. It is mostly made up of sand and silt, having lost most of its minerals and clay as water drips through the soil (in the process of eluviation).

B Horizon

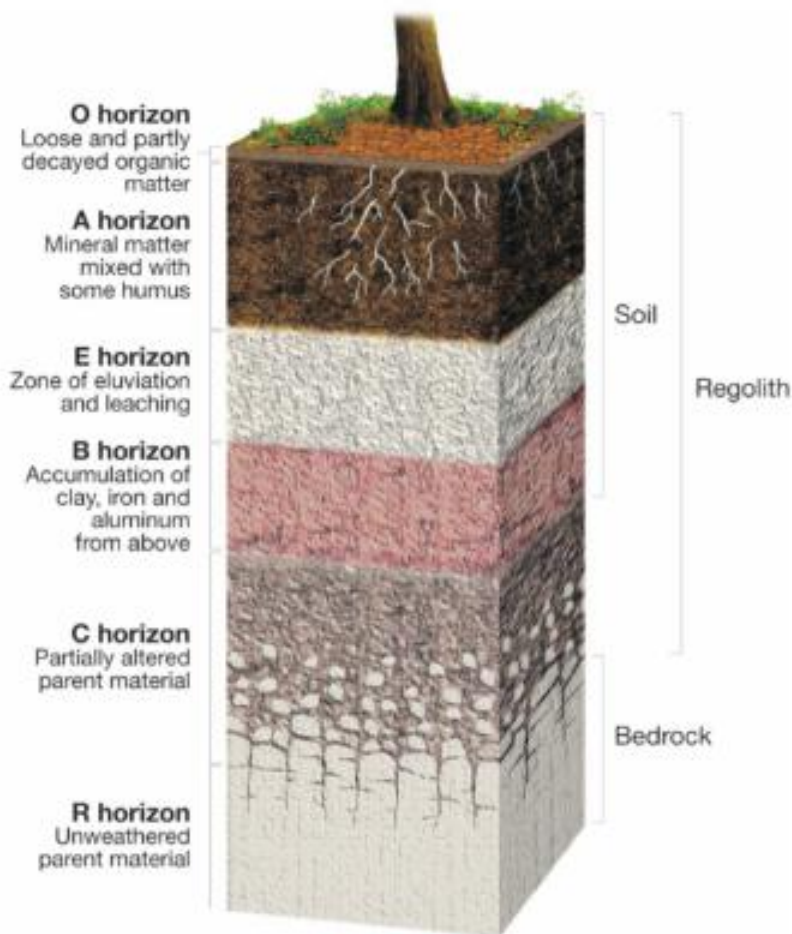
Also called the subsoil, this layer is beneath the E Horizon and above the C Horizon. It has mineral deposits (like iron, aluminium oxides and calcium carbonate) that it receives from layers above it when mineralized water drips from the soil above.

C Horizon

Also called regolith, the layer beneath the B Horizon and above the R Horizon. It consists of slightly broken-up bedrock. Plant roots do not penetrate into this layer; very little organic material is found in this layer.

R Horizon

The unweathered rock (bedrock) layer that is beneath all the other layers.



Soil Profile

TYPES OF SOIL

Soil is our prime natural and economic resource. Soils in India differ in composition and structure.

1. Alluvial Soils

These are formed by the decomposition of sediments by rivers. They are rich in humus and very fertile. They are found in Great Northern plain, lower valleys of Narmada and Tapi and Northern Gujarat. These soils are renewed every year.

2. Black Soils

These soils are made up of volcanic rocks and lava-flow. It is concentrated over Deccan Lava Tract which includes parts of Maharashtra, Chhattisgarh, Madhya Pradesh, Gujarat, Andhra Pradesh and Tamil Nadu. It consists of lime, iron, magnesium and also potash but lacks in phosphorus, nitrogen and organic matter.

3. Red Soils

These are derived from weathering of ancient metamorphic rocks of Deccan Plateau. Its redness is due to iron composition. When iron content is lower, it is yellow or brown. They cover almost the whole of Tamil Nadu, Andhra Pradesh, Chhattisgarh, Karnataka, Maharashtra and parts of Odisha.



Red Soil

4. Laterite Soils

These soils are formed due to intense leaching and are well developed on the summits of hills and uplands. They are commonly found in Kerala, Tamil Nadu, Maharashtra, Chhattisgarh and hilly areas of Odisha and Assam.

5. Mountain Soils

These soils are formed as a result of the accumulation of organic matter derived from forest growth. They are found in the Himalayan region and vary in different regions according to altitude. Tea is grown in those areas which receive sufficient rainfall.

6. Desert Soils

In the desert regions of Rajasthan, soils are not well developed. As evaporation is in excess of rainfall, the soil has a high salt content and saline layer forms a hard crust. These soils are generally sandy and deficient in organic matter.

The sizes of the particles in soil determine its properties.

- (i) Sand particles are so small that cannot fit closely to each other, so there are some spaces between them. These spaces are filled with air. Thus the sand is well aerated. Water can drain quickly through the spaces between the sand particles. So, sandy soils tend to be light, well aerated and rather dry.



Sandy Soil

- (ii) Clay particles are extremely small so that they closely pack together, leaving little space for air. Water can be held in the tiny gaps between the particles of clay. So clayey soils have little air. They are heavy as they hold more water than the sandy soils.



Loamy Soil



Clayey Soil

- (iii) Loam is a mixture of sand, clay and silt. Silt occurs as a deposit in river beds. The loamy soil also has humus in it. It has right water holding capacity for the growth of plants. Loamy soil has enough air, i.e., it is reasonably porous to allow aeration of roots. Thus loamy soil is the best topsoil for growing plants.

PROPERTIES OF SOIL

Characteristics or properties of soil are percolation rate of water, moisture and absorption of water.

Percolation Rate of Water in the Soil

Percolation rate is the gradually movement of water through the spaces or pores in the soil usually expressed as inches per hour or inches per day. A soil with a greater percolation rate can usually absorb more water. More water can pass through sand than clay.

Absorption of Water by the Soil

Different soils absorb different amount of water.

Percentage of water absorbed

$$= \frac{\text{Weight of water absorbed}}{\text{weight of soil}} \times 100$$

(Weight of 1 ml water = 1 g)

SOIL AND CROPS

The crops commonly grown in any area depend on many factors. One major factor is the kind of soil available in that area and its properties.

Soil composition along with climatic factors determine the various types of crops that might grow in any region.

Suitability of soils for some typical crops	
Alluvial Soil	Rice, wheat, sugarcane, cotton and jute all grow well in these soils.
Black Soil	This type of soil is also used for rice, wheat, sugarcane and cotton. It is additionally used to grow groundnut, millet and oilseeds.

Desert Soil	Generally only drought-resistant crops such as barely and millet can grow in this type of soil.
Laterite Soil	It is used to grow more tropical crops such as cashew, rubber, coconut, tea and coffee.
Mountain Soil	They are somewhat acidic, but suitable for growing tea, coffee, spices and may types of tropical fruits.
Red and Yellow Soil	Red and yellow soils are used to grow rice, wheat, sugarcane, millet, groundnut, ragi and potato.

SOIL EROSION

Removal of topsoil by weathering agents; such as wind, water and ice is known as **soil erosion**. Soil erosion affects the fertility of soil, as humus present in the soil is eroded along with soil. It is important to prevent the soil erosion.



Soil Erosion

Preventive measures for soil erosion :

- (i) Deforestation should be prevented at all cost to prevent soil erosion.
- (ii) Shelter belt plantation of trees is good for preventing soil erosion.
- (iii) Terrace farming in hilly areas is effective in preventing soil erosion.

SOIL POLLUTION

Soil pollution is considered as a major challenge for healthy environment. The weathering of earth's crust forms soil over the centuries that supports the variety of microscopic and macroscopic life-forms. The top most layer of soil is composed of minerals of various sizes and organic matters along with pores filled with air and water. A number of waste products, chemicals and pesticides pollute the soil. Polythene bags and plastics pollute the soil. They also kill the organisms living in the soil.



Soil Pollution

Control of Soil Pollution

- (i) Use of pesticides should be minimised. Biological control should be known and implemented.
- (ii) Use of fertilizers should be highly judicious.
- (iii) Cropping techniques should be improved so that weeds are unable to take foothold in the fields. This would automatically reduce the use of weedicides or herbicides.

- (iv) Special pits or low lying areas are selected for dumping of industrial wastes.
- (v) Improvement in mining techniques and transport of extracted materials so that spread of mine dust should be minimised. The area should not be left barren and dry. Instead, afforestation should be carried out as soon as it becomes feasible.
- (vi) Controlled grazing and proper forest management.
- (vii) Wind breaks and wind shields in areas exposed to wind erosion.
- (viii) Planting of soil binding grasses and other perennials along the banks and slopes prone to rapid erosion.
- (ix) Afforestation and reforestation.

◀ REWIND

- ▶ Soil is essential for growing trees and crops to provide food to all the humans and animals.
- ▶ During the past millions of years, the rocks had been going on broken by violent earthquakes.
- ▶ Soil is important for life on the earth.
- ▶ Soil profile is a section through different layers of the soil. Various layers are called horizons.
- ▶ Soil is of different types: clayey, loamy and sandy.
- ▶ Percolation rate of water is different in different types of soil. It is highest in the sandy soil and least in the clayey soil.
- ▶ Different types of soils are used to cultivate different types of crops. Clay and loam are suitable for growing wheat, gram and paddy. Cotton is grown in sandy loam soil.
- ▶ Soil holds water in it, which is called soil moisture. The capacity of a soil to hold water is important for various crops.
- ▶ Clayey soil is used to make pots, toys and statues.



ASSIGNMENTS



COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

a. The rotting dead matter in the soil is called

(i) Humus

(ii) Salt

(iii) Sugar

(iv) Protein

b. Breaking down of rocks by the action of wind and climate is called

(i) Weathering

(ii) Soil vegetation

(iii) Cultivating

(iv) Percolation

c. The soil containing greater portion of big particles is called

(i) Sandy soil

(ii) Loamy soil

(iii) Clayey soil

(iv) Alluvial

- d. The soil containing greater portion of fine particles is called
 (i) Sandy soil (ii) Loamy soil (iii) Clayey soil (iv) Alluvial soil
- e. The soil containing same amount of large and fine particles is called
 (i) Sandy soil (ii) Loamy soil (iii) Clayey soil (iv) Alluvial soil
- f. Water can drain easily through
 (i) Sandy soil (ii) Loamy soil (iii) Clayey soil (iv) Alluvial soil

2 Fill in the blanks :

- _____ is a mixture of sand, clay and silt.
- A vertical section through different layers of the soil is called _____.
- The mixture of rock particles and _____ is called the soil.
- _____ rate of water is different in different types of soil.
- The best top soil for growing plants is _____.
- Soil pollution is considered as a major challenge for healthy _____.

3 Write T for true and F for false :

- Sandy soil retains more water.
- The C-horizon is the layer of soil which is rich in humus.
- Humus is important because it makes the soil fertile.
- Soil is affected by climatic factors.
- The soil which is light, well aerated and rather dry is clayey soil.
- In deserts, soil erosion occurs through wind.

4 Match the following :

- | | |
|------------------------------|-----------------------------|
| a. Upper layer of soil | (i) Large particles |
| b. Middle layer of soil | (ii) All kinds of soil |
| c. Sandy soil | (iii) Small particles |
| d. A home of living organism | (iv) Lesser amount of humus |
| e. Clayey soil | (v) Humus |
| f. Loamy soil | (vi) Dark colour |

5 Answer these questions in short :

- Name two agents of weathering.
- What is the uppermost layer of soil?
- Which type of soil can hold more water?
- Name the soil which is a mixture of sand, clay and silt.

- e. Write a short note about soil pollution.
- f. How do farmer stops soil erosion?

6

Give answers in detail :

- a. Write a detailed note on soil formation.
- b. List the differences between clayey soil and sandy soil.
- c. Describe the different layers of soil with the help of a labelled diagram.
- d. Explain how soil erosion and soil pollution can be prevented.
- e. Explain with an activity, the percolation rate of water through the soil.
- f. How can we check soil pollution?

NEP QUERY



PROJECT WORK

- ◆ In the laboratory : To find out the moisture content of a soil sample, weight 100 g. Place the soil on a newspaper in the afternoon sun and allow it to dry for two hours. Weight the soil again. The difference in the weights gives the amount of moisture contained in 100 g of soil. This is called the percentage moisture content.

(Learning by Experiment)

COMMUNICATION SKILLS

- ◆ Hold a group discussion in your class on the topic- " **Which is more important inside the soil : moisture or air?** "

(Discussion)

CRITICAL THINKING

Name the type of soil that is most suitable for each of the following :

(Self-Awareness)

- a. Growing cotton _____
- b. Making pot _____
- c. Growing groundnut _____
- d. Growing coffee, tea _____

Animal Fibres

3

- Wool Fibres
- Rearing of Sheep

- Silk Fibres
- Sericulture or Rearing of Silk Moths

The records of man's use of textiles, dates back thousands of years before the birth of Christ. Our knowledge of the early development in textiles is very meagre as they are not evident through proper records. Textiles were first developed as a means for carrying food and as mats in shelter. Only in later stages it is used as clothing.

Clothes protect us from natural calamities like storm, heat, cold, sunlight, etc. There are clothes for every season. For example: woollen clothes for winters, muslin and viol for summers, raincoat for rainy season, fireproof clothes for protection against fire and so on. They protect us from minor injuries and help to maintain the body at a suitable temperature. The type of clothes worn by people depends on culture, occupation and daily needs. We wear special dresses at parties and while going out but we wear casuals at home.

WOOL FIBRES

Animals are well protected from heat and cold with the fur (hair) on their skin. Humans have no such protection. The early men used to protect their bodies by wearing animal furs (skin with hair). Gradually men learnt to use animal's hair for making fabrics and blankets.

All mammals have hair. Only when the hair is very thick, fine and silky, it is called **fur**. Often, fur grows in a kind of double coat. The inside coat is short and airy and extra-fine. It is covered by longer, coarser hair called guard hair. The underfur keeps the animals warm and the guard hair which shed water easily, help it to keep it dry.

The wool is obtained from hair of all the animals with fur . It is also called **fleece** or fur collectively. It is mainly a protein. Good quality wool is obtained from Merino sheep. They are specially reared. The fleece is 3-5 inches long and very fine and most valuable. A merino sheep may yield about 5 to 18 kg of wool per year.



Types of Sheep

Sheep's wool is one of the oldest animal fibres. Apart from sheep, wool also comes from the angora goat, yak, llama, alpaca, and even camels. All these animals have a thick coat of hair or fleece, which keeps them warm. Wool is obtained from this fleece.

Sheep wool is more commonly available in India. In Tibet and Ladakh, yak wool is more common, whereas in Jammu and Kashmir, it is the angora wool. The famous Pashmina shawls come from the soft underfur of the Kashmiri goat. In South America, the llama and the alpaca yield wool. Hair of camel are also used to make clothes. Camels are reared in deserts. The Bactrian camels give us the best quality wool. They are two humped camels found in Siberia, Mongolia and China. Alpaca and llama also belong to the camel family.

Finer and softer wool is used for making shawls. Good quality wool is used for making fabrics and hosiery. Coarse wool is used for making blankets and carpets.

REARING OF SHEEP

Sheep rearing is a branch of animal husbandry. Sheep are reared mainly for their meat, wool, and milk. In India, sheep are reared generally in the sates of Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Arunachal Pradesh and Sikkim, or the plains of Haryana, Punjab, Rajasthan and Gujarat.

Sheep require less place for living. In domestic rearing, sheep can be kept with other livestock animals. You must have seen the flocks of grazing sheep in the fields. Sheep are herbivores and prefer grass and leaves. However, shepherds also feed them mixtures of pulses, corn, jowar, oil cakes and minerals. In winter, they are fed leaves, grain and dry fodder.



Sheep Farm

In India, sheep are reared in the hills of Himachal Pradesh and Arunachal Pradesh, and in the plains of Haryana, Punjab, Rajasthan and Gujarat.

The animals need proper care and feed. So they should be washed and cleaned on regular basis. They should be protected from mosquitoes and flies. They should give proper treatment in case of illness.

Vicunas are being killed on a large scale to obtain fleece. Its fibre is rare and very expensive, costing a few thousand rupees per yard. The vicuna is being protected by rigorous conservation measures.

Obtaining Wool

Wool is obtained by shearing or shaving the hair off the skin by a machine. The shearing is done once in a year in the spring or early summer. After shearing, the animal is washed in a water tank containing antiseptic so as to prevent infection.

Processing of Wool

Shearing

The fleece of the sheep along with a thin layer of skin is removed from its body. This process of removing fleece from the body of the sheep is known as shearing. Machines similar to those used by barbers are used to shear off fleece from the body of the sheep.

Scouring

The sheared skin with hair is thoroughly washed in tanks to remove grease, dust and dirt. This process of removing grease, dust and dirt from the sheared skin is known as **scouring**. In the old days it is done manually, but now a days it is done by machines.



Scouring

Sorting

After scouring, sorting is done. The hairy skin is sent to the factory where hair of different textures are separated or sorted. This process of separation of wool is known as sorting.

Dyeing

The natural fleece of sheep and goats being black, brown or white, the fibres are dyed in various colours. This process is known as **dyeing**.

Rolling

The process by which fibres are straightened, combed and rolled into yarn is called **rolling**. The longer fibres are made into wool for sweaters and the shorter fibres are spun and woven into woollen clothes.



Rolling

Types of Woollen Fabrics

There are two types of wool yarn : woollens and worsteds.

Woollens

Woollen is a general term describing various fabrics woven from woollen yarn that is spun from the shorter wool fibres. These shorter fibres are not combed to lie flat as in the worsted yarn. This results in soft surface textures and finishes and the weave of individual yarns does not show as clearly as in worsted fabrics. Fabric woven from rough yarn having mixed lengths of fibres is called **tweed**. It is bulky fabric having rough texture.

Worsteds

Worsted is a general term for fabrics woven from worsted yarns that contain longer fibres spun from combed wool. Worsted wool refers to tightly woven, smooth, clear finished goods in a variety of twill and other stronger weaves. Fabric woven from smooth long fibres of worsted yarn is called worsted fabric. It has smooth texture.

Some trade names of woollen fabrics are erge, gabardine, flannel and crepe. Serge is a strong woollen cloth used for making suits, trousers etc. Gabardine is a strong cloth which does not allow water to go through and is often used to make coats. Flannel is a soft cloth. Crepe is light soft thin cloth with very small folded lines on its surface.



Wool Mark

Quality Marks on Woollen Cloth

The best quality woollens bear a trademark Woolmark. It is not a brand label, it is a label originally issued by the International Wool Secretariat, Ilkley, North Yorkshire (U.K.) after passing and maintaining several standards under constant inspection by the secretariat.

SILK FIBRES

Silk is a fabric that is used throughout the entire world. It is a very elegant fabric that many people pride to have. Silk fabric shines with a soft light that looks as if it shakes slightly. It is because the silk fibres have triangular prism-like structure.



Silk

History of Silk :

Silk was discovered in China; around 3500 BC. Chinese legend has it that empress Si-lung-Chi was worried about the damaged mulberry leaves in her garden. Emperor Huang-ti found out that the white worms were eating up the mulberry leaves and spinning shiny cocoons. A cocoon accidentally dropped into the empress's cup of hot tea, and the delicate tangle of threads separated from the cocoon, thus leading to the discovery of silk.

This was kept secret for hundreds of years. Later this was introduced in other countries by traders and travelers much later. The route they travelled was called the silk route.

As per historians, silk was produced in India also. Proof of use of silk in during the Indus Valley Civilization has also been found.

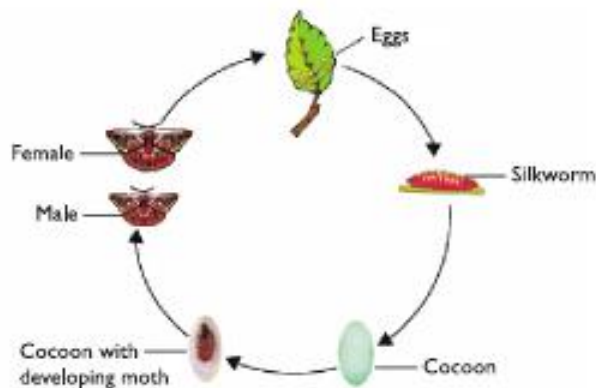
Properties of Silk

Some special properties of the silk fibre are the following:

Silk is one of the strongest natural fibres, but it loses up to 20% of its strength when wet. Silk is lustrous. It reflects light and is caused by the silk filament and gives shimmering appearance. Silk can be spun to be the longest of all natural fibres. Therefore it is produced in very thin threads. Silk is light weight. Silk is highly elastic (next to wool). Silk keeps warm in winter and cool in summer. Silk resists wrinkling. It can be ironed easily. Silk has a smooth, soft texture that is not slippery, unlike many synthetic fibers. Silk is resistant to moths and dust mite.

SERICULTURE OR REARING OF SILK MOTHS

Sericulture, or silk farming, is the rearing of silkworms for the production of raw silk. Silkworms are reared under suitable conditions of temperature and humidity to obtain silk threads from their cocoons. The female silk moth lays hundreds of eggs. These are stored on



Life Cycle of Silkworm

strips of cloth or paper. Mulberry leaves are the staple diet of silkworms. When the mulberry tree bears a fresh crop of leaves, the eggs are warmed suitably so that the larvae hatch from them. The larvae, caterpillars or silkworms are then stored in clean bamboo trays and are fed freshly chopped mulberry leaves. They eat day and night, and grow to enormous sizes. The bamboo trays are provided with small racks or twigs to which the cocoons can be attached. This happens usually after 25 to 30 days when the caterpillars stop

feeding and move to the twigs to spin cocoons. The silk moth develops inside these cocoons. Silk is obtained by heating these cocoons. First of all, cocoons are boiled and then silk fiber is separated out; using machines. Machine unwinds the silk thread from cocoons. The process by which silk fiber is obtained is called **reeling the silk**.

Spinning of Silk Thread

A single larvae can produce about 700 m to 1500 m of silk filament. This filament is very thin. Raw silk is twisted into a strand sufficiently strong for weaving or knitting. This process of creating the silk yarn is called "throwing," and prevents the thread from splitting into its constituent fibres.

Four different types of silk thread may be produced from this procedure: crepe, tram, thrown singles and organzine. Crepe is made by twisting individual threads of raw silk, doubling two or more of these together and then twisting them again. Tram is made by twisting two or more threads in only one direction. Thrown singles are individual threads that are twisted in only one direction. Organzine is a thread made by giving the raw silk a preliminary twist in one direction and then twisting two of these threads together in the opposite direction.



Spinning of Silk

Weaving of Silk Cloth

To weave cloth a yarn (called the weft) is passed crosswise through multiple lengthwise yarns (called the warp). The warp yarns are separated by lifting one harness at a time which allows the weft shuttle to be passed between the warp yarns. Organzine thread is used as the multiple lengthwise yarn, warp and tram thread is used as the crosswise weft or filling.

CULTIVATED SILK AND WILD SILK

Cultivated Silk

Cultivated silk has certain properties where it is fine, smooth, lustrous and has a soft handle and elegant drape. This has become a highly technical and controlled industry where silk worms are raised and fed mainly on mulberry leaves to produce a near white coloured silk.

Wild Silk

This is usually not white and is from the tannins from eating plants other than the mulberry tree. It can be coarser than cultivated silk, making it better for high wear items. It is cheaper. Wild silk cocoons are usually gathered after the moth has emerged, therefore the staples or fibres will have been cut, making it only suitable for spinning. It is also known as Tussar Silk. Muga and Endi are other examples of wild silk.

ROUGHNESS AND SMOOTHNESS OF SILK

We observe different kinds of fibres around us. We can observe these fibres or threads with a hand lens. Fibres of cotton and wool have many crimps or folds and scales. These crimps and scales make them rough in appearance and touch. Silk fibres have very small crimps and the surface is not very uneven. Thus silk is soft to touch. Synthetic fibres have no folds and their surface is smooth and softer to touch.

◀ REWIND

- ▶ In the beginning cotton and flax were used to make clothes.
- ▶ The type of clothes worn by people depends on culture, occupation and daily needs.
- ▶ Animals are well protected from heat and cold with the fur (hair) on their skin.
- ▶ The wool is obtained from hair of all the animals with fur.
- ▶ The famous Pashmina shawls come from the soft underfur of the Kashmiri goat.
- ▶ Sheep is the main animal reared for wool.
- ▶ Silk is a fabric that is used throughout the entire world.
- ▶ Sericulture, or silk farming, is the rearing of silkworms for the production of raw silk.
- ▶ Silk moth produces fibre which it wraps around its body and forms a ball-like cocoon.
- ▶ Silk obtained from the silk moth larvae reared on silk farm is called cultivated silk.
- ▶ Silk produced by undomesticated silk moths in the forest is called wild silk.



ASSIGNMENTS



COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

a. Silkworm is a

(i) Caterpillar

(ii) Larva

(iii) Egg

(iv) Adult moth

b. Which of the following do not yield wool?

(i) Yak

(ii) Camel

(iii) Goat

(iv) Woolly dog

c. Sheep are reared for getting

(i) Cotton

(ii) Jute

(iii) Wool

(iv) Rayon

d. The silk and wool are made of a kind of

(i) Fats

(ii) Carbohydrates

(iii) Proteins

(iv) Vitamins

2 Fill in the blanks :

- Silk was discovered in _____.
- A weaver weaves silk thread into _____.
- Cotton and jute are _____ fibres.
- A female silk moth lays _____ eggs.
- Gabardine is a tradename of _____ fabric.

3 Write T for true and F for false :

- Wool is a synthetic fibre.
- Camel is a wool giving animal.
- Silk was discovered in China.
- Animal fibres include mohair, angora and cashmere.
- Silk is made by the caterpillar of the silkworm.

4 Answer these questions in short :

- Why is wool fibre so popular?
- Why are the silk fibres lustrous?
- What are silk cocoons?
- What do you mean by sheep breeding?
- What is sericulture?

5 Give answers in detail :

- Write a detail description of wool processing.
- Name and describe any two types of silk threads used in the making of silk cloth.
- Write about popularity of wool and silk.
- What do you mean by roughness and smoothness of fibres?
- Differentiate between cultivated silk and wild silk.

NEP QUERY



PROJECT WORK

- In the laboratory :

Take threads of natural silk, artificial silk, natural wool and synthetic wool. The threads should be about 10 cm long. Burn each of the thread by holding it from one end with tongs and the other end over a candle flame. Observe the flame, smell and the residue formed in case of each thread. Write the observations in a table.

(Learning by Experiment)

Nutrition in Animals

4

- Nutrition in Animals
- Modes of Nutrition

- Nutrition in Amoeba
- Nutrition in Humans

- Nutrition in Ruminants

Nutrition can be defined as the process by which an organism obtains food which is used to provide energy and materials for its life sustaining activities. Plants make their food by the process of photosynthesis, but animals cannot make their food themselves. Animals get their food from plants. Some animals eat plants directly while some animals eat plant, eating animals. Thus, animals get their food from plants either directly or indirectly. Animals nutrition covers nutrient requirements, how the food is consumed, and how the body utilises the food.

NUTRITION IN ANIMALS

Animals eat readymade food, be it a plant or an animal. Thus, they are heterotrophic in nutrition. They consume solid organic matter as their food. This type of heterotrophic nutrition is called **holozoic**.

The nutrition in animals is performed through five steps :

Ingestion : The intake of food is called **ingestion**. Method of ingestion, i.e., taking of food, varies from one animal to another.

Digestion : The process of breaking down of complex component of food into simpler substances is called **digestion**.

The process of digestion is different in humans, grass eating animals, amoeba, etc.

Absorption : The process of passing of digested food into blood vessels in the intestine is called the **absorption**.

Assimilation : The conversion of absorbed food in complex substances such as proteins and vitamins required by body is called **assimilation**.

In other words, assimilation is the conversion of absorbed food (nutrients) into living tissues. Through the process of assimilation our cells are supplied with oxygen and nutrients.

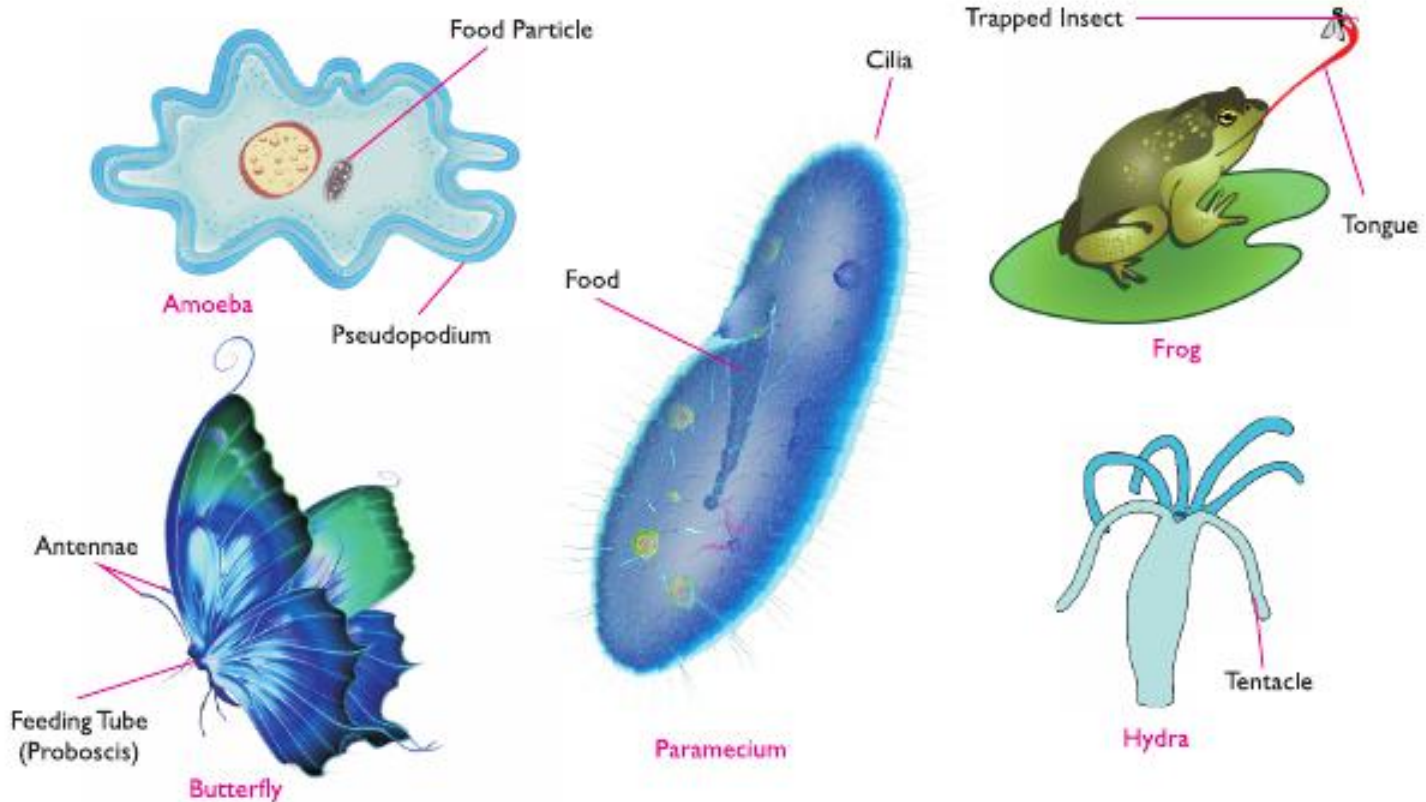
Egestion : Removal of waste materials from the body is called **egestion**. The faecal matter is removed through the anus from time-to-time.

In the unicellular (single-celled) animals such as amoeba all these steps in the process of nutrition are performed by the single cell. Higher organisms have special organs in their bodies for each of these steps of ingestion, digestion, absorption, assimilation and egestion.

MODES OF NUTRITION

Ingestion

The first activity of the digestive system is to take in food through the mouth. This process, called **ingestion**, has to take place before anything else can happen. It varies in different animals.



Unicellular organisms like an amoeba do not have mouth, therefore food intake may take place from any part of the body.

Hydras extend their body to maximum length and then slowly extend their tentacles to ingest the food.

Cockroaches have special mouth parts for sucking, catching and cutting the food.

The mode of ingestion of a butterfly is through the long proboscis.

The mosquitoes suck the blood of animals through their feeding tubes.

Houseflies use their saliva to dissolve their food and form a solution. Then they suck the solution through their feeding tubes.

A starfish doesn't eat through its mouth. Starfish captures its food with the help of tube feet present on its body. It actually shoots its stomach out of its body to capture food through its mouth and wraps the stomach around the captured animal. Then it withdraws stomach into the body.

A frog projects its long tongue to catch its food.

Birds have beaks to pick, cut, crush and take in their food.

The mammals have well developed organs to take in their food.

Humans ingest their food in their mouth with the help of their hands. They cut and grind the food with their well developed jaws and then swallow it.

Digestion

The process by which food is converted into substances that can be absorbed and assimilated by a living organism is called **digestion**. Every part of food we eat has to be broken down into nutrients that can be absorbed by the body, which is why it takes hours to fully digest food. Organisms secrete enzymes to help digestion. Enzymes are special molecules of proteins produced in the body of the organism for this purpose. They break down food particles into simpler molecules through chemical reaction.

In lower organisms, food particles directly enter the cell and get digested. In complex higher organisms such as frog and man, the food passes through special organs of the digestive system for the digestion of food, which secrete enzymes. Such mode of digestion is called extracellular (outside the cells).

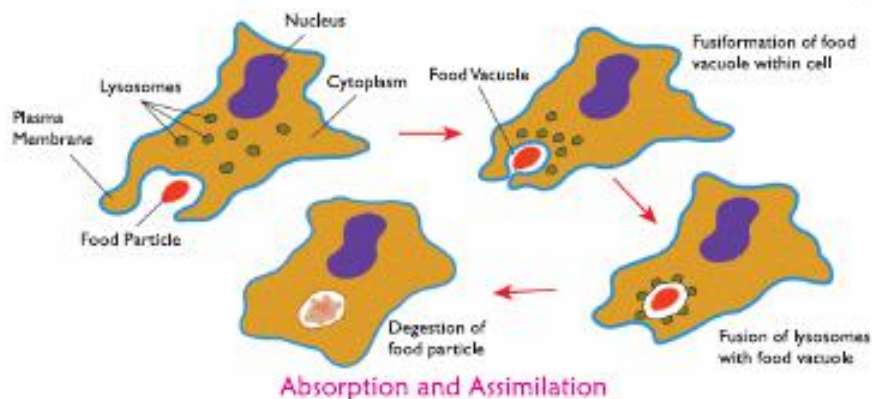
Absorption and Assimilation

Absorption : The process of passing of digested food into blood vessels in the intestine is called the **absorption**.

Assimilation : The conversion of absorbed food in complex substances such as proteins and vitamins required by body is called **assimilation**.

In other words, assimilation is the conversion of absorbed food (nutrients) into living tissues. Through the process of assimilation our cells are supplied with oxygen and nutrients.

Egestion : Removal of waste materials from the body is called **egestion**. The faecal matter is removed through the anus from time-to-time.



NUTRITION IN AN AMOEBA

The different processes involved in the nutrition of amoeba are:

1. **Ingestion** : Amoeba is a unicellular animal, so it doesn't have a mouth for ingestion of food. Amoeba ingests the food by encircling it by forming pseudopodia. When the food is completely encircled, the food is engulfed in the form of a bag called **food vacuole**.

2. **Digestion** : In amoeba, several digestive enzymes react on the food present in the food vacuoles and break it down into simple and soluble molecules.

3. **Absorption** : The food digested by digestive enzymes is then absorbed in the cytoplasm by the process of diffusion. While the undigested food remains in the food vacuole.

4. **Assimilation** : In this process, the absorbed food is utilized for obtaining energy, repair and growth is called **assimilation**.

5. **Egestion** : When a sufficient amount of undigested food gets collected in the food vacuole, it is thrown out of the body by rupturing cell membrane.

NUTRITION IN HUMANS

In humans, the food passes through a food canal (called alimentary canal or digestive tract). Alimentary canal is a long, muscular and coiled tube. It starts from the mouth and ends at anus.

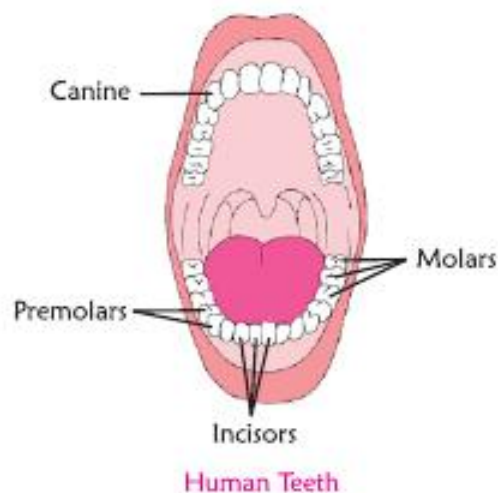
Ingestion

Mouth

In fact, digestion starts here as soon as you take the first bite of a meal. Chewing breaks the food into pieces that are more easily digested, while saliva mixes with food to begin the process of breaking it down into a form your body can absorb and use.

Teeth : Humans have two sets of teeth, primary (or baby) teeth and then permanent teeth, which develop in stages. Teeth are of three types based on their functions biting, tearing and grinding. But depending upon their shape and size they are of four types.

The four front teeth in each arch are called **incisors**, and their function is to cut food with their sharp thin edges. On each side of the incisors, at the corners of the mouth, are the **canines**. These teeth are very strong, very sharp and help in tearing of food. Behind the canines are the **premolars**, which are designed for holding food like the canines because they have cusps, but they also function to crush food. The teeth farthest back in the mouth are the **molars**. These teeth have broad chewing surfaces with four or five cusps, and are designed for grinding food. The incisors and canines are called **anterior teeth**, because they are located in the front of the mouth, while the premolars and molars are called **posterior teeth** because they are located in the back of the mouth.



Milk Teeth and Permanent Teeth : At birth a human infant has no teeth. First set of 20 teeth appears by the age of 2 years. These teeth are called milk teeth. Primary teeth are very important for maintaining good nutrition by permitting the child to chew properly, are involved in speech development, and help the permanent teeth by saving space for them and guiding them into position. By the age of 12 a new permanent set of 32 teeth is formed. They are called permanent teeth and last for the rest of the human life.

Structure of a Tooth

Each tooth has these external parts:

Crown : visible part of the tooth.

Neck : division between the crown and root of a tooth.

Root : part of the tooth implanted in the gum.

Apex : point of a tooth.

The internal structure of the tooth is the following :

Enamel : Enamel is the visible substance that covers the tooth crown. Harder than bone, enamel protects the tooth from decay.

Dentine : Underneath the enamel you find dentine, which is calcified and looks similar to bone. Dentine is not quite as hard as enamel, so it is at greater risk for decay and tooth get spoiled.

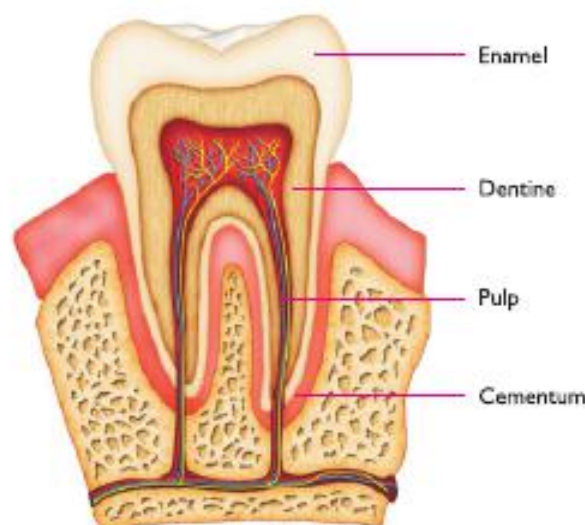
Cementum : This tissue covers the tooth root and helps anchor it (cement it) into the bone. It is softer than enamel and dentine; the best way to protect this softer tissue from decay is by taking good care of our gums.

Pulp : Pulp is found at the centre of our tooth and contains the blood vessels, nerves, and other soft tissues that deliver nutrients and signals to our teeth.

Tooth Decay : When we do not clean our teeth properly, some harmful bacteria start growing in the mouth. Germs in our mouth grow on the teeth every day and help in the formation of plaques. Plaque is the sticky deposits that collect, in particular, around the gum line, the edges of fillings and the grooved surfaces of the teeth. The plaque uses the food and drink that we eat to make acid. This acid attacks the tooth's outer layers and eats them away . Over time, a small hole known as a cavity can develop on the surface of the tooth . Once cavities have formed in the enamel, the plaque and bacteria can reach to the root of tooth. At this stage, nerves will be exposed to the bacteria, making tooth very painful and after some time, the tooth breaks away from jaw.

Tongue : The tongue is a muscular structure attached to the floor of the mouth. It is the main sensory organ of the taste sense. The upper surface of a tongue is covered with taste buds which contain taste receptors. The human tongue has on average 3,000 - 10,000 taste buds. It also helps swallowing the food. Humans also use the tongue for speech where it helps with changes in sound. The tongue also works as a natural way of cleaning teeth after eating.

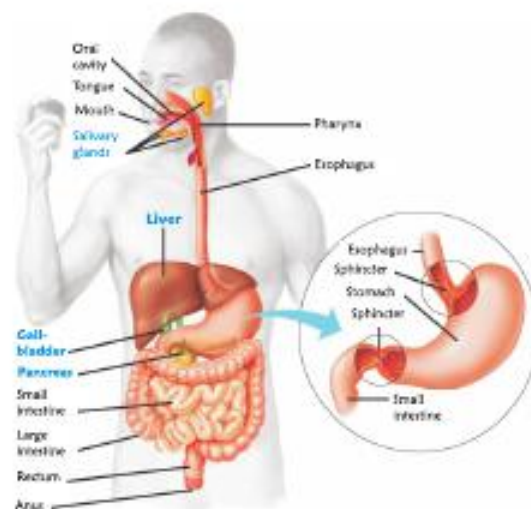
Salivary Glands : The salivary glands produce saliva, which keeps the mouth and other parts of the digestive system moist. There are three pairs of salivary glands in our mouth. Saliva contains an enzyme called amylase (also called ptyalin). Amylase acts on starch and changes it into maltose, a kind of sugar. This sugar is sweet and soluble in water.



Structure of Teeth

Digestion

The process of digestion starts as soon as the food is mixed with saliva. Saliva contains amylase (ptyalin) enzyme which helps in the breakdown of starch into sugar. Saliva also helps the food to be easily swallowed. Then the food passes from the mouth into a long tube called the oesophagus or food pipe. Muscles in the oesophagus tighten and relax to create a wave-like process called peristalsis which pushes food down the small tube, which is why your food never falls back out if you happen to be eating and swallowing upside down. This movement helps to move the food into a large sac-like muscular organ called stomach. The adult stomach has a very small volume when empty but can expand to hold up to 1.5 litres of food when full. Main function of the stomach is mastication of food. The inner wall of the stomach secretes hydrochloric acid to help kill bacteria and, along with proteases enzymes, aids in the digestion of food. To protect itself from the corrosive acid, the stomach lining must create a thick coating of mucus. Now food is pushed along this tube by contractions of muscles in the intestinal walls. In the small intestine, muscle contractions mix chyme with enzymes produced by the pancreas, liver and intestinal lining. The bile, secretion from the gall bladder of the liver helps in the digestion of fats to fatty acids and glycerol. The secretion from the pancreas helps in the break down of carbohydrates into sugars, proteins into amino acids and fats into fatty acids and glycerol. Small intestine also releases some enzymes which further digests the carbohydrates, proteins and fats. It is here that nutrients are absorbed into the bloodstream.



Digestive System

Absorption

Digestion of all components of food is carried out and completed here. The end products of digestion are ready for absorption. The inner wall of the small intestine is covered with millions of microscopic, finger-like projections called villi. The villi are the vehicles through which nutrients can be absorbed into the body. Each villus has a network of fine blood vessels. Nutrients are absorbed into the blood present in these fine blood vessels.

The undigested food passes into the large intestine which absorbs water and minerals.

Assimilation

The absorbed food is carried by blood to different parts of the body for blending into cell components. The final products of digestion are glucose, amino acids, fatty acids and glycerol. Glucose is broken down in the cells with the help of oxygen into carbon dioxide and water to release energy. Amino acids are used for repairing worn out cells and tissues. Fatty acids and glycerol are stored as energy reserve for future use.

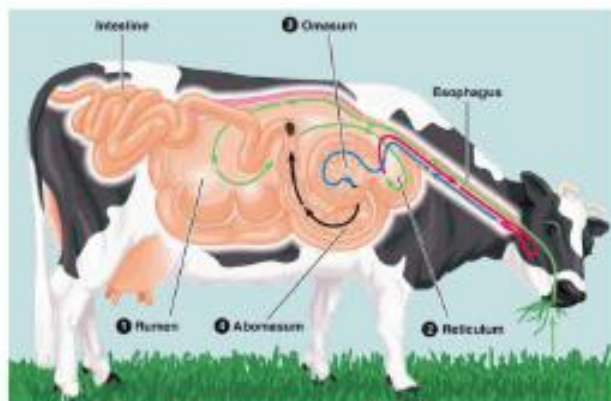
Egestion

The undigested food is stored in the rectum and is excreted out through anus as faeces from time to time.

NUTRITION IN RUMINANTS

Ruminants are also called as grass-eating animals. These are hooved animals whose body stands on four limbs. They include cows, buffaloes, sheep, giraffe, camel, deer etc. They also possess special type of digestive tract which extracts nutrition from the food they eat.

➤ Molars are very strong as these animal chew the cud.



Digestive System of Cow

Grass eating animals swallow the food quickly and store it in the rumen. All animals who chew are called **ruminants**. Their stomach is divided into four chamber the rumen, reticulum, omasum and abomasum.

First of all, half chewed food is swallowed and it then goes from mouth to the rumen, the first chamber of the stomach. Here, it is acted upon by bacteria. These microorganisms digest the cellulose. This half digested food goes to the second muscular chamber; the reticulum. From the reticulum the food is sent back to the mouth; as cud; to be chewed again. Chewing of the cud is called **rumination** and such animals are called **ruminating animals or ruminants**. Cow, goat, buffaloes, sheep, bison, etc. are good example of ruminating animals. The re-chewed food is swallowed for the second time. After passing the first two chambers it enters the third chamber; the omasum. Here the food is further broken down into smaller pieces and finally enters the fourth chamber, the abomasum. Here, all enzymes act upon the food and the digestion is completed.

From the fourth chamber, the food enters the small intestine, where the absorption of nutrients occur.

◀ REWIND

- ▶ Animal nutrition includes nutrient requirement, mode of intake of food and its utilisation in the body.
- ▶ The human digestive system consists of the alimentary canal and secretory glands.
- ▶ The stomach wall and the wall of the small intestine also secrete digestive juices.
- ▶ The modes of feeding vary in different organisms.
- ▶ Digestion is a complex process involving: (i) ingestion, (ii) digestion, (iii) absorption, (iv) assimilation and (v) egestion.
- ▶ The absorbed substances are transported to different parts of the body. Water and some salts are absorbed from the undigested food in the large intestine.
- ▶ The undigested and unabsorbed residues are expelled out of the body as faeces through the anus.
- ▶ The grazing animals like cows, buffaloes and deer are known as ruminants. They quickly ingest, swallow their leafy food and store it in the rumen. Later, the food returns to the mouth and the animal chews it peacefully.
- ▶ Amoeba ingests its food with the help of its false feet or pseudopodia. The food is digested in the food vacuole.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- The breakdown of complex components of food into simpler substances is called
 (i) Digestion ☐ (ii) Nutrition ☐ (iii) Breathing ☐ (iv) Respiration ☐
- The set of teeth that exists in the mouth during the period from six to eight years of a human being are called
 (i) Permanent teeth ☐ (ii) Premolar teeth ☐ (iii) Molar teeth ☐ (iv) Milk teeth ☐
- The function of the digestive juices present in the stomach is to breakdown
 (i) Proteins into simpler substances ☐ (ii) Starch into sugars ☐
 (iii) Fats into juices ☐ (iv) Food into gases ☐
- The process of digestion taking place in grass-eating animals is called
 (i) Egestion ☐ (ii) Rumination ☐ (iii) Assimilation ☐ (iv) Absorption ☐
- The intake of food is called
 (i) Digestion ☐ (ii) Absorption ☐
 (iii) Ingestion ☐ (iv) Assimilation ☐
- Single _____ celled organism among the following is
 (i) Human being ☐ (ii) Yeast ☐ (iii) Cow ☐ (iv) Amoeba ☐

2 Fill in the blanks :

- Animals eat _____ food.
- Four front teeth in each is called _____.
- The adult stomach can hold _____ litres of food.
- Liver secretes bile that helps to digest _____.
- _____ is found in centre of our tooth.
- The final products of digestion are glucose, amino acids, fatty acids and _____.

3 Write T for true and F for false :

- Saliva helps in the breakdown of proteins.
- Bile is stored in the liver.
- The modes of feeding vary in different organisms.
- Glycerol is stored as energy reserve.
- Ruminants are plant-eating animals.
- The function of molar teeth is grinding of food.

4 Match the following :

- | | |
|--------------------|---------------------------------|
| a. Salivary gland | (i) Bile juice secretion |
| b. Stomach | (ii) Storage of undigested food |
| c. Liver | (iii) Saliva secretion |
| d. Rectum | (iv) Acid release |
| e. Small intestine | (v) Digestion is completed |
| f. Large intestine | (vi) Absorption of water |

5 Answer these questions in short :

- What do you mean by saliva?
- Give an account of teeth structure of human.
- Where is bile produced? Which component of food does it digest?
- Describe the nutrition in an amoeba.
- What do you mean by assimilation? Why is it important?

6 Give answers in detail :

- Describe digestive system with help of a labelled diagram.
- Describe in detail that ingestion of food in amoeba and hydra.
- What is the function of pancreatic juice?
- Describe the various types of teeth in humans, and their roles in digestion.
- Write a note on digestion in ruminants.
- What are the different parts of nutrition in animals?

NEP QUERY



PROJECT WORK

- Make a model of the digestive system using rubber tube to make the oesophagus and small intestine. (Learning by Experiment)

CRITICAL THINKING

- Find out various factors which do not allow the proper functioning of digestive system. (Self-Awareness)

COMMUNICATION SKILLS

- In the classroom : Role of different types of teeth in digestion. (Discussion)
Hold a debate on above topic in classroom.

Nutrition in Plants

5

- Modes of Nutrition in Plants

- Other Modes of Nutrition in Plants

We already know that the food is essential for all living organisms to perform life processes such as respiration, growth, repair and replacement of damaged body cells or tissues, excretion, reproduction and response to stimuli, etc. The components of food which are necessary for our body are called **nutrients**.

Plants can make their food themselves but animals including humans cannot. Animals including human beings take ready-made food from plants or animals (that eat plants) because they cannot prepare their own food. Thus the green plants are called **producers** of food and animals as **consumers**. **Primary consumers** directly eat the plants while **secondary consumers** eat the primary consumers. Thus the secondary consumers also indirectly depend upon plants.

MODE OF NUTRITION IN PLANTS

It is the mode of nutrition in which the organisms manufacture their own food. Plants are the only organisms to do that. They use water, carbon dioxide and minerals. The raw materials are present in their surroundings.

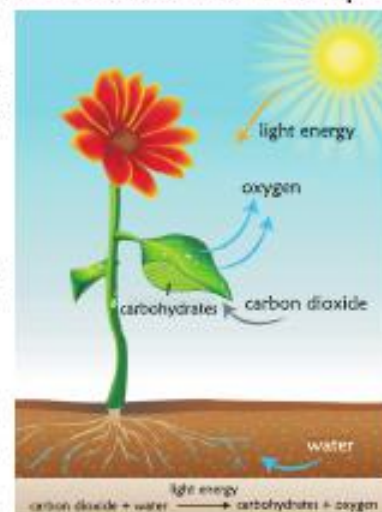
The nutrients enable living organisms to build their bodies, to grow, to repair damaged parts of their bodies and provide the energy to carry out life processes. **Nutrition** is the mode of taking food by an organism and its utilisation by the body. Plants obtain their nutrition by various modes. The mode of nutrition in plants can be autotrophic mode or heterotrophic mode.

The mode of nutrition in which organisms make food themselves from simple substances is called **autotrophic** (auto = self; trophos = nourishment) nutrition. Therefore, plants are called **autotrophs**. Animals and most other organisms take in readymade food prepared by the plants. They are called **heterotrophs** (heteros = other).

Photosynthesis - Food Making Process in Plants

The word "photosynthesis" is a combination of two Greek words: photo and synthesis. "Photo" means light, and "synthesis" means to make.

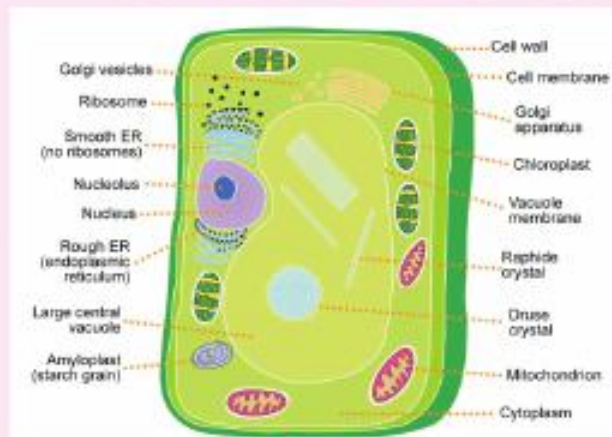
Leaves are the food factories of plants. The synthesis of food in plants occurs in leaves. Photosynthesis is a process by which green plants can synthesis glucose from raw materials like carbon dioxide and water in the presence of sunlight and chlorophyll.



Photosynthesis

Cells

A building is made up of many bricks, which are its basic units of structure. Similarly, every living thing is made up of tiny living structures called **cells**. Cells can be seen only under the microscope. Some organisms are made of only one cell. They are called **unicellular**. For example: amoeba, paramecium etc. Living organisms made up of many cells are called **multicellular** like human, tree etc. The cell is enclosed by a thin outer boundary, called the **cell membrane**. This membrane encloses a jelly-like substance called **cytoplasm** and a small object called a **nucleus**.



Cells

Water and minerals absorbed by roots are conducted upwards to the leaves, flowers and other parts of the plant by the vessels which run like pipes throughout the root, the stem, the branches and the leaves. They form a continuous path or passage for the nutrients to reach the leaf.

In the absence of photosynthesis, there would not be any plants. The survival of almost all living organisms directly or indirectly depends upon the food made by the plants. Besides, oxygen which is essential for the survival of all living organisms is produced during photosynthesis. In the absence of photosynthesis, life would be impossible on the earth.

Green plants make food in the form of carbohydrates by combining carbon dioxide and water using energy from sunlight. Carbohydrates are chemicals containing only the elements carbon, hydrogen and oxygen.

We can summarise the process of photosynthesis as follows -



During this process, oxygen is released. The carbohydrates ultimately get converted into starch. The presence of starch in leaves indicates the occurrence of photosynthesis. The starch is also a carbohydrate.

Following are necessary for photosynthesis to take place :

- Presence of sunlight
- Presence of water
- Presence of chlorophyll
- Presence of carbon dioxide

We often see slimy, green patches in ponds or in other stagnant water bodies. These are generally formed by the growth of organisms called **algae**. Algae are plants. Often algae grow near the shallow waterlogged areas such as near tube-wells, taps, etc. Algae can also prepare their own food by photosynthesis.



Activity

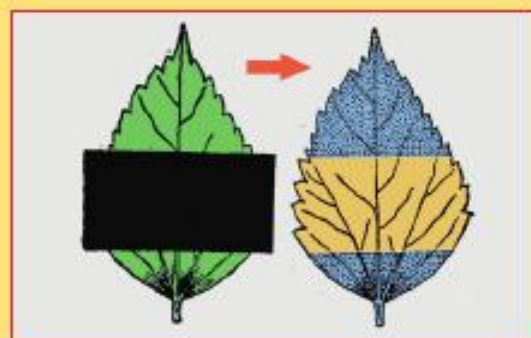
- ▶ To show that starch is formed during photosynthesis.

Materials Required : Leafy plant, Black paper, Alcohol, Iodine solution

Procedure : Identify a healthy green leaf on a plant and cover a portion of it on both sides with two uniform pieces of black paper, either in the morning before the sun rises or the previous evening so that the experiment is performed with a starch-free leaf. Or, keep a healthy, green pot plant in a dark room for 1 or 2 days so that its leaves become starch-free, and then cover a portion of a leaf of this plant as above.

Fix the pieces of paper properly with clips. To make sure that there is no starch, collect a few neighbouring leaves in the morning, decolourize them with alcohol and dip them into iodine solution.

Observation : The exposed portion turns blue or black, showing the presence of starch in it. The screened portion, on the other hand, turns yellowish brown as no starch formed in it, the yellow colour is due to the action of iodine solution on protoplasm and cellulose.



Synthesis of plant food other than carbohydrates

You have just learnt that plants synthesise carbohydrates through the process of photosynthesis. The carbohydrate group includes sugars, starches, and cellulose. These are used to synthesise other components of food. Plants need proteins and fats besides the carbohydrate. But proteins are nitrogenous substances which contain nitrogen.

Nitrogen is present in abundance in gaseous form in the air. However, plants cannot absorb nitrogen in this form. Certain types of bacteria called rhizobium, are present in soil which convert gaseous nitrogen into a usable form and release it into the soil. These soluble forms are absorbed by the plants along with water. Sometimes farmers add nitrogenous fertilizer to their field to fulfil the need of nitrogen. In this way plants get fulfillment of nitrogen along with other nutrients.

OTHER MODES OF NUTRITION IN PLANTS

There are some plants which do not have chlorophyll. They cannot synthesise their food. Like humans and animals such plants depend on the food produced by other plants. They use the **heterotrophic mode** of nutrition. The word Heterotrophic is the combination of two words i.e. Hetero + Trophos. Hetero means 'others' and 'trophos' means nourishment. If organisms depend on others for their food, it is called the **Heterotrophic Nutrition**.

According to the mode of nutrition, heterotrophic plants are of four types : Parasitic, Symbiotic, Saprophytic and Insectivorous.

Parasitic Plants

Do you see yellow tubular structures twining around the stem and branches of a tree? This is a plant called cuscuta (amarbel). It does not have chlorophyll. It takes ready made food from the plant on which it is climbing. The plant on which it climbs is called a **host**. Since it deprives the host of valuable nutrients, it is called a **parasite**.



Parasitic Plants

Some plants are total parasite while some are partial parasite.

A total parasite fully depends on other plants for their nutrition. For example - cuscuta. Partial parasite is a parasite that receives a part of its nutrients from host. For example: mistletoe bears green leaves. It synthesises its own food, but receives water and mineral from host plant.

Symbiotic Plants

Some organisms live together and share shelter and nutrients. This is called **symbiotic relationship**. For example: certain fungi live in the roots of trees. The tree provides nutrients to the fungus and, in return, receives help from it to take up water and nutrients from the soil. This association is very important for the tree.

In organisms called **lichens**, a chlorophyll-containing partner, which is an alga and a fungus live together. The fungus provides shelter, water and minerals to the alga and, in return, the alga provides food which it prepares by photosynthesis.

Roots of leguminous (dicotyledon) plants such as peas and beans contain bacteria called rhizobium which convert atmospheric nitrogen to nitrate, ammonia etc. The plant can use them easily. Plants need nitrogen to make proteins. Bacteria get nutrients from the plant for their growth.

Insectivorous Plants

Have you seen or heard of plants that can eat animals? Such plants may be green or of some other colour. In a pitcher plant, the leaf lamina is modified to form a pitcher-like structure and the leaf tip is modified into a lid. The bright colour of the pitcher makes it very attractive to insects. Inside the pitcher, there are several hair-like structures. These hairs direct the trapped insects downwards. When a flying insect happens to enter the pitcher, the lid is closed upon it. The insect growth is digested inside. The biggest insectivorous plants are tropical pitcher plants in the island of Borneo (Southeast Asia) that catch frogs and venus flytrap. Such insect-eating plants are called **insectivorous plants**.

Saprotrophs

Saprotrophs are non-green plants e.g. agaricus (mushroom) fungi, yeasts and bacteria. Saprotrophs get their food from dead or decaying organic matters. They grow on decaying organic matters such as cow dung, wood, bread, etc.



Saprotrophs

Saprophytic decay bacteria and fungi are essential decomposers which recycle vital nutrients back into the ecosystem. This mode of nutrition in which organisms take in nutrients in solution form from dead and decaying matter is called **saprotrophic nutrition**. Plants which use saprotrophic mode of nutrition are called **saprotrophs**. The terms 'Sapros' refers to rotten and 'trophic' refers to food. In this mode of nutrition, digestive enzymes are secreted on the food. The digested food is then absorbed by the organism. In saprotrophs, digestion takes place outside the body of the organism.

How nutrients are replenished in the soil

We have learnt that plants absorb mineral nutrients from the soil. So, their amounts in the soil keep on declining. Hence, to replenish those nutrients, fertilizers are added to the soil.

Plants require a lot of nitrogen to grow and to be healthy. Although, a lot of nitrogen is available in the atmosphere, but plants cannot absorb nitrogen in gaseous form. There is a bacterium called Rhizobium, which lives in the root nodules of legumes, such as gram, moong, etc. These bacteria can absorb nitrogen from the atmosphere and make it available for the plants. Rhizobium cannot make its food, hence, it provides nitrogen to the legumes and in return legumes provide them food and shelter. This is an example of symbiotic relationship. This has great importance for farmers. Fields in which leguminous plants are grown from time to time, do not need addition of nitrogenous fertilizers.

◀ REWIND

- ▶ Nutrition, movement, respiration, growth, excretion, reproduction and response to stimuli are some of the basic life processes.
- ▶ The whole process of taking food by an organism, digesting and using it for growth and producing energy in its cells through respiration is called nutrition. In the presence of sunlight and chlorophyll, plants prepare their own food from carbon dioxide and water and release oxygen. This process is called photosynthesis.
- ▶ Non-green plants take their nutrition (i) from other living plants (ii) share with other living plants (iii) from dead and decaying organisms (iv) from insects.
- ▶ Fields in which leguminous plants are grown from time to time, do not need addition of nitrogenous fertilizers.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- The components that are necessary for our body are called
 (i) Vitamins (ii) Nutrients (iii) Pulses (iv) Minerals
- The mode of nutrition in which organisms make food themselves from simple substances is called :
 (i) Autotrophic nutrition (ii) Heterotrophic nutrition
 (iii) Saprotrophic nutrition (iv) All of these
- The food making process of plants is known as
 (i) Respiration (ii) Nutrition (iii) Photosynthesis (iv) Symbiosis
- Some organisms live together and share shelter and nutrients. This is called :
 (i) Predation (ii) Symbiotic relationship
 (iii) Autotrophs (iv) None of these
- _____ is an insectivorous plant.
 (i) Neem (ii) Venus flytrap (iii) Moulds (iv) Mushroom
- Which of these plants live off rotting material?
 (i) Parasitic (ii) Saprophytic (iii) Insectivorous (iv) Symbiotic

2 Fill in the blanks :

- The process of taking or consuming and utilizing food is called _____.
- _____ derive their food from other organisms.
- _____ plants lack chlorophyll.
- _____ use carbon dioxide and water to prepare food.
- During photosynthesis, the gas released is _____.
- The presence of _____ can be tested by using iodine solution.

3 Write T for true and F for false :

- Fungi and bacteria are autotrophs.
- Saprotrophs are non-green plants.
- All the plants have chlorophyll.
- Stomata are small openings present on the upper side of the leaves.
- Cuscuta is a fully parasite plant.
- Plants require a lot of nitrogen to grow and to be healthy.

4 Match the following :

- | | |
|--|---------------------|
| a. Tiny pores in leaves | (i) Water, Sunlight |
| b. Needed by plants to make food | (ii) Stomata |
| c. A plant's food factory | (iii) Symbiosis |
| d. Live on other organisms and derive food from them | (iv) Pitcher Plant |
| e. Two different organisms live together and both are beneficial | (v) Leaf |
| f. Plant feeding on insects | (vi) Parasitic |

5 Answer these questions in short :

- From where does carbon dioxide enter a plant?
- Mention the role of chlorophyll in photosynthesis.
- List out the main life characteristics of living things.
- Name two insect-eating plants.
- Define nutrition. Name the major modes of nutrition.
- What is Rhizobium?

6 Give answers in detail :

- Explain different modes of nutrition.
- Define these with an example :
(i) Saprophytic nutrition (ii) Symbiosis (iii) Parasitic nutrition
- Which are the raw materials required for photosynthesis?
- What is the role of stomata?
- How can you say a green plant has starch in it?
- What are the ways by which the nutrient content of the soil can be replenished?

NEP QUERY



PROJECT WORK

- Grow mushroom and study their characteristics like mode of nutrition, conditions required for their growth and survival. (Learning by Experiment)

ACTIVITY

- Look around you . Collect the samples of saprophytic plants. (Art Integration)

COMMUNICATION SKILLS

- Hold a group discussion in your class on the topic- "Our earth will become a waste basket without saprophytic plants". (Discussion)

Reproduction in Plants

6

- Kinds of Reproduction
- Asexual Reproduction

- Sexual Reproduction
- Pollination

- Dispersal of Fruits and Seeds

All living beings reproduce young ones of their own kind. The ability of the living organisms to produce new living beings similar to themselves is called **reproduction**. Reproduction helps all living organisms multiply (increase in number). But an organism of one species can never produce an organism of another species.

KINDS OF REPRODUCTION

Plants can reproduce in different ways. Non-flowering plants reproduce by producing spores. Flowering plants produce seeds.

The type of reproduction in which a cell from only one parent develops into offspring is called **asexual reproduction**. Various methods of asexual reproduction are commonly found in plants.

When two gametes one from each parent combine to form a zygote, the process is called **sexual reproduction**. Flowers are responsible in plants for sexual reproduction. The zygote formed in this process transforms into seed.

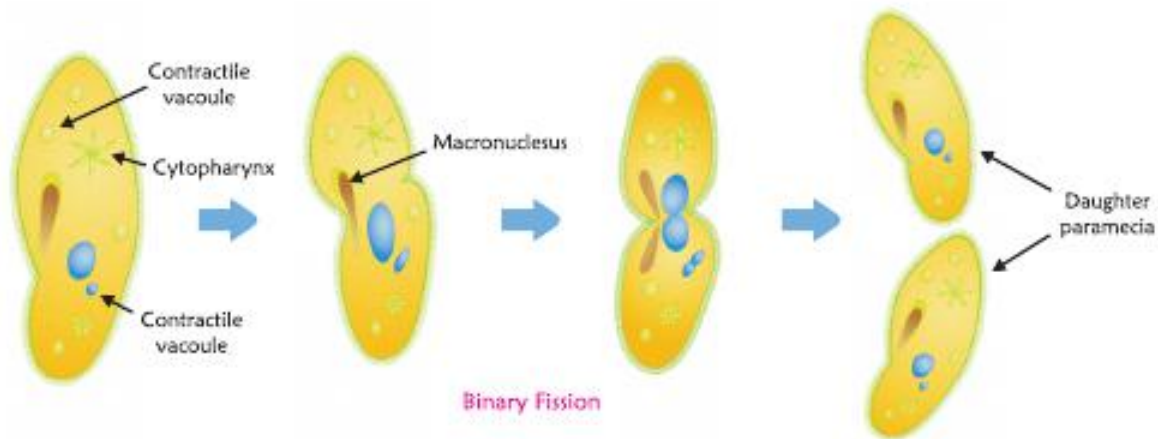
In asexual reproduction, plants can give rise to new plants without seeds. In sexual reproduction, new plants are obtained from seeds.

A SEXUAL MODE OF REPRODUCTION

It is the growth of a new plant from a part of the plant other than the seed. Following means of asexual reproduction are used by plants :

Binary Fission

It is the simplest and most common method of asexual reproduction. The whole parental body acts as the reproductive unit. The nucleus of the unicellular parent organism divides into two. This is the most common method of reproduction in bacteria, algae and fungi under the most favourable environmental conditions, that is, when there is plenty of food, water and a suitable temperature.



Spore Formation

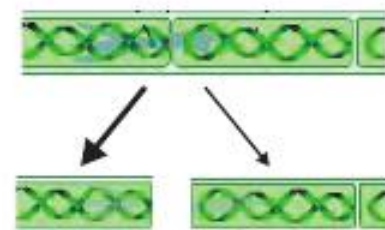
Spores are produced by bacteria, fungi and green plants. Spores serve a function analogous to that of seeds. Spores are most conspicuous in the non-seed-bearing plants, including algae, liverworts, mosses, and ferns. Produced and released by specialized fruiting bodies, such as the edible portion of the familiar mushrooms, fungal spores germinate and grow into new individuals under suitable conditions of moisture, temperature and food availability.



Spore Formation

Fragmentation

In some organisms like spirogyra and oscillatoria the body breaks into several parts. Each part then develops into a complete organism. When water and nutrients are available algae grow and multiply rapidly by fragmentation. An algae breaks up into two or more pieces called **fragments**. Each fragment then grows into a new plant.



Fragmentation

Budding

A bud develops as a small outgrowth on the parent's body. Each bud enlarges, develops the characteristics of the parent organism. A bud may become separated from the parent body and then develop into a new individual, or it may separate only after the completion of development. In some cases the buds never separate and as a result, colonies of interconnected individuals are formed. Yeast is a single celled organism. It grows and multiplies every few hours if sufficient nutrients are made available to them.



Budding

Vegetative Propagation

When plants make seeds and pollen, they often use a significant amount of time and energy. Additionally, remember that not all plants make seeds. For plants that do not make

seeds, there are many options for making new offspring. When reproduction takes place only from the vegetative parts of a plant (roots, stems, leaves and buds), it is known as **vegetative reproduction**.

1. Vegetative Propagation by Modified Roots

The modified (fleshy) roots of sweet potato and dahlia give rise to new plants when buried into the soil.

2. Vegetative Propagation by Modified Underground Stems

The underground stems, by being situated below the surface of the soil, protect themselves against unfavourable conditions of weather and the attack of animals, and serve as store houses for reserve food, and in vegetative propagation. Their stem nature can be distinguished by the presence of nodes and internodes, scale leaves at the nodes, axillary buds in axils of scale leaves and a terminal bud. The buds look like eyes and are also called as eyes. Each bud develops into a new plant. For examples: ginger, turmeric, etc.



Sweet Potato

3. Vegetative Propagation by Leaves

Some plants produce adventitious buds on their leaves. These buds usually remain dormant, when the leaf is attached with plant. However, the leaves when come in contact with moist soil develop new plantlets along the margins. These buds after falling on the ground grow into new plants. For example: bryophyllum and begonia.



Vegetative Propagation by Leaves

4. Artificial Methods of Vegetative Propagation

Besides natural methods of vegetative propagation, artificial modes of propagation are also being used. There are several ways in which man can interfere with plants so as to get a desired offspring. We can grow many plants from one plant by using the man-made methods. The process of growing many plants from one plant by man-made methods is called **artificial propagation of plants**. A number of methods of artificial propagation of plants are used in agriculture (for raising crops), and horticulture (cultivation of vegetables, fruits and flowers). The three common methods for the artificial propagation of plants are:

Cutting : Cuttings refer to regeneration of pieces of roots, shoots etc. and formation of a complete plant from them. Many plants like rose, bougainvillea, croton, coleus, money plant, tapicoa, sugarcane are grown through cuttings from their stem. Cuttings of some of their plants can be grown even in water, where they strike roots and develop adventitious buds. In some cases as in lemon even root cuttings can yield a new plant.



Cutting

Grafting : Grafting is an ancient and valuable horticultural practice. This consists of inserting a small branch into a rooted plant. In this stock a branch is inserted which is known as the scion or graft. Usually, the grafting ends of stock and scion fit well with each-other and are bound firmly with tape or rubber band, until their tissues unite and vascular continuity is established. Grafting is mostly practised in dicot plants. It is especially important for propagation of seedless varieties of plants. This is common in plants such as mango, guava, apple, pear, etc.



Grafting

Layering : In this method a lower branch of a plant is bent down and covered with moist soil, leaving the growing tip above the soil. A ring of bark is removed from the stem before it is bent down. In a few weeks time when enough roots have developed on the underground portion above the ringed part, it is clipped off from the parent plant and grown separately as an independent plant. This is a common practice with plants like jasmine, strawberry, grapewine, bougainvillea.

Advantages of Vegetative Propagation

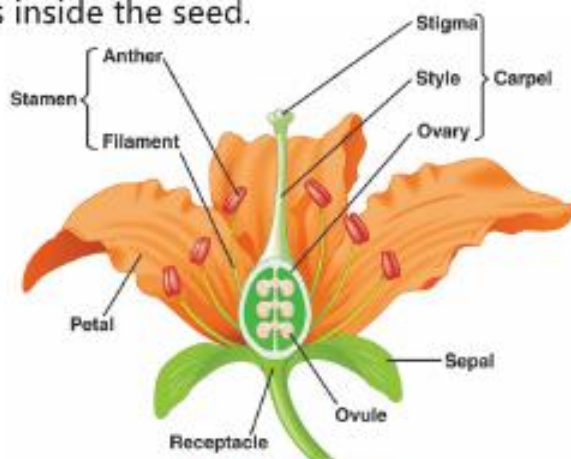
Plants produced by vegetative propagation take less time to grow and bear flowers and fruits earlier than those produced from seeds. The new plants are exact copies of the parent plant, as they are produced from a single parent.

SEXUAL REPRODUCTION

Two parents are needed in sexual reproduction. In plants, seeds are responsible for obtaining new plants. Seeds are formed from flowers. Flowers are important for sexual reproduction by plants. They produce male sex cells and female sex cells.

For sexual reproduction, one male and one female parents are required. Plants reproduce sexually by producing male gametes in the form of pollen and the female gametes in the form of eggs. The fusion of male germ cell with the female germ cell resulting in the formation of a single cell, the zygote. After fertilisation, the zygote divides repeatedly to form an embryo which resides inside the seed.

Parts of a Flower



Parts of Flower

FACT

In 1995, potato plants were taken into space with the space shuttle Columbia. This was the first time any food was ever grown in space.

Most flowers have four main parts, i.e. sepals, petals, stamens and carpels. Green sepals protect the flower from the Sun and rain in bud form. The coloured petals attract insects and other animals for the pollination of the flower. Stamens are the male parts of a flower. Each stamen has a filament and an anther. Pollen grains are produced in anthers. Carpels are the female parts of a flower. Each carpel has a sticky stigma, a style and an ovary. Ovules are present in the ovary.

Unisexual and Bisexual Flowers

A flower that contains both the male and female reproductive structure (stamen and pistil) is called a **bisexual flower**. Examples of plants bearing bisexual flowers are water lily, rose, hibiscus, mustard and petunia.



Cross Pollination

POLLINATION

As the pollen is not capable of locomotion, this process either involves some agent for this transference or the anther must be placed in the flower right above the stigma so that pollens



Self Pollination

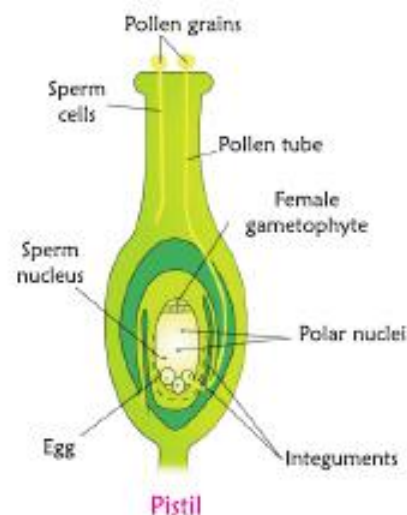
may drop directly on the stigma. The transference of the pollen from the anther to the receptive stigma, whether of the same flower or of a different flower, is known as **pollination**.

Types of Pollination

If the pollen falls on the stigma of the same flower or to the stigma of another flower of the same plant, it is called **self-pollination**. When the pollen of a flower falls on the stigma of a flower of another plant of the same kind (species), it is called **cross pollination**.

Fertilization

The surface of the stigma in a flower is sticky and pollen grains stick to it. Here, a pollen tube grows out from each pollen grain. Two sperms are present in this pollen tube. The tube grows downward through the style and enters the ovary. Pollen tube finally enters an ovule and releases its sperms in it. One of the sperms combines with the egg to form zygote. The other sperm combines with another cell to make the store of food. The process of fusion of sperm with the egg is called **fertilization**.



Changes after Fertilization

After fertilization, several changes take place in the flower. The sepals, petals and

stamens dry up and fall off. The fertilized egg inside the ovary develops into embryo. Ovules become seeds. The ovary grows large and develops into a fruit. The fruit protects the seed or seeds.

Fruit and Seed Formation

After fertilisation, the ovary grows into a fruit and other parts of the flower fall off. The fruit is the ripened ovary. The seeds develop from the ovules. The seed contains an embryo enclosed in a protective seed coat.

DISPERSAL OF SEEDS

The dispersal of fruits and seeds is the advantage of plants, as it enables them to flourish in a peaceful manner and find new localities where conditions are favorable for their growth.

If all the seeds of a plant were to fall at the same place and grow there, this would create severe competition for sunlight, water, minerals and space among the young growing plants. As a result, the seeds would not grow into healthy plants. Plants are benefited by seed dispersal. It prevents competition between the plant and its own seedlings for sunlight, water and minerals. It also enables the plants to reach new habitats for wider distribution.

Seeds are usually carried away, normally by wind, water and animals or by their own dispersal.

The seeds are dispersed by different agencies; the chief agencies are wind, water, animals and explosive mechanisms.

1. Dispersal by Wind

The seeds dispersed by wind are light in weight and this enables them to remain in air for a long time, e.g., seeds of orchids. The mode of wind dispersal is observed in seeds of pinus, sunflower, dodak, cotton, dandelion and datura.

2. Dispersal by Water

Water dispersal is observed in seeds of aquatic plants and the plants which grow along the banks of rivers and lakes and by the shores of the sea. The seeds and fruits of such plants develop spongy outer covering to keep them floating for some time, e.g., lotus, coconut, etc.

3. Dispersal by Animals

The seeds and fruits in this case may develop outgrowths like hooks, spines and sticky glands by means of which they stick to the bodies of animals or clothing of mankind, e.g., seeds of saffron, and mulberry are eaten by birds and animals and are passed out unaffected along with the faeces at some other places.

4. Dispersal by Explosive Mechanism

In this case, the ripened fruits burst out by slight disturbance with such a force that the

seeds are scattered and dispersed out far away from the parent plant, e.g., balsam, castor, oxalis and squirting gourd.



Dispersal by Wind



Dispersal by Water



Dispersal by Animal



Dispersal by Explosive Mechanism

◀ REWIND

- ▶ Living organisms reproduce their own kind by asexual and sexual methods.
- ▶ Asexual reproduction does not involve sex organs or gametes.
- ▶ Asexual reproduction is of several types: binary fission, spore formation, budding, fragmentation and vegetative propagation.
- ▶ Vegetative propagation is done by modified roots, stems, cutting, grafting, layering, etc involving buds at nodes.
- ▶ Plants with seeds reproduce by sexual method.
- ▶ Flower is the reproductive part of the plant. It may be unisexual or bisexual.
- ▶ The male gametes are found in the pollen grains and female gametes are found in the ovules.
- ▶ Transfer of pollen grains from the anther of one flower to the stigma of the same or another flower of the same kind is called pollination.
- ▶ The fusion of male and female gametes is called zygote which develops into an embryo.
- ▶ Fruit is the ripened ovary where as ovule develops into a seed which contains the developing embryo.
- ▶ Seed dispersal helps the plants to avoid competition for sunlight, water and minerals.

🔍 ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

a. Binary fission do not takes place in

(i) Amoeba

(ii) Paramecium

(iii) Yeast

(iv) Bacteria

b. Fertilization is

(i) Fusion of male gametes

(ii) Fusion of female gametes

(iii) Fusion of male and female gametes

(iv) All of these

c. Which one is a dicotyledonous plant?

(i) Banana

(ii) Sugarcane

(iii) Gram

(iv) Wheat

- d. The mode of vegetative reproduction in which scion and stock are tied is called
(i) Grafting (ii) Cutting (iii) Layering (iv) Tissue culture

2 Fill in the blanks :

- _____ develops into a new plant.
- _____ is the outermost part of the fruit.
- Yeasts reproduce by _____.
- Fusion of two gametes is called _____.
- Fruit is generally ripened _____.

3 Write T for true and F for false :

- Bacteria reproduce asexually by binary fission.
- Potato used as vegetable is modified root.
- Female gametes in plants are produced inside the ovary.
- The shoot system emerges from radicle.
- Bryophyllum reproduce by leaf.

4 Answer these questions in short :

- Write differences between cutting and grafting.
- Why is reproduction necessary for living organisms?
- What are the unique features of flowers that are pollinated by wind and insects?
- Describe budding as a method of asexual reproduction.
- How does the process of fertilisation take place in flowers?

5 Give answers in detail :

- Define reproduction. Name the type of reproduction in which two parents are involved.
- What is vegetative reproduction? Write its advantage.
- What is pollination? Name two types of pollination.
- Describe the various ways by which seeds are dispersed.
- What is fertilization? Where fertilization takes place in plants?

NEP QUERY



PROJECT WORK

- In the laboratory : Make your own flower garden by collecting pieces cut from different kinds of roses. Grow the variety in one single flat container or in separate pots.

(Learning by Experiment)

COMMUNICATION SKILLS

- Hold a group discussion in your class on the topic- "Importance of vegetative propagation in plants".
- Some disadvantages of asexual reproduction when compared to sexual reproduction.

(Discussion)

Weather, Climate and Adaptation

7

- Weather Forecast
- El-Nino Effect
- Adaptations of Animals to Climate
- Adaptation in shelter
- Climate
- Climate of Our Country, India
- Adaptation in Food

Temperature, pressure, wind, humidity and precipitation interact with each other. That is why we hear the words, 'It is hot and humid today,' 'It is cold and dry,' 'It is hot and dry' or 'It is likely to rain.' They influence the atmospheric conditions like the direction and velocity of wind, amount of isolation, cloud-cover and the amount of precipitation. These are known as the elements of both weather and climate. The influence of these elements differs from place to place and time to time. It may be restricted to a small area and for a short duration of time. We very often describe this influence in the name of weather as sunny, hot, warm, cold, fine, etc. depending upon the dominant element of weather at a place and at a point of time. Therefore, weather is the atmospheric condition of a place for a short duration with respect to its one or more elements. Two places even a short distance apart may have different kind of weather at one and the same time.

The Sun, wind, clouds and rain make the weather.



WEATHER FORECAST

It is important to know by some means the coming weather in advance. You may be planning to go on a hike without knowing that the particular day may be rainy. Farmers, sailors, aviators, tourists and many others are interested to know the weather conditions in advance for their own benefits. That is why newspapers publish weather reports and weather forecasts along with a map showing this information. Now, better weather forecasts are available with the use of weather satellites. Weather conditions are televised every day. When a cyclone or dangerous weather is expected, warnings are issued over the radio, television and newspapers so that people can prepare to save themselves and their property from its hazard.

Role of the Sun in changing the weather

The Sun's energy is the driving force for the weather. It is the source of heat and light on the earth. Though its distance from the earth is very long, the Sun heats the earth, in turn

the air near the ground gets heated and moves. Wind is moving air and is caused by differences in air pressure within our atmosphere. Air under high pressure moves toward areas of low pressure. The Sun heats the water of the water bodies to turn into water vapour which rises up high in the air. At height, it changes into tiny water droplets which together form clouds. Clouds bring rains.



Earth and Seasons

The Sun's energy is absorbed and reflected by the earth's surface, oceans and the atmosphere play an important role in determining the weather at any place. The weather at a place near the sea is different from that of a place in a desert, or near a mountain.

Days are shorter in winter than in summer. In shorter days, the place gets less heat energy from the Sun and thus the place is less warmer. When the days are longer, the place gets heat energy from the Sun for a longer time and the place is more warmer or hotter.

CLIMATE

The average weather conditions, prevalent from one season to another in the course of a year, over a large area is known as **climate**. The average of these weather conditions is calculated from the data collected for several years (about 35 years) for a larger area. Rajasthan, for example, experiences hot and arid climate, Kerala has tropical rainy climate, Greenland has cold desert climate and the climate of Central Asia is temperate continental. Climate of a region is considered more or less permanent.

To find the weather pattern of a place, averages are taken of the weather records of that place over a long period of time, (say 25 years). The mean temperature for a given month is found in two steps. First, we find the average of the temperatures recorded during the month. Second, we calculate the average of such average temperatures, over many years. This is called the **mean temperature**.

The rainfall average for a specific period is also calculated in the same way. The numbers for the mean total rainfall are usually rounded off.

Differences between Weather and Climate

Weather	Climate
1. Weather is the study of atmospheric conditions for short duration of a limited area.	1. Climate is the study of the average weather conditions observed over a long period of time for a larger area.
2. Weather is influenced by anyone of its predominant elements i.e., temperature or humidity.	2. Climate is the collective effects of all its elements.

3. The weather changes very often.	3. It is more or less permanent.
4. It is experienced over small areas of country.	4. It is experienced over large area of the continent.
5. A place can experience different types of weather conditions in a year.	5. A place can experience only one type of climate.

Geographical Factors Affecting the Climate

The climate of a place depends on the following factors—

1. The places near the equator are warmer than the places which are far away from it. This is because the rays of the sun fall vertical on the equator and slanting in the temperate and polar regions.
2. We all know that mountains are cooler than the plains. Shimla situated on a higher altitude is cooler than Jalandhar, although both are almost on the same latitude. The temperature decreases with the height of a place.
3. The water is a bad conductor of heat i.e., it takes longer time to heat and longer time to cool. Due to this moderating effect of the sea, places near the coast have low range of temperature and high humidity. The places in the interior of the continent do not experience moderating effect of the sea.
4. **Direction of the winds** : The winds blowing cold or hot or monsoon winds, coming from the sea make other places cool or hot or wet.
5. In areas generally of cloudless sky as in deserts, temperature even under shade are very high because of the hot daytime sunshine.
6. Rainfall makes the air cooler and humid.

EL NINO EFFECT

Weather conditions in India are also influenced by El-Nino which causes widespread floods and droughts in tropical regions of the world. El-Nino is a narrow warm current which sometimes appears off the coast of Peru in South America. It is a temporary replacement of the cold Peru current which normally flows along this coast. Sometimes, becoming more intense, it increases the surface water temperatures of the sea by 10° C. This warming of tropical Pacific waters affects the global pattern of pressure and wind systems including the monsoon winds in the Indian Ocean. It is believed that the severest drought of 1987 over India was caused by El-Nino.

Effect of Human Activities on Climate

One of the bad sides of rapid population growth is environment pollution. For various reasons the environment becomes harmful for survival and unfavourable for habitation. The

burning of gasoline by automobiles releases carbon dioxide and other types of air pollution that contribute to global climate change. Large scale of deforestation lead to increased amount of carbon dioxide and a subsequent increase in atmosphere temperature. Acid rain is also caused due to this.



Pollution

CLIMATE OF OUR COUNTRY, INDIA

India is a big country. The average climate of India is said to be of tropical monsoon type. This, however, doesn't take into consideration extreme local variations that range from alpine climate in the Himalayan region in the north to the tropical climate in the south. In the west is a vast stretch of desert known as the Thar while the east of the country boasts of plenty of flora and fauna and receives heavy rains each year. Couched between them, the climate of Indo-Gangetic plains is of continental type; very hot summers and freezing winters. The climate in India varies from tropical climate in the south to temperate climate in the north. Certain parts in the Himalayas have very cold climate. The western part of the country has hot and dry climate. In general, most regions in India have tropical monsoon climate, with significant seasonable variations in rainfall and temperature.

India actually has two monsoons – the southwest monsoon and the northeast monsoon. The southwest monsoon, which is the main monsoon, comes in from the sea and starts making its way up India's west coast in early June. By mid July, most of the country is covered in rain. This gradually starts clearing from most places in northwest India by October.

The northeast monsoon affects India's east coast during November and December. It's a short but intense monsoon. The states of Tamil Nadu, Karnataka and Kerala receive most of their rainfall from the northeast monsoon, while the rest of the country receives most of its rainfall from the southwest monsoon.

ADAPTATIONS OF ANIMALS TO CLIMATE

We know that light, temperature, air, soil and water are abiotic factors of the environment. Changes in these factors bring changes in the populations of a habitat. Some other natural factors and humans also cause changes in habitats.

Climate has an intense effect on all living organisms. Animals adapt themselves to survive various conditions in which they live. Animals and plants; living in such places; develop certain features which help them survive the harsh climate. To develop certain features in order to survive the given climate is called adaptation. Animals that live in extreme

climates acquire some special features to protect themselves from it.

Example : The camel has a hump to store food in the form of fat, and hooves to tread on the desert sand with ease. At several places, such as desert, polar region, etc. climate is very harsh. Many organisms live and thrive at these places, in spite of the harsh climate.

Some countries belonging to the polar regions (near the poles) are Canada, Greenland, Iceland, Norway, Sweden, Finland, Alaska in USA and Siberian region of Russia. Some countries having the tropical rainforests are India, Malaysia, Indonesia, Brazil, Republic of Congo, Kenya, Uganda and Nigeria.



Camel

Animals of the Polar Regions

Polar Regions are characterized by extreme cold and long snowy winters. Organisms living here have to adapt themselves to extreme cold. The animals in these regions are usually white or light coloured, e.g., polar bear, penguin. Animals living there have adapted to these severe conditions of extreme climate by developing certain specific features like thick and white fur, fat under skin, etc.

Adaptations in Polar Bears

- (i) Their white fur helps them blend in with the snow and ice. It protects them from their predators and helps to hide while catching their prey.
- (ii) A polar bear has a layer of fat under its skin which helps it stay warm. It also has a thick layer of fur.
- (iii) The wide, large paws help a polar bear to walk in the snow.
- (iv) The polar bear has a strong sense of smell. This helps it to catch its prey for food and locate its enemy.
- (v) During winter, polar bear goes for a long sleep called **hibernation** in their dens. During this period, they live on the stored body fat.
- (vi) When a polar bear swims under water it closes its nostrils so no water can get in.



Polar Bear



Penguins

Adaptations in Penguins

- (i) Penguins have white fronts and black backs. This colouration acts as a form of camouflage.
- (ii) Penguins have a special system of feathers with the parts of the feather close to the body trapping a lot of air and so making for good insulation against the cold, while the outer tips interlock with one another to provide a waterproof outer layer.

- (iii) Penguins also have special adaptations to allow them to drink sea water and excrete the salt from a special gland near their eyes. They also have adaptations that allow them to dive to great depths and stay submerged for long periods without the need to breathe.
- (iv) Some species of penguin (those that live in the coldest climates) also have adaptations of their circulatory systems that allow them to keep their main organs warm, but let other parts of their body operate at lower temperatures.

Other animals of the polar regions : Other animals living in the polar regions are many types of fishes, musk oxen, reindeers, foxes, seals, whales and birds.

Adaptations in Polar Birds

While fish can remain under cold water for long, birds must remain warm to survive. They migrate to warmer regions when winter sets in. They come back after the winter is over. This temporary change of habitat (living place) is called migration. The primary aim of migration is to take advantage of the longer days of the northern summer for breeding and to feed their young ones and to avoid harsh winters. Between 1500 and 4000 species of birds are known to migrate in India.

The birds' feathers provide a lot of insulation. They fluff their feathers to trap air between their feathers and bodies. The result is a natural layer of insulation, like a down sleeping bag. Also, when they sleep or rest, the birds tuck their bills under their wing feathers in order to breathe in this warmer air.



Birds

Some migratory birds travel as much as 15000 km to escape the extreme cold conditions at poles. Generally, they fly high where the wind flow is helpful and the cold conditions help them to disperse the heat generated by their flight muscles. But how these birds travel to the same place year after year is still a mystery.

Animals of the Tropical Rainforests



Leopard

Rainforests are very dense, warm, wet forests. They are having for millions of plants and animals. Rainforests are extremely important in the ecology of the Earth. The plants of the rainforest generate much of the Earth's oxygen. The tropical region is located around the equator so it has generally a hot climate. Even in the coldest month, the temperature is generally higher than about 15°C. During hot summers, the temperature may cross 40°C. Duration of days and nights are almost equal throughout the year. These regions get plenty of rainfall. Because of continuous warmth and rain, rainforests have grown here with wide variety of plants and animals.

The major types of animals living in the tropical rainforests are monkeys, apes, gorillas, lions, tigers, elephants, leopards, lizards, snakes, birds and insects.

The climate of rainforests is hot and humid and supports very large number and variety of animals. They have hard competition for food and shelter.

ADAPTATIONS IN FOOD

Some animals are adapted to get food not easily reachable. For example: the bird toucan has a long, large beak to reach the fruits on branches too weak to support its weight if it sits on them.



ADAPTATIONS IN SHELTER

Many animals are adapted to live on trees. For example: the feet of monkeys can easily hold on to the branches like their hands. Their tails are long for grasping branches. Red eyed frog has developed sticky pads on its feet to help it climb trees on which it lives.

Adaptations in sense organs and body colour

Many animals have sensitive hearing, sharp eyesight, thick skin and a skin colour which blends with the surroundings (to camouflage). Thus they cannot be seen by their predators and preys.

Animals of Indian Tropical Rainforests

The tropical rainforests of India are found in Western Ghats and Assam. Two well-known animals of the Indian tropical rainforests are the following.

Lion-tailed Macaque (or Bearded Ape)

Lion-tailed macaque also called as Wanderoo is a species of primate confined to Western Ghats Mountains of South India. The most important characteristic of Lion-tailed macaque is the silver-white mane which surrounds the head from the cheek down to its chin. The hair of the Lion-tailed macaque is black. It is a good climber and spends a majority of its life in the upper canopy of the forests. Macaque feeds mainly on fruits. It also eats seeds, young leaves, stems, flowers, buds and insects under the bark of the trees. Like many primates, lion-tailed macaques have a number of ways of communicating with one another. This communication may come in the form of gestures, screams, growls or other vocalizations and could be used for a variety of reasons, such as warning other macaques of danger.



Macaque

Elephant

It has a strong sense of smell because of its long nose called trunk. It is an extremely useful part which not only help elephants to reach food high up in branches, it also allows



Elephant

them to get food and water on the ground. It can pluck leaves and twigs with its trunk and put them in its mouth. It can fill its trunk with water and pour this water into its mouth to drink.

The tusks of the elephant are modified teeth. These can tear the bark of trees that elephant loves to eat. Their tusks can also help them to survive through dry spells, as they use them to bore down into the earth and find water in dry riverbeds. Large ears of the elephant help it to hear even very soft sounds. Elephants live in hot conditions and need to be able to cool themselves down. They also help it to keep cool in the hot and humid climate: the thin skin cover on the ears can lose heat quickly.

◀ REWIND

- ▶ The weather is generally not the same on any two days and week after week.
- ▶ The times of sunrise and sunset also change during the year.
- ▶ All the changes in the weather are driven by the sun.
- ▶ The average weather pattern taken over a long time, say 25 years, is called the climate of the place.
- ▶ The tropical and the polar regions are the two regions of the earth, which have severe climatic conditions.
- ▶ Animals are adapted to the conditions in which they live.
- ▶ Animals in the polar region are adapted to the extremely cold climate by having white fur, strong sense of smell, a layer of fat under the skin.
- ▶ Migration is another way to escape the harsh, cold conditions.
- ▶ Special characteristics developed in an animal that enable them to live in a particular climate successfully are called adaptations of animals.
- ▶ Because of the hospitable climatic conditions, huge populations of plants and animals are found in the tropical rainforests.
- ▶ Animals in the tropical rainforests are adapted such that they eat different kinds of food to overcome the competition for food and shelter.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

a. All changes taking place in the atmosphere are caused by the

(i) Moon

(ii) Sun

(iii) Earth

(iv) Stars

b. The average weather pattern taken over a long time is called the

(i) Climate of the place

(ii) Temperature of the place

(iii) Humidity of the place

(iv) Rainfall of the place

- c. The animal living in polar regions is
 (i) Monkey (ii) Donkey (iii) Polar bear (iv) Gorilla
- d. The short term variation which occurs in the atmosphere is called
 (i) Climate (ii) Relative humidity
 (iii) Weather (iv) Adaptation
- e. Penguins are adapted to live in
 (i) Desert climate (ii) Equatorial region
 (iii) Polar region (iv) Rainforests
- f. How does the polar bear keep itself warm in cold weather?
 (i) It has two thick layers of fur (ii) It has a coat of wool
 (iii) It covers its body with a woollen coat (iv) It sits by the side of the fire

2 Fill in the blanks :

- a. The average weather taken over a long time is called _____.
- b. A place receives very little rainfall and the temperature is high throughout the year, the climate of that place will be _____ and _____.
- c. The two regions of the earth with extreme climatic conditions are _____ and _____.
- d. Elephant has a strong sense of smell because of its long nose called _____.
- e. _____ are very dense, warm, wet forests.
- f. Trees take carbon dioxide and produce oxygen during _____.

3 Match the following :

- | | |
|------------------------------|---|
| a. Feeling very hot | (i) Temperature below 20°C or temperature above 40°C |
| b. Sweating too much | (ii) Expected rain |
| c. Very pleasant weather | (iii) Temperature above 40°C or temperature above 20°C |
| d. Too cold | (iv) Temperature less than 50°C or temperature more than 15°C |
| e. Early morning temperature | (v) Maximum temperature of a day |
| f. Humid with cloud cover | (vi) Humid weather or Dry weather |

4 Write T for true and F for false :

- a. The day to day condition of the atmosphere at a place with respect to the temperature, humidity, rainfall, wind-speed, etc., is called the climate at that place.
- b. The polar region present an extreme climate.
- c. The tropical region has generally a hot climate because of its location around the poles.
- d. Weather conditions in India are also influenced by El-Nino.
- e. Siberian Crane is a migratory bird.
- f. Rainfall makes the air cooler and humid.

5

Answer these questions in short :

- What is migration?
- Write some of the geographical factors affecting the climate of a place.
- Describe the weather condition of the polar region.
- Why we find animals of certain kind living in particular climatic conditions?
- How does a penguin living in the cold place adapt itself?
- Name three elements that determine the weather of a place.

6

Give answers in detail :

- How do elephants living in the tropical rainforest adapt itself?
- Write the adaptive features of polar bear to live in extremely cold conditions.
- What is climate? How in it differ from weather?
- Why tropical rainforests have larger population of animals?
- Why do some animals develop the ability to camouflage?

NEP QUERY



PROJECT WORK

- In the laboratory : Study the construction and working of a maximum and minimum thermometer. Note down the maximum and minimum temperature.

(Learning by Doing)

COMMUNICATION SKILLS

- Hold a group discussion in your class on the topic-“**Effect of climate on our daily activities**”.

(Discussion)

CRITICAL THINKING

- Following are the characteristics of some animals :
 - White fur
 - Long tails and silver-white mane
 - Need to migrate
 - Sticky pads on feet
 - Long and large beak
 - Layer of fat under skin

(Self-Awareness)

Indicate whether it is adaptation for tropical rainforest or polar regions or both the regions.

Physical and Chemical Changes

8

- Types of Changes
- Rusting of Iron
- Reversible and Irreversible Changes
- Crystallisation

Changes in materials are going on around us all the time. Leaves change their colour, trees shed their leaves, milk changes to curd, and iron nails rust in moisture. There are certain factors that influence these changes and there are reasons for every change. Among the changes we observe in our daily life some changes are slow and some are fast. There are some changes that are temporary and some are permanent. There are many changes which take place naturally but for some we need to initiate or intervene in some way for the change to occur.

TYPES OF CHANGES

Most of the changes in materials are of two main types, i.e., physical changes and chemical changes.

Physical Change

Change in which only physical properties of a substance change and no new substance is formed is called **Physical Change**. Most of the physical changes are reversible, this means we can get the substance back even after the change. A paper can be folded or unfolded and hence it is an example of physical change. In melting wax, only the state of the wax changes; from solid to liquid. Solid wax can be obtained from molten wax. It means only the state of the wax changes. This is an example of physical and reversible change.



Activity-1

- ▶ (To be performed under teacher's guidance)

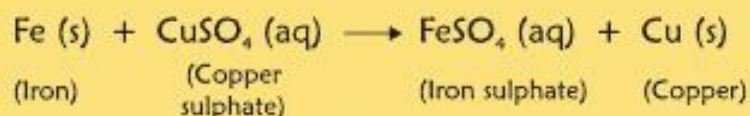
Things required : Beaker, copper sulphate crystals, an iron blade, sulphuric acid, water.

Method :

1. Take some copper sulphate crystals (also called blue vitriol because of their bright blue colour in the beaker).
2. Add half cup water and a few drops of a dilute sulphuric acid in the beaker.
3. Stir the mixture well. You will get a blue coloured CuSO_4 solution.
4. Put an iron blade into the solution and keep it undisturbed for about half an hour.
5. Do you find any change? Is the colour of the solution same as before?

Observation : A chemical reaction takes place in the beaker. Iron being more reactive than copper has a tendency to remove copper from its salt solution and form its own salt. In other words, it displaces copper from its salt solution (displacement reaction). After this reaction, blue colour of CuSO_4 changes to green due to the formation of iron sulphate (FeSO_4). Copper is left behind as red deposit on the iron blade.

The reaction is



Based on the above activity, fill in the blanks.

Before reaction	After reaction
Iron is _____ in colour.	Iron gets _____ deposits.
The solution is _____ in colour	The solution is _____ in colour.

This type of change means that no new substances are made, but there is a change in the appearance of a chemical. Examples of physical change include state changes and dissolving. All substances can be solid, liquid or gas.



Physical Change

Some Typical Physical Changes



Evaporation and Condensation



Ice Melting



Breaking of a Wooden Stick

1. Dissolution of a solid in a liquid is a physical change. For example: if one decided to mix sugar into water to make sugar water, a solution of sugar in water also tastes sweet. Thus, sugar on dissolving does not lose its basic properties. So the original substance sugar is recovered back from the solution by a simple method. Thus the dissolution of sugar in water is a physical change.
2. Melting of a solid and freezing of a liquid are physical changes.
3. Evaporation of a liquid and condensation of its vapour are physical changes.

Chemical Change

A chemical change is one in which a new substance is formed. Chemical changes are permanent and are not easy to reverse. There is one or more new substances are formed. A chemical change is also called a **chemical reaction**.

Burning of paper, rusting of iron, turning of milk into yogurt, cooking of food, etc. are some examples of chemical changes.

Coal is carbon. When we burn coal, it changes into smoke, energy and ash. So, burning of coal is a chemical change because new substances are formed during this process.

Some Clues to Chemical Changes

- › Formation of gas bubbles
- › Change of colour
- › Releasing and absorbing of energy

If any one of the above clues takes place, it shows the chemical change.

We see that the chemical composition and chemical properties of the reacting substances are changed.

Changes that Accompany a Chemical Change

With new products, some other features may be included in a chemical change.

- › Heat, light or any other radiation (ultraviolet, for example) may be given off or absorbed.
- › Sound may be produced.
- › A change in smell may take place or a new smell may be given off.
- › A colour change may take place.
- › A gas may be formed.

Some common examples of Chemical Change

1. Plants apply a chemical reaction called photosynthesis to convert carbon dioxide and water into food (glucose) and oxygen.
2. Every time we strike a match, burn a candle, build a fire, or light a grill, we see the combustion reaction. Combustion combines energetic molecules with oxygen to produce carbon dioxide and water.



Chemical Change

3. Burning any fuel produces heat and light. Explosion of a fire work produces heat, light, sound and unpleasant gases.
4. Spoiled food produces a foul smell.

Importance of Chemical Changes

Chemical changes are extremely important to our everyday life. Infact chemical changes or reactions are key to several of our daily tasks. It helps us in our kitchen with cooking and baking. Chemical reaction of baking soda while baking cookies. It also helps us with our cleaning. For example: chemical reaction of detergent particles on stains. Thus, without chemical reactions we will not be able to perform our daily routine tasks.

Differences Between Physical and Chemical Change	
Physical Change	Chemical Change
1. Physical changes only effect physical properties.	1. Chemical changes will effect both physical and chemical properties.
2. Physical changes produce no energy.	2. Chemical change produce energy generally in the form of heat, light, or sound.
3. Physical changes produce no new substances the atoms are arranged the same way in the products and reactants.	3. Chemical changes produce new substances the atoms rearrange and form new compounds.
4. Physical changes are generally easy to reverse.	4. Chemical changes are not easily reversible without an additional chemical reaction.

RUSTING OF IRON

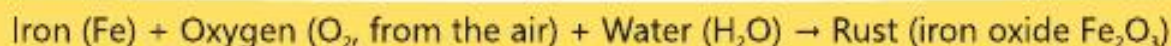
The upper layer of the iron articles exposed to moist air get converted into a brown powdery substance called **rust**. The rust slowly separates out of the iron articles and falls down. The conversion of the upper layer of the iron articles exposed to moist air into a brown powdery substance is called **rusting**.



Rust

Rusting slowly destroys iron articles. Bridges, ships, cars, truck bodies, buckets, pipes and many other articles are made of iron. They all get rusted and gradually destroyed.

The process of rusting can be represented by the following equation :



For rusting, the presence of both oxygen and water (or water vapour) is essential. In

fact, if the content of moisture in air is high, which means if it is more humid, rusting becomes faster.

Prevention of Rusting

Rusting is heavy loss to the iron articles, small to big. To prevent rusting, prevent iron articles from coming in contact with air or water or both. One simple way is to apply a coat of paint or grease. In fact, these coats should be applied regularly. Another way is to deposit a layer of a metal like zinc or chromium on iron. In this method, surface of iron is covered with layer of more active metal like zinc. This active metal loses electrons (undergoes oxidation) in preference to iron and hence, prevents the rusting of iron. Zinc metal is generally used for protecting iron and the process is called **galvanisation**. Galvanised iron sheets maintain their shine due to the formation of a thin protective layer of basic zinc carbonate (ZnCO_3), Zn(OH)_2 due to the reaction between zinc, oxygen, CO_2 and moisture in air. Galvanised iron sheets are used for roofing and for making buckets, drums, etc. Auto parts, and bathroom fittings are electroplated with chromium or nickel to prevent rusting.



Galvanisation

Iron is alloyed (mixed) with carbon and metals like chromium and nickel to make stainless steel which does not rust.

REVERSIBLE AND IRREVERSIBLE CHANGES

A change that can go forward or backward is called a **reversible change**. It is a temporary change. We can get the same thing again. Melting of ice into liquid water, switching on a tube light, increase of heartbeat during running, mixing of salt in water, wetting a dry cloth, etc. are reversible changes.



Changes

A change that cannot go back is called a **irreversible change**. It is a permanent change. We cannot again get the thing in its original form. Turning of milk

into yogurt, mixing of Plaster of Paris with water, burning of paper and wood, rotting egg or fruit, etc. are examples of irreversible changes. How can we relate reversible and irreversible changes to physical and chemical changes?

CRYSTALLISATION

In Class VI, we have learnt that salt can be obtained by the evaporation of sea water. The salt obtained in this manner is not pure and its crystals are small. The shape of the crystals

cannot be seen clearly. However, large crystals of pure substances can be formed from their solutions. The process is called **crystallisation**. It is an example of a physical change.

Pure crystalline substance can be obtained from the solution of its impure sample. For example: common salt is obtained by the evaporation of sea water. The salt shape of the crystals cannot be seen clearly.

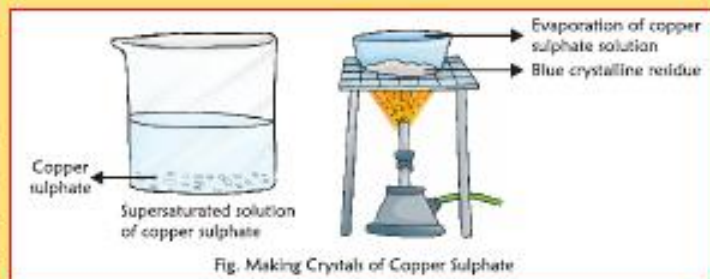
Large crystals of pure substances can be formed from their solutions.



Activity-2

- ▶ To make pure crystals of copper sulphate.

Take some water in a beaker and add a teaspoonful of copper sulphate in it. Stir the solution till it cannot dissolve more salt in it. Few crystals of copper sulphate may remain at the bottom of the beaker. This is the supersaturated solution of copper sulphate. Transfer the solution to a china dish. As the solvent evaporates, you will find that blue coloured crystalline substance is left behind as a residue in the china dish. Now, filter the solution. Allow this solution to cool at room temperature by placing the solution where it cannot be disturbed. Observe the solution after a few hours. You will get the crystals of copper sulphate.



REWIND

- ▶ Changes can be of two types, physical and chemical.
- ▶ Physical changes are changes in which only physical properties of substance change. No new substances are formed in these changes. These changes may be reversible.
- ▶ In chemical changes new substances are produced.
- ▶ Some substances can be obtained in pure state from their solutions by crystallisation.
- ▶ Rusting can be prevented by depositing a thin layer of zinc, chromium, etc. on iron articles.
- ▶ In reversible change the product formed can again go back to its original form. In an irreversible change, the product formed cannot again go back to its original form.
- ▶ Boiling or freezing of water, melting of wax, etc. are examples of reversible changes.
- ▶ Rotting egg or fruit, turning of milk into yogurt, etc. are examples of irreversible changes.
- ▶ Some solids can be obtained in pure solid geometrically shaped particles from their solution by crystallisation.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. Which of the following is a chemical change?
- | | |
|----------------------------------|---|
| (i) Melting of ice | (ii) Rusting of iron |
| (iii) Changing of water into ice | (iv) Dissolving of common salt into water |

- b. Physical changes are
- | | | | |
|------------------|---------------|------------------|-------------------|
| (i) Irreversible | (ii) Periodic | (iii) Reversible | (iv) All of these |
|------------------|---------------|------------------|-------------------|
- c. Chemical changes are characterised by
- | | |
|--------------------------------|------------------------|
| (i) Formation of new substance | (ii) Release of energy |
| (iii) Absorption of energy | (iv) Reversible |
- d. Which of the following is an example of chemical change?
- | | |
|---------------------------|--------------------------------------|
| (i) Melting of wax | (ii) Burning of candle wick |
| (iii) Heating of iron rod | (iv) Woollen yarn to knitted sweater |

2 Fill in the blanks :

- Burning of magnesium is a _____ change.
- The changes that occur in chemical reactions are _____.
- _____ metal is used in galvanisation.
- The process of protecting iron from rusting is called _____.

3 Write T for true and F for false :

- Formation of manure from dead animals is not a chemical change.
- Making curd is a chemical change.
- Condensation of steam is not a physical change.
- Coal and diamond are the same substances.

4 Answer these questions in short :

- Why is burning a paper a chemical change?
- Name the change when sugar is dissolve in water and how?
- Is a physical change reversible or irreversible change?
- Define galvanisation.

5 Give answers in detail :

- Differentiate a physical and a chemical change with an example of each.
- What do you mean by rusting? What are the conditions required for rusting to take place?
- Write a note on galvanisation.
- What do you understand by crystallisation?

NEP QUERY



PROJECT WORK

- In the laboratory : Have you seen LCD screens of televisions and computers. LCD screens contain thousands of tiny crystal units. They flash on and off very quickly to build up moving pictures. Find out more about LCD and their uses.

(Learning by Experiment)

COMMUNICATION SKILLS

- Hold a group discussion in your class on the topic- "Alloying is better technique to prevent rusting than galvanisation technique to mankind".

(Discussion)

Acids and Bases

9

- Indicators
- Acids
- Bases
- Neutralisation
- Salts
- Neutralisation in Everyday's Life

We use in our daily life a large number of substances such as lemon, tamarind, common salt, sugar and vinegar. Some are sweet, some are sour and some are bitter. Some are tasteless too.

Curd, lemon juice, orange juice and vinegar taste sour. These substances taste sour because they contain acids. The chemical nature of such substances is acidic. Substances like baking soda are basic in nature.



Acids and Bases

INDICATORS

It is not always possible to know the acidic or basic nature of substances by tasting them. Tasting a substance in laboratory is not also advisable because it may be harmful. Thus, to test the chemical nature (acidic or basic nature) of substance a special kind of substance is used. This special kind of substance which is used to test the acidic or basic nature of anything is known as **indicator**.

Indicator is a substance that shows the acidic or basic nature of a substance by change in its colour. They are used extensively in studying substances and their properties.

Natural Indicators

In nature, leaves, flowers and fruits of many plants have very attractive colours. It is interesting that these colours may change. Many indicators can be extracted from plants. Naturally occurring indicators are turmeric, litmus, china rose and red cabbage.



Turmeric



China Rose

China rose is another natural indicator. It has pink petals. China rose solution gives dark pink (magenta) colour with acid and green colour with base. Turmeric is used as another natural indicator. Turmeric is of yellow colour. Turmeric paper turns into red when it is dipped into basic solution. Turmeric paper does not change its colour with acid.

Litmus

The most commonly used natural indicator is litmus. It is extracted from **lichens**. It has a mauve (purple) colour in distilled water. When added to an acidic solution, it turns red and when added to a basic solution, it turns blue. It is available in the form of a solution, or in the form of strips of paper, known as litmus paper.



Litmus



Activity

- To perform litmus test.

Materials required : Test tube, water, lemon, knife, red litmus paper, blue litmus paper.

Method : Take some water in a test tube. Add a few drops of lemon juice to it. Shake well. Put a drop of the above solution on a strip of the red litmus paper with the help of a dropper (or dip a strip of red litmus in this solution). Repeat the experiment with a strip of blue litmus. Record your observations.

Repeat the experiment with baking soda solution or soap solution. Record your observations.

Observation : Lemon water does not change the colour of the red litmus. But it changes the blue litmus to red. And soda or soap solution turns the red litmus to blue. But it does not change the colour of the blue litmus.

Conclusion : Acidic solutions turn blue litmus to red. Basic solutions turn red litmus to blue.



Phenolphthalein is a synthetic organic dye. It is dissolved in alcohol. Its solution is colourless. When a few drops of phenolphthalein are added to a basic solution, such as soap solution, washing soda solution or shampoo solution, it changes to deep pink colour. But it remains colourless in acidic solution.

Solution of methyl orange changes to pink colour in acidic solution and yellow in basic solution.

ACIDS

The word acid is derived from Latin word 'acidus' means sour. In chemistry, the term acid has been used to name a group of compounds that have sour taste.

Sources of Common Acids

Generally, acids are obtained from two different sources. Some acids are obtained from minerals. While others occur in substances that we use daily in our homes. Acids obtained from minerals are called **mineral acids** while those occur in plants and animals are known as **organic acids**.

Examples :

- (i) **Common Acids** : Hydrochloric acid (HCl), sulphuric acid (H_2SO_4) and nitric acid (HNO_3).
- (ii) **Organic Acids** : Some of these along with their sources are listed in table as follows :

Name	Source	Name	Source
Formic Acid	Ants	Tartaric acid	Tamarind, grapes
Acetic Acid	Vinegar	Lactic acid	Curd
Oxalic Acid	Tomatoes	Malic acid	Apples
Citric Acid	Citrus fruits	-	-

General Properties Of Acids

Let us now study the properties, which are common to all acids.

- (i) All acids have a sour taste.
- (ii) All acids turn blue litmus solution and methyl orange solution red.

Uses of Acids

- Mineral acids are used in different industries, like manufacture of fertilizers, dyes, drugs, explosives, etc.
- Hydrochloric acid is used for cleaning tiles, wash basins and toilets, cleaning iron sheets before galvanisation (depositing zinc on them).
- Nitric acid is used for cleaning gold and silver ornaments. It is used for the manufacture of explosives and in production of fertilisers. It is also used in the preparation of dyes and drugs.
- Sulphuric acid is used in lead acid batteries and in the manufacture of paints, drugs, dyes and artificial silk. It is also used in preparing detergents, plastics, synthetic fibres and for refining in petroleum industries.

BASES

The substances that are bitter in taste and their solutions have soapy (slippery) touch are called **bases**. The bases which are soluble in water are called **alkalies**. Alkalies are corrosive in nature and harmful to skin.

Strong bases are very corrosive and can burn the skin. For example: caustic soda or sodium hydroxide (NaOH) and caustic potash or potassium hydroxide (KOH). Magnesium hydroxide [$\text{Mg}(\text{OH})_2$], calcium hydroxide [$\text{Ca}(\text{OH})_2$] and ammonium hydroxide (NH_4OH) are weak bases. They are less corrosive.

Uses of Bases

- › Sodium hydroxide is used in the manufacture of soaps, detergents, paper and textiles (such as rayon). In homes it is occasionally used to unblock drains.
- › Calcium hydroxide is used in whitewash and mortar, in the manufacture of bleaching powder, for softening hard water, in leather industry and for neutralising acidity of the soil. Ammonium hydroxide is used to manufacture fertilizer.
- › Magnesium hydroxide is used as an antacid or laxative. It helps to correct excess acidity in the stomach.

NEUTRALISATION

The reaction of an acid with a base to give salt, water and heat is called **neutralisation**.

Example :

Hydrochloric Acid + Sodium Hydroxide $\rightarrow$ Sodium Chloride + Water + Heat



The heat evolved in the neutralisation reaction raises the temperature of the reaction mixture.

During the mixing of dilute sulphuric acid with potassium hydroxide, the test tube heats up. The resulting solution is neither acidic nor basic. In neutralisation reaction, heat is always produced or evolved. The evolved heat raises the temperature of the reaction mixture.

In neutralisation reaction a new substance is formed. This is called salt. A salt may be acidic, basic or natural in nature. Thus, the reaction between an acid and base is known as neutralisation. Salt and water are produced in this process with the evolution of heat.



For example,

Hydrochloric acid + Sodium hydroxide $\longrightarrow$ Sodium chloride + Water



Sulphuric acid + Lime water (calcium hydroxide) $\rightarrow$ Calcium sulphate + water



SALT

A salt is a compound formed by the neutralization between an acid and a base.

Preparation of Salts

- By the neutralization of an acid with a base. The reaction between an acid and a base is called neutralisation reaction.
- By the action of metallic oxide and acid.
- By the direct combination of two elements.

Uses of Salts

(i) Role of salts in human body

Salts of sodium, potassium, calcium, magnesium and iron are needed for the normal working of our body. They perform the following functions :

- Haemoglobin present in blood contains compounds of iron.
- Sodium and potassium salts are needed for the proper functioning of muscles and the nervous system.
- Salts of calcium are present in bones and they are also responsible for the strength of bones. These salts are also responsible for preventing heart attacks.
- Potash alum is used to coagulate the blood coming out of a wound. It is also used for the purification of water.
- Salts of iodine are needed for the proper functioning of thyroid glands. They are also used for the treatment of goitre.

(ii) Uses of salts in our daily life

- In our daily life, we use common salt for seasoning food as a preservative for fish and in pickles.
- Baking soda is used for giving softness to bread and cake.
- Washing soda is used for washing clothes.
- Sodium potassium tartrate is used as a laxative.

(iii) Uses of salts in industries

- Sodium chloride is used for the manufacture of chlorine, hydrogen chloride, caustic soda, washing soda and sodium hydrogen carbonate.
- Sodium carbonate is used for softening hard water, for the manufacture of glass and soap.
- Potassium nitrate is used for the manufacture of gun powder and fire works.
- It is also used as a fertilizer. Potash alum is used for purification of water, in dyeing cloth and for tanning hides.
- Copper sulphate is used as a fungicide, in calico printing and in electroplating.

Neutralisation in Everyday Life

Neutralisation of an acid by a base and of a base by an acid is very important in our everyday life. Some common examples are as follows:

In digestion

The stomach contains hydrochloric acid, which helps in the digestion of food. If this acid is produced in excess, it causes indigestion, which is painful. An antacid such as milk of magnesia neutralises the excessive acid in the stomach, and provides relief from the pain due to indigestion.

Insect bite

Ant sting contains formic acid. When an ant bites, it injects this formic acid into the skin. This causes pain. It can be neutralised by rubbing the ant bite with moist baking soda, which contains sodium hydrogen carbonate (NaHCO_3), or with a solution of calamine, which contains zinc carbonate (ZnCO_3).

Soil treatment

Excessive use of chemical fertilisers makes the soil acidic. When the soil is too acidic, it is treated with bases like quick lime (calcium oxide) or slaked lime (calcium hydroxide). Organic matter releases acids which neutralise the basic nature of the soil. Gypsum (Calcium sulphate) is also added to the basic soil.

Industrial wastes

The wastes of many factories contain acids. If they are allowed to flow into the water bodies, the acids will kill fish and other organisms. The factory wastes are, therefore, neutralised by adding basic substances.

◀ REWIND

- ▶ Acids are sour in taste. Bases are bitter in taste and soapy to touch.
- ▶ Acid turns blue litmus red. Base turns red litmus blue.
- ▶ Substances which are neither acidic nor basic are called neutral.
- ▶ Solutions of substances which show different colours in acidic, basic and neutral solutions are called indicators.
- ▶ An acid and a base neutralise each other and form a salt. A salt may be acidic, basic or neutral in nature.



ASSIGNMENTS



COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- Which of the following is a natural indicator
(i) Methyl orange ☐ (ii) Phenolphthalein ☐ (iii) Turmeric ☐ (iv) Oxalic acid ☐
- In case of indigestion, we use
(i) Antacids ☐ (ii) Antipyretic ☐ (iii) Antibiotic ☐ (iv) Alcohols ☐
- All acids contain
(i) Oxygen ☐ (ii) Nitrogen ☐ (iii) Carbon ☐ (iv) Hydrogen ☐
- The acid present in curd/sour milk is
(i) Formic acid ☐ (ii) Maleic acid ☐ (iii) Lactic acid ☐ (iv) Tartaric acid ☐

2 Fill in the blanks :

- When acid and base react together a new compound called _____ is formed.
- All acids contain _____.
- Metal on heating or burning produce _____ of metals.
- Litmus is extracted from _____.

3 Write T for true and F for false :

- Sulphuric acid is a strong acid.
- Sodium chloride is a base.
- Litmus is a natural indicator.
- Water soluble bases are called alkalies.
- The common name of sodium carbonate is baking soda.

4 Answer these questions in short :

- What are alkalis? Give two examples.
- What are organic acids? Give two examples.
- What are natural indicators? Give two examples.
- Why is baking soda applied on insect stings?
- What is a neutralisation?

5 Give answers in detail :

- What are indicators? Give two examples.
- State differences between acids and bases.
- Ammonia is found in many household products, such as window cleaners. It turns red litmus blue. What is its nature?
- Name the source from which litmus solution is obtained. What is the use of this solution?
- Is the distilled water acidic/basic/neutral? How would you verify it?
- Describe the process of neutralisation with the help of an example.

NEP QUERY



PROJECT WORK

- laboratory : Study the behaviour of acids and bases.

(Learning by Experiment)

COMMUNICATION SKILLS

- Identify common house hold acids and bases.
- Hold a group discussion in your class on the topic. Effect of acid and bases in our daily activities.

(Discussion)

CRITICAL THINKING

- Identify the following properties as : acid, base or both :

(Self-Awareness)

- | | |
|---|------------------------------|
| a. Bitter in taste _____ | b. Sour in taste _____ |
| c. Soapy to touch _____ | d. Conduct electricity _____ |
| e. Turns blue litmus paper to red _____ | |
| f. Usually found in citric juices _____ | |

Temperature

10

- Heat : A Form of Energy
- Unit of Heat
- Types of Thermometer
- Heat
- Conduction
- Convection
- Radiation
- The Vacuum Flask

We are familiar with the feelings of hot and cold. We wear different clothes in different seasons. We wear woollen and dark coloured clothes during winter when it is cold outside. Woollen and dark coloured clothes keep us warm. We prefer to wear light coloured cotton clothes when it is hot. They give us a feeling of coolness. You might have wondered why a particular type of cloth is suitable for a particular season. When we sip hot tea, we feel hot. When we eat an ice cream we feel cool. To be more clear when we go in the sun, we feel hot. When we go inside an air-conditioned room we feel cool or cold.

The water from a tube-well appears cool in summer. But the water from the same tube-well appears warm in winter. It means our sense of touch sometime is misleading.

In winter, we feel cold inside the house. If we come out in the sun, we feel warm. In summer we feel hot even inside the house.

We see that some objects are cold and some are hot. We also know that some objects are hotter than others while some are colder than others.

Temperature is a measure of the degree of hotness or coldness of an object.

HEAT : A FORM OF ENERGY

Heat is a form of energy which causes the sensation of hotness and coldness. When a body is heated, its energy increases and when it is cooled, its energy decreases. Heat is measured by the change in the degree of hotness or coldness it produces in material bodies.

We cannot depend on our sense of touch to judge hotness or coldness. We always need a reference body relative to which we can say whether a given body is hot or cold. So we need an instrument to measure the degree of hotness or coldness which is not affected by the hotness or coldness of our sense of touch.

Measurement of Temperature

If you stand close to fire, you feel warm. When a warm object is placed close to a cooler object, heat energy moves from the warmer object to the cooler one until both objects attain the same temperature. Often we think that heat and temperature are the same things; this is wrong. Temperature is a measure of the heat energy in a body and which indicates the ability of a body to give heat to another body or absorb heat from another body.

We use thermometer to measure temperature. Thermo is a Latin word meaning heat and meter means a measuring device.

Some liquids increase in volume with the increase in temperature. This property of expansion is used to measure temperature by measuring the increase in the volume of the liquid. Mercury is used in conventional thermometers. Mercury thermometer is commonly used because it has a good expansion rate. Different units of temperature are used so there are different temperature scales. But in all scales, the normal boiling and freezing points of water serve as reference points. The liquid of thermometer expand between these point so the tube divided into certain number of divisions corresponding to these temperatures.

UNIT OF HEAT

There are three units which are used to measure the temperature: Degree Celsius, Fahrenheit and Kelvin.

Degree Celsius

Degree Celsius is written as $^{\circ}\text{C}$ and read as degree Celsius or simply Celsius. For example: 20°C is read as twenty degree Celsius. On the Celsius scale, the melting point of ice is taken as 0°C and the boiling point of water as 100°C . The difference between these two points is divided into 100 equal divisions called degrees Celsius. This is the SI unit of temperature. This was also called as centigrade.

Fahrenheit

Fahrenheit is written as $^{\circ}\text{F}$ and read as degree Fahrenheit. For example: 25°F is read as twenty five degree Fahrenheit. On the Fahrenheit scale, 32°F is taken as the melting point of ice and 212°F as the boiling point of water. The difference between these two points is divided into $212-32=180$ equal divisions called degrees Fahrenheit.

Kelvin

Kelvin is written as K. For example: 100K is read as hundred Kelvin.

To convert Celsius into Fahrenheit and vice versa are as follows :

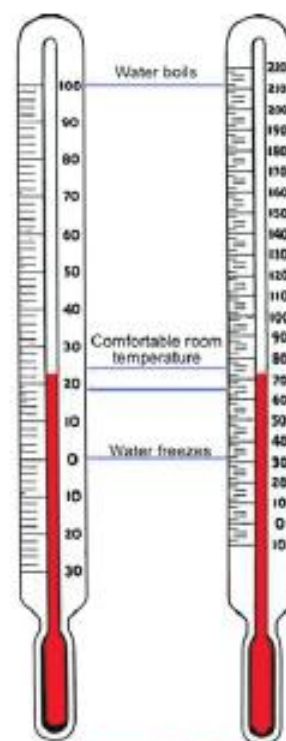
$$F = \frac{9}{5} (C) + 32 \text{ and } C = \frac{5}{9} (F-32)$$

Example 1 : Convert 86°F to $^{\circ}\text{C}$.

Solution : $C = \frac{5}{9} (F - 32) = \frac{5}{9} (86 - 32) = \frac{5}{9} \times 54$

$$= 5 \times 6 = 30$$

$$\text{Thus, } 86^{\circ}\text{F} = 30^{\circ}\text{C}$$



Temperature on Different Scales

Example 2 : Convert 50°C to $^{\circ}\text{F}$.

Solution :
$$F = \frac{9}{5} C + 32 = \frac{9}{5} \times 50 + 32$$
$$= 9 \times 10 + 32 = 90 + 32 = 122$$
Thus $50^{\circ}\text{C} = 122^{\circ}\text{F}$.

Why is mercury considered an ideal liquid for thermometers:

Properties of Mercury

Its expansion is uniform. (For equal amounts of heat it expands by equal lengths). It is opaque and shining. It does not stick to the sides of the glass tube. It is a good conductor of heat. It is easily available in pure state. It can measure a wide range of temperatures from -40°C to 356°C , and up to 570°C under pressure, in liquid state. It expands and contracts uniformly with a rise and fall in temperature. Mercury freezes at 39°C and boils at 357°C under normal pressure. Therefore, it can be used over a wide temperature range (say 10°C to 300°C). Thermometers are of two types, one used in laboratories and the other by doctors.

TYPES OF THERMOMETERS

Laboratory Thermometer

Lab thermometers are used to measure the temperature in school and other laboratories for scientific research. They are also used in the industry as they can measure temperatures higher than what clinical thermometers can record. The stem and the bulb of a lab thermometer are longer when compared to that of a clinical thermometer and there is no kink in the lab thermometer. The scale of temperature is graduated generally from 10°C to 110°C over the glass tube. Each division of temperature scale is further divided into 10 parts to read fraction of temperature.

Proper Use of a Laboratory Thermometer

To find the temperature of an object, the bulb of the thermometer needs to be in close contact with that object. Watch the shiny line of mercury in the tube. The highest point on the scale, at which the rise of mercury stops, shows the temperature of the object. To learn the use of a laboratory thermometer first of all, hold the thermometer from the top-end. Observe the calibration on the stem of the thermometer. To create a scale, we need two fixed points. Let us choose the melting point (0°C) and boiling point (100°C) as two fixed points for the scale of thermometer. Now divide the distance between these two points on the thermometer into 100 equal parts. Think about how we can achieve this. Each of the 100 equal parts represents 1°C .

We further divide 1°C into 10 small divisions. It can be read as $1/10 = 0.1^{\circ}\text{C}$.



Laboratory Thermometer

Use of Laboratory Thermometer

Take some water in a beaker. Take a laboratory thermometer and immerse its bulb end in water; holding it vertically. Ensure to dip whole portion of bulb end. The bulb end should not touch the bottom or side of the beaker. Observe the movement of rise of mercury. When it becomes stable, take the reading of the thermometer. Repeat this with hot water and take the reading.

Precautions while using laboratory thermometers

The thermometer should neither be placed in very hot liquid nor in ice cold liquid. It can break. The temperature of the liquid should be within the range of the thermometer. The thermometer should be kept upright not tilted. The bulb of the thermometer should be surrounded from all sides by the substance of which the temperature is to be measured. The bulb should not touch the surface of the container. The reading of temperature should be taken without taking it out of the liquid or substance. The eye should be kept at the level of the end of the mercury thread. Wait till the mercury thread becomes steady and note the temperature.

Clinical Thermometer

A clinical thermometer is a tool for measuring temperature, designed for clinical use in humans or animals. It has narrow kink (or constriction) in the capillary tube just above the bulb. Kink is a one-way valve. Mercury will move up past the kink due to the pressure of the expanding bulb of mercury in the bottom of the thermometer, as temperature goes up. But as the temperature cools, the tiny weight of the mercury above the kink is not enough to push it back down past the kink, so it remains until shaken down. So the doctor can read the patient's temperature while the thermometer is not in the patient's mouth. The mercury can be brought back into the bulb by giving it a jerk, after the thermometer has been read.



Clinical Thermometer

Reading a Clinical Thermometer

Determine the value of one small division on the thermometer scale. For this, count the number of small divisions between two bigger marks i.e., a difference of 1°C . If the number of small divisions is 5, value of one small division will be $1 \div 5 = 0.2^{\circ}\text{C}$.

Reading of thermometer and measuring of temperature

Take a clinical thermometer and hold it horizontally with reading scale towards your eye. Do not hold the thermometer from the bulb. Rotate the thermometer slightly clockwise and anticlockwise. By doing this you will see a shiny thin silvery thread. The end of the silvery thread shows the reading of temperature. If mercury lining ends at 37, the reading is 37°C .

Wash the bulb end of thermometer with an antiseptic solution.

Give two or three jerks slightly. By doing this the mercury level would fall. When it falls to 35°C or below, place the bulb of the thermometer under the tongue or under the armpit. It is safer to place the thermometer under the armpit. Take out the thermometer and read the temperature. Temperature would be near 37°C . The normal body temperature is 37°C . This can differ from person to person. In Celsius scale the thermometer is marked 35°C to 42°C .

Let the reading is 2 small divisions more than 37°C . If the number of small divisions is 5 in a difference of 1°C ,

$$\text{Value of each small division} = 1 \div 5 = 0.2^{\circ}\text{C}$$

$$2 \text{ small divisions} = 2 \times 0.2^{\circ}\text{C} = 0.4^{\circ}\text{C}$$

$$\text{Reading} = 37^{\circ}\text{C} + 0.4^{\circ}\text{C} = 37.4^{\circ}\text{C}$$

HEAT

Kinetic energy is the energy of any matter in motion. The small particles that make up matter are constantly moving. They have kinetic energy. The energy of the motion of particles in matter is called thermal energy. When thermal energy is transferred, it is known as heat. The word thermal means 'heat'. Heat is the thermal energy that flows from an object. Heat flows from an object of higher temperature to an object of lower temperature.

Heat is measured in Calories or Joules. In the international standard system of units, heat is measured in Joules.

Transfer of Heat

Imagine your teacher has brought packets of dates to distribute in your class. The teacher has three options.

Option 1 : The teacher can give a packet of dates to first student and ask him/her to pass it to the next student and then next student passes on further. In this way, each one of you will receive a packet without moving from your place.

Option 2 : The teacher can ask students to line up and come to the table one by one. After receiving the packet every student moves back to his/her seat.

Option 3 : The teacher can just throw a packet towards each one of you without any movement on your part.

Much the same way heat energy transfers from a hot body to a cold body. The transfer of heat from one object to the other is called **transmission of heat**. Heat energy transfers in three ways, i.e., conduction, convection and radiation.

CONDUCTION

If one end of a metal spoon is heated with a flame, the other end will also get heated up after a while.

The heat energy is transferred from one end of the spoon to the other without the actual movement of particles (atoms or molecules) of the spoon. Such a mode of transmission of heat is called **conduction**. It resembles to the example given above in option 1. The transfer of heat through matter without the actual movement of particles from their position is called conduction. Conduction occurs in solids, liquids and gases, but solids usually conduct heat better than liquids or gases.



Conduction of Heat

In solids, the particles are held very close to each other. They vibrate constantly. When we heat one part of a solid, the particles gain heat energy and start vibrating faster. During their vibration, they bump into other particles and cause them to vibrate fast. In this way, the particles of hot part of a solid transfer heat to those in the colder parts.

The gradual flow of heat in a substance from the hot end to the cold end is called **conduction of heat**.

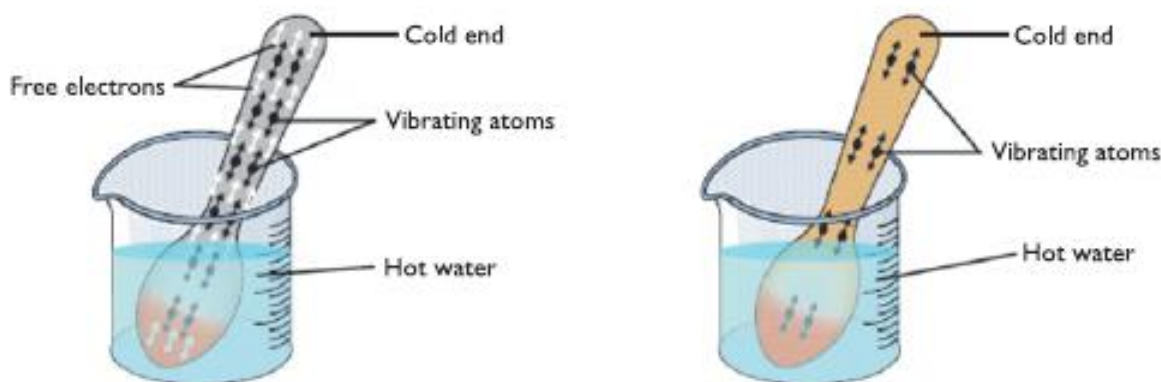
Good and Bad Conductors

Different materials conduct heat at different rates. Materials which allow heat to flow through them easily are called **good conductors of heat**. Solids such as metals are good conductors of heat. Materials which do not allow heat to flow through them easily are called **bad conductors of heat** or **heat insulators**. Solids such as wood, glass, plastic, styrofoam, etc. are bad conductors of heat. All liquids (except mercury which is a liquid metal) and gases are also bad conductors of heat.

Heat Conductors and Insulators	
Good Conductors	Insulators
silver	air or any gas
copper	cork
aluminium	glass
iron	plastic
mercury	wood

How does Conduction of Heat takes place?

All solids (metals and non-metals) are made of tiny particles called **molecules**. But, some solids conduct heat better than others. Let us suppose, a metal and a plastic spoon be in hot water. Which one is a good conductor of heat, metal or plastic? When both spoons receive heat energy, the particles (atoms or molecules) at the hot end of each spoon vibrate fast and bump into neighbouring particles. These particles transfer heat energy to next neighbouring particles.



Conduction of Heat

Conduction in Liquids and Gases

The process of conduction in liquids and gases is very slow as compared to solids (metals). The particles in liquids and gases are not held closely together. The particles have less chances to bump into other particles in liquids and even lesser in gases. That is why, the transfer of heat energy from fast-moving molecules to neighbouring molecules is **slow**. Water and air are bad conductors of heat.

Everyday Applications of Conduction of Heat

Conduction plays an important role in our lives.

1. Cooking utensils, electric kettle, iron, soldering iron, etc. are made of metals to conduct heat quickly. Their handles are made of plastic or wood which are bad conductors.
2. Birds have feathers which keep their bodies warm because feathers are bad conductor of heat.
3. Woollen clothes and blankets slow down the transfer of heat. It happens so because the wool traps air in it. The air is a bad conductor of heat.
4. Ice is covered with jute rugs to reduce its melting speed. Jute is a bad conductor of heat.



Bird



Woollen Cloth

5. An insulating material (e.g. styrofoam) is filled between the double walls of a refrigerator. It reduces the transfer of heat across the walls of the refrigerator.
6. Double-pan windows are used in buildings to slow the transfer of heat. Air between the two layers of glass acts as an insulator.
7. Thermos bottles use air or a vacuum to slow the transfer of heat by conduction.

CONVECTION

Unlike particles of solids, particles in liquids and gases move from one place to another. Take a beaker and put small pieces of paper in it. Fill half of the beaker with water. Heat the beaker by a spirit lamp. We shall see that pieces of paper rise to the top of water, move sideways and sink to the bottom. The water in the beaker also gets warm. The molecules of water absorb heat energy from the bottom of the beaker and rise to the top. Other surrounding molecules of water come to the bottom to absorb heat energy. The transfer of heat in which molecules of a medium actually move to the source of heat energy to absorb heat and then move away from it, is called **convection**. Convection occurs in liquids and gases only because their molecules can move freely. The molecules of a solid are held closely together. They cannot move freely, therefore, convection is not possible in solids. The upward and downward movement of molecules of water or air is called a **convection current**.



Activity

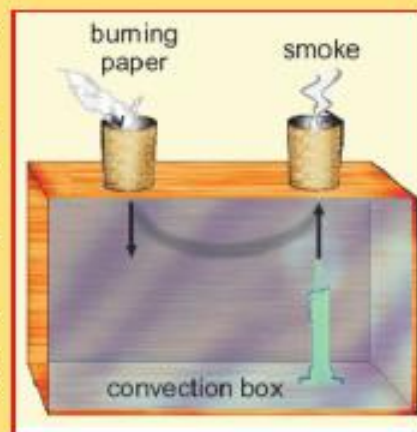
- ▶ To draw convection currents in air.

Materials required : A rectangular wooden box with sliding glass in front, candle, paper, glass chimney.

Method : Make two big holes in the top of the box and place a glass tube on each hole. Place a lighted candle below the left hole. Bring a burning paper just above the right hole.

Observation : We see that the smoke of burning paper enters through left chimney and comes out through the right chimney.

Conclusion : On heating air by the candle flame, gas particles move upward and the colder air from the left chimney enters the box. Thus convection currents are set up in gases and gases transfer heat by convection.



Wind and Ocean Currents

We know that convection is the transfer of heat by the actual movement of the particles in materials. Wind and ocean currents are examples of effects of convection. The heat of the Sun heats up the surface of the Earth and the air near it also gets hot. The air expands and gets lighter. So, it rises up and cool air from the neighbouring regions moves in to fill its space. The rising warm air reaches upper colder layers of the air and cools



Ocean Currents

down. Cool heavy air sinks to the Earth in cold regions to blow again to take the place of the rising air. Thus, convection currents are set up and the wind-system goes on. Ocean currents are also set up due to convection of heat. Water of the hot regions of an ocean gets hot, it expands and gets lighter, but water in the colder regions remains cold and heavy. Hot water moves along the surface of the ocean towards the colder regions. The cold water flows below the surface of the ocean towards the hot regions. In this way, ocean currents are set up.

Convection and Gliding Flight of Birds

Convection currents also take place in the atmosphere. The heat from the Sun warms the air near the ground. The warm air expands and becomes lighter in weight. As warm air rises, colder air rushes in to fill its place near the ground. This process continues. Birds like eagles, hawks, vultures and gulls take advantage of this phenomenon. They enjoy gliding. During gliding flight a bird does not move its wings, but glides on air currents. A lot of energy of birds is saved during gliding.



Flight of a Bird

Sea Breeze and Land Breeze

Sea breeze and land breeze are actually convection currents. During the day, the land warms up more than the sea. The air over the land becomes hotter and rises upwards. To take its place, the cooler air from the sea moves towards the land. This convection current from the sea to the land is called the **sea breeze**.

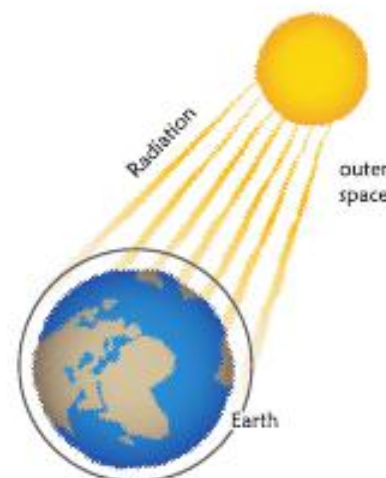
After sunset, the land loses heat faster than the sea water. As a result, the air over the sea is warmer at night. The air over the sea being warmer rises up and to take its place, cooler air from the land starts moving towards the sea. The convection current from the land to the sea is called **land breeze**.



Land Breeze and Sea Breeze

RADIATION

The transfer of heat energy from a hot body to a cold body directly, without heating the space in between the two bodies is called **radiation**. When we sit in the Sun or in front of a heater, we feel warmth. Heat energy reaches us by radiation. This heat cannot reach us by conduction because air is a bad conductor of heat. Similarly, this heat cannot reach us by convection, as the hot air rises upward, rather than sideways. If we put a cardboard or a plastic sheet between us and the source of heat, we no longer feel warmth. So, we can say that heat from the Sun or a heater reaches us by radiation which requires no medium.



Radiation

FACT In hot countries, houses are painted with light-coloured paints. The light colour of the paint absorbs little heat and reflects most of the radiation from the Sun. What colour of paint would you suggest for the houses in very cold countries?

Everyday Applications of Radiation of Heat

Every object emits or radiates some amount of heat. Knowledge of radiation can help us in many ways.

1. When we sit beside a fire, the heat of fire reaches us by radiation.
2. The cooling fans at the back of our refrigerator need to radiate its heat quickly to the surroundings. Its surface is made rough and painted black.
3. During hot summer days, it is advised to wear white or light-coloured clothes. White colour absorbs less heat than dark colours.

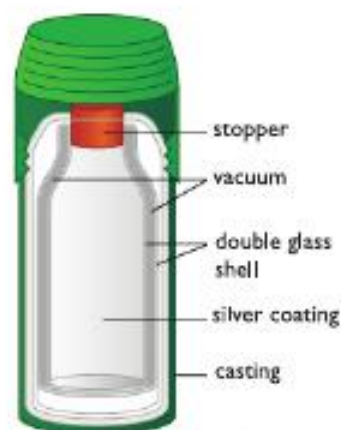


Green-house

4. In cold areas, a greenhouse is used for better growth of plants. Radiation from the Sun passes through the glass or plastic and warms up the soil and plants. Plants and soil absorb and emit radiation and increase the temperature in the greenhouse. Plants grow well in increased temperature of the greenhouse .

THE VACUUM FLASK

The vacuum flask is a container which can keep hot things hot and cold things cold. The vacuum flask reduces the rate of transfer of heat by all the three ways, i.e. conduction, convection and radiation. The vacuum flask (thermos flask) is actually two thin glass or metal bottles, one inside the other. Air between the glass walls is removed to create vacuum. The vacuum prevents the transfer of heat by conduction and convection. The walls of both bottles are coated with aluminium on the vacuum side. These silvered (like a mirror) and smooth glass walls prevent transfer of heat by radiation. The mouth of the flask is made from a bad conductor such as cork or plastic. A little amount of heat may be lost by conduction through the mouth. The thin walled glass bottle is protected by keeping it in a metal or plastic container.



Vacuum Flask

REWIND

- ▶ Heat is a form of energy. Heat always moves from an object of higher temperature to an object of lower temperature.
- ▶ Conduction is the transfer of heat through matter without the actual movement of particles from their positions.
- ▶ Convection is the transfer of heat in which molecules of a medium actually move to the source of heat energy to absorb heat and then move away from it.
- ▶ Radiation is the transfer of heat from a hot body to a cold body directly, without heating the space in between the two bodies.
- ▶ Conduction occurs in solids, liquids and gases. But metals are better heat conductors.
- ▶ Convection occurs only in liquids and gases.
- ▶ Radiation needs no material medium to transfer heat energy.
- ▶ Good and bad conductors of heat play very important role in our lives.
- ▶ Convection causes wind and ocean currents.
- ▶ Heat from the sun reaches us by radiation.
- ▶ Some birds take advantage of convection currents in air and glide in the air for hours.
- ▶ A good radiator of heat is also a good absorber of heat.
- ▶ A vacuum flask reduces the transfer of heat by conduction, convection and radiation to keep things hot or cold.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. The bottom of stainless steel pan have copper because
- | | | | |
|----------------------------------|--------------------------|--------------------------------|--------------------------|
| (i) Copper is more durable | ☐ | (ii) Copper is more attractive | ☐ |
| (iii) Copper is better conductor | ☐ | (iv) Copper is easier to clean | ☐ |
- b. Heat is a form of
- | | | | | | | | |
|-----------------|--------------------------|-------------|--------------------------|-------------|--------------------------|-----------|--------------------------|
| (i) Temperature | ☐ | (ii) Energy | ☐ | (iii) Power | ☐ | (iv) Work | ☐ |
|-----------------|--------------------------|-------------|--------------------------|-------------|--------------------------|-----------|--------------------------|
- c. Method of heat transfer in steel rod is
- | | | | | | | | |
|----------------|--------------------------|-----------------|--------------------------|-----------------|--------------------------|-------------------|--------------------------|
| (i) Conduction | ☐ | (ii) Convection | ☐ | (iii) Radiation | ☐ | (iv) All of these | ☐ |
|----------------|--------------------------|-----------------|--------------------------|-----------------|--------------------------|-------------------|--------------------------|
- d. Which of the following is as insulator
- | | | | | | | | |
|----------------|--------------------------|--------------------|--------------------------|-----------|--------------------------|---------------|--------------------------|
| (i) Copper rod | ☐ | (ii) Aluminium rod | ☐ | (iii) Air | ☐ | (iv) Graphite | ☐ |
|----------------|--------------------------|--------------------|--------------------------|-----------|--------------------------|---------------|--------------------------|

2 Fill in the blanks :

- a. The S.I unit of heat is _____.
- b. Heat is a form of _____.
- c. Mercury level falls easily in a _____ thermometer.
- d. Cooking vessels should be made up of _____ conductor.
- e. Range of clinical thermometer is from _____ °C to _____ °C.
- f. The heat radiation can pass through _____ without heating it.

3 Write T for true and F for false :

- a. The S.I unit of temperature is Joule.
- b. Kink is present in clinical thermometer.
- c. Mercury is used in thermometer as it expands uniformly.
- d. Convection is the fastest mode of heat transfer.
- e. 100°C is equal to 373° K.
- f. Light coloured clothes are better absorbers of heat.

4 Answer these questions in short :

- a. Why mercury is used in a clinical thermometer?
- b. What is the role of bad conductors of heat in our daily life?
- c. How does a blanket keeps you warm in winter but prevents ice from melting?
- d. Solids do not favour convection. Why?
- e. Why do people wear light coloured clothes in the summer?
- f. What is temperature?

5 Give an example where the following energy changes would take place :

- Electrical to thermal
- Chemical to Thermal
- Electrical to Mechanical

6 Give answers in detail :

- State similarities and differences between the laboratory thermometer and the clinical thermometer.
- During winter mornings why do people stand in the Sun? Explain.
- Why do we use mercury in the thermometer? Can water be used instead of mercury? What are the problems in using it?
- List the three ways in which transfer of heat takes place explaining each method briefly.
- Describe in your own words : Conduction, Convection and Radiation.
- Explain the function of a vacuum flask.

NEP QUERY



PROJECT WORK

- In the laboratory : At what temperature does water boil in your town? Take some water in a beaker. Fix a thermometer in a stand and let its bulb dip in the water. The bulb should not touch the base or side of the beaker. Heat the beaker on a wire gauge on a tripod stand. When the water boils, record the temperature. Is it 100°C or less? At sea level, the water boils at 100°C . The higher your town is above sea level, the lower will be the boiling temperature of water.

(Learning by Experiment)

COMMUNICATION SKILLS

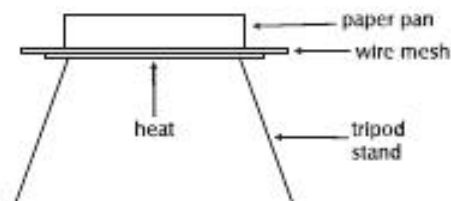
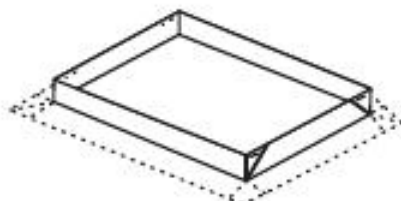
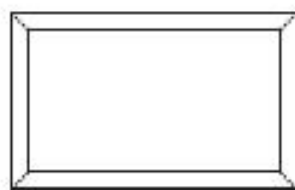
- Hold a group discussion in your class on the topic- "Transfer of heat through convection is the most important type of heat transfer for human survival".

(Discussion)

ACTIVITY

- Work in small groups. Try this amazing experiment with help from an adult. Make a paper pan as shown in the figure. Borrow a tripod stand from your school laboratory. Place a wire mesh over it. Pour some water into this paper pan. Place it on the tripod stand. Start heating the paper pan. You will find that in a little while the water will start to boil.

(Creativity)



Our Forest

11

- Importance of Forests
- Types of Forest
- Interdependence of Animals and Plants
- Food Chain
- Deforestation
- Conservation of Forests

The word forest derived from Greek word 'foris' means outside. Forests are precious national resource which not only play significant role in national condor but help in pollution control and maintaining balance. An area with high density of trees is called a **forest**. A forest is a system which is composed of plants, animals and microorganisms. Forests serve as the "Green Lungs" and "Water Purifying Systems" in nature.



Forest

IMPORTANCE OF FORESTS

Forests conserve soil, water and air. Forests are rich in medicinal plants. They offer biotic check for pests and diseases. Forests are rich sources of biodiversity and gene pools. They serve as sanctuaries for wildlife. They prevent frequent floods and droughts. They induce rainfall. They provide fuel to millions and raw materials to industries. They generate employment and income. The list is endless. So the forests must be maintained properly and conservation approaches are adopted for upkeep of forests.

FACT

World Forest Day is celebrated on 21 March every year.

TYPES OF FORESTS

There are different types of forests in our vast country due to different climatic conditions. The chief types of forests in India are the following–

Tropical Rain Forests

Tropical rainforests may be called the lowland equatorial evergreen rainforests. Rainforests are those forests which are characterised by high rainfall between 1750 mm and 2000 mm. These forests incur heavy showers of 100-600 cm a year, and hence the name, rainforests. Rainforests are dominated by the broad-leaved evergreen trees, which form a leafy canopy over the forest floor. Coffee, chocolate, banana tree, mango tree, papaya tree, avocados and sugarcane all originally came from tropical rainforests.

Temperate Deciduous Forests

Temperate deciduous forests are those, which consist of predominantly broad-leafed trees. The tropical monsoon deciduous forests are found in areas receiving an annual rainfall of 100 to 200 cms in India, with a distinct dry and rainy season and minimum temperature. The deciduous forest can further be divided into moist and dry.



Moist Deciduous Forests

The moist deciduous forests are scattered throughout India except in the western and the north-western regions. The trees have broad trunks, are tall and have branching trunks and roots to hold them firmly to the ground. These forests are dominated by sal and teak, along with mango, bamboo and rosewood.



Dry Deciduous Forests in India

Indian dry deciduous forests are found throughout the northern part of the country except in the Northeast. The canopy of the trees does not normally exceed 25 metres. The principal trees of these forests are Indian teak tree, sal, sandalwood, mahua, khair, mango, jackfruit, wattle and arjun, semal, myrobalan and banyan tree.



Deciduous or Monsoon Forests

These forests are found in monsoon areas (100 to 200 cm annual rain). Rainforests are dominated by the broad-leafed evergreen trees, which form a leafy canopy over the forest floor. The common trees are teak, sal, sesum and sandal.



Evergreen Forests

The tropical evergreen forests usually occur in areas receiving more than 200 cm of rainfall and having a temperature of 15 to 30 degrees Celsius. The trees are 50 to 60 metres tall. They make a canopy of broad leaves on the top. The common trees are rosewood, ebony and mahogany.



Mountain or Alpine Forests

The alpine forests occur all along the Himalayas at altitudes ranging between 2,900 to 3,500 m or even up-to 3800 m above sea level, depending upon the location and the variety of species. These trees have soft wood. The common trees are chir (pine) and deodar. These forests are found in the Himalayas above 1500 metre height.



Tidal Forests

Tidal forests are one of the most productive and bio diverse wetlands on earth. These forests are found near the sea in the large deltas of rivers. The trees have the roots covered by salt water at high tide. Common trees are mangroves. In the delta of the river Ganga, these forests are called the sunderban because of sundri trees.



Dry Forests or Thorny Shrubs

They grow in scanty rainfall areas. Common trees are babul and kikar. These are found in the north-western part of the Indian Desert.



FACT

In India, there are sixteen types of forests recognized.

INTERDEPENDENCE OF ANIMALS AND PLANTS

Animals and plants are interdependent on each other for basic survival needs like food, protection, shelter and propagation. Plants provide food for animals and humans, who cannot make their own foods like plants do. Plants and trees also provide shelter and protection. As green plants produce their own food, they are called **producers (or autotrophs)**. Animals consume plants or other animals, so they are called **consumers (or heterotrophs)**.

Many plants and animals that live on earth choose special places to live. The place where a plant or animal lives is called its **habitat**.

Plants and trees also provide shelter and protection for different animals like birds, squirrels and insects. Birds also build nests high up in trees away from predators. Similarly, plants also benefit from animals in that animal can aid in seed dispersal ensuring the survival of plants in other regions of an ecosystem. Bees also help in the pollination of plants in a habitat.

Some organisms derive their food from the dead and decaying animals/plants. They are called **saprotrophs** or **saprophytes**.

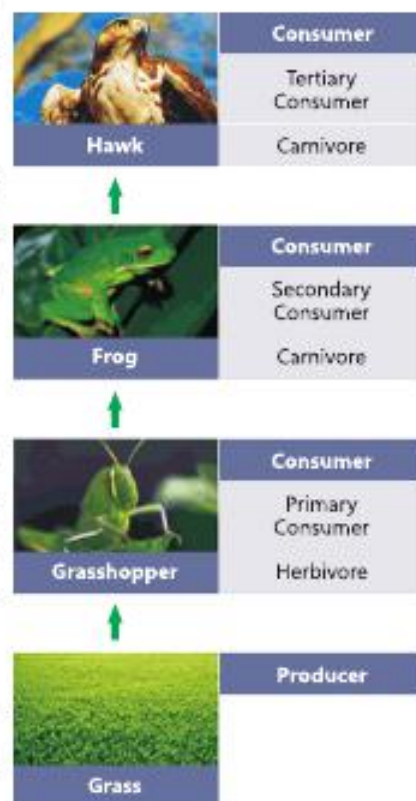
Some organisms decompose the dead remains of plants and animals to simple substances. They are called **decomposers**. They help in maintaining the supply of nutrients to the growing plants in the forests.

FOOD CHAIN

A food chain shows the different species of an organism in an ecosystem, and what eats what. A food chain includes producers, herbivores and carnivores. Grass is eaten by grasshopper, grasshoppers eaten by frog, frog is eaten by snake and snake is eaten by eagle.

A food chain always starts with a producer, an organism that makes food. This is usually a green plant, because plants can make their own food by photosynthesis.

A food chain ends with a consumer, an animal that eats a plant or another animal.



A food chain from grass to hawk

Animals Depend upon Plants

Plants provide food and shelter for animals, and as they photosynthesize, regulate the levels of oxygen and carbon dioxide in the atmosphere.

As food producers, plants are eaten by herbivores, which in turn become food for the omnivores and carnivores. Plants are also the homes of many animals, small and big. Plants provide shelter from predators and harsh factors of the environment, like the hot sun, cold snow and torrential rain. When plants photosynthesize, they take in carbon dioxide and give out the fresh oxygen that all the animals need for respiration. Plants are crucial for the health of all animals.



Plants Depend upon Animals



Plants depend on animals for nutrients, pollination and seed dispersal, and as the animals consume plants, they regulate the numbers of different species of plants. While plants provide oxygen for the animals as they photosynthesize, animals respire and give out carbon dioxide for plants to make food with. It is an interdependent relationship here. This is not to say that plants do not respire themselves. They do, it is just that the amount of carbon dioxide they give out is not enough for the plants to make enough food with. As such, plants need animals.

Also, when animals die, they decompose and become natural fertilizers for plants. Being pretty much immobile, plants also depend on animals to pollinate them for reproduction. And when the fertilized plants eventually produce seeds encased in fruits, animals eat them or carry them along on their fur, dispersing the seeds far and wide, ensuring the continuity of the plant species.

DEFORESTATION

Large scale cutting of forests is called **deforestation**. The main cause of the present day crisis is increasing population. Majority of people living in rural areas depend on forest for their firewood for cooking. Plant derivatives such as leaves, saw dust, charcoal, etc. play an important role in providing domestic energy.

Timbers, bamboos for thatched houses are main ingredients of rural areas. Bullock-carts and agricultural implements are still largely manufactured out of timber and poles. As our population is increasing in a geometric progression, the pressure of population brings more and more area under cultivable land, the total deforested area may gradually increase.

Removal of vegetation cover brings about marked changes in the local climate of the area. Thus deforestation, overgrazing, etc. bring about changes in rainfall, temperature, wind, velocity,



etc. and also lead to soil erosion. Such changes then lead to desertification of the area. Due to lowering of water table, a large number of trees die.

The forest area of a country should be at least 33 per cent of the total area. But careless cutting down of trees has reduced our forest area to about 22 percent.

The commercial use of forests nowadays has reached such an extent that it has become a threat to the environment in the form of:

- (i) Increase in temperature,
- (ii) Lesser precipitation,
- (iii) Increased rate of soil erosion,
- (iv) Increase in frequency and volume of floods,
- (v) Loss of soil productivity,
- (vi) Extinction of several species,
- (vii) Non-availability of several essential forest products, and
- (viii) Imbalance in ecosystem.

The harmful effects of deforestation are so much that all over the world people and authorities have realised that forest resources must be conserved properly in order to protect the ecosystem.

Deforestation and its Effect : Large scale cutting of trees to clear land for human use is called **deforestation**. Due to growing human population, vast tracts of forest have been removed. This has created many problems. Some of them are given below:

- (i) Reduced forest cover has led to soil erosion and loss of soil fertility at many places.
- (ii) Reduced forest cover has disturbed the natural process of recharging of groundwater. This has resulted in shortage of drinking water at many places.
- (iii) Reduced forest cover means there is loss of habitat for many animals. Existence of most of the animals is in danger because of this.
- (iv) Reduced forest cover means there is higher percentage of carbon dioxide in the atmosphere. This is leading to global warming. The average temperature of the earth is on the rise.

CONSERVATION OF FORESTS

Keeping in view the importance of forests in our life, the people all over the world have stopped unnecessary deforestation. Our Government has also made laws to prevent unnecessary felling of trees. Government has decided to declare certain forests as protected areas. These protected areas are called Reserves or wild Life Sanctuaries. Here no one is allowed to hunt animals. There are about 150 wild life sanctuaries in India. People come from all over the world to see these sanctuaries. They consist of some of the most beautiful and rare animals.

The Government has made plans to grow more trees. Vanmahotsava is celebrated every year to plant more trees. Thousands of school children participate in the Vanmahotsava and plant trees every year. Chipko Movement and Narmada Bachao Andolan are some of the movements



◀ REWIND

- ▶ We obtain various things and materials from the forest.
- ▶ A forest is a good habitat for many plants and animals.
- ▶ Forest helps in binding of soil and protecting it from erosion.
- ▶ People living in forests depend on its products for their livelihood.
- ▶ Social forestry could help overcome deforestation.
- ▶ Forests are lungs of our earth.
- ▶ Destroying forests poses threat to life of animals and plants living there as well our own survival.



ASSIGNMENTS

(Learning by Doing)

COGNITIVE SKILLS

1 Tick (✓) the correct one :

a. The item that is not a product of forest

- (i) Sealing wax ☐ (ii) Honey ☐ (iii) Wooden statue ☐ (iv) Salt ☐

b. The branchy part of a tree above the stem is known as

- (i) Canopy ☐ (ii) Crown ☐ (iii) Herb ☐ (iv) Shrub ☐

c. Find the missing word from the following equation :

Grass → Insects → Frog → _____ → Eagle.

- (i) Peacock ☐ (ii) Snake ☐ (iii) Koel ☐ (iv) Crow ☐

d. Micro-organisms act upon the dead plants to produce

- (i) Sand ☐ (ii) Mushrooms ☐ (iii) Humus ☐ (iv) Wood ☐

2 Fill in the blanks :

- a. Deforestation leads to _____ in nature.
- b. Plants are natural air _____ of atmosphere.
- c. Judicious use of forest is called _____.
- d. Saprophytes are also known as _____.
- e. Quinine is obtained from _____ tree.
- f. _____ are lungs of our earth.

3 Write T for true and F for false :

- a. The forest cover is increasing in our country.
- b. Forests help in regulating the water cycle.
- c. Overgrazing is one of the reasons for depletion of forest.
- d. In a forest, only one food chain can exist.
- e. Plants do not depend on animals.
- f. A forest is a good habitat for many plants and animals.

- 4 Answer these questions in short :**
- How forests prevent floods?
 - Forests maintain the balance between oxygen and carbon dioxide in the atmosphere. How?
 - Name different products we get from forests.
 - How are animals important for plants?
 - Why are forests being cut these days?
 - What are decomposers?

- 5 Give answers in detail :**
- Why are world worry about issue of forest these days?
 - What are decomposers? Name any two of them. What do they do in the forest?
 - Write a note on the interdependence of plants and animals. Give examples.
 - Explain a food chain.
 - How can forests be conserved?

NEP QUERY



PROJECT WORK

- In the laboratory : Collect different forest products like rubber, gum, rasin, lac, spices, medicines. Put each in a separate transparent packing and label it. Write the plant part from which it is obtained.

(Learning by Experiment)

COMMUNICATION SKILLS

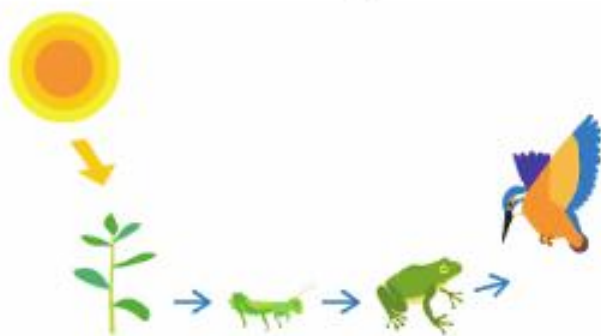
- Hold a group discussion in your class on the topic- "Deforestation and Development".

(Discussion)

ACTIVITY

- Look at the food chain and give the answer of the questions.

(Creativity)



- Select the living thing from the food chain that is producers _____.
 - Select the living things from the food chain that are consumers _____.
 - Select the primary consumer _____.
 - All plants and animals depend on energy from the _____.
- e. Organisms that make their own food are _____.
- f. The arrows in a food chain of web show _____.

Importance of Water

12

- Sources of Water
- Uses of Water
- Recharging of Ground Water
- Forms of Water
- Natural Distribution of Water

International World Water Day is held annually on 22nd March as a means of focussing attention on the importance of freshwater. Life, both animal and plant, is impossible without water. Without water men die of dehydration. About 70% of the human body is water. Of the total surface of the earth, 75% is water and just below the surface of the land in most areas there is a saturated layer known as the water table. Animals and plants have large percentages of water in their make-up and some animals regulate their movements according to the water sources.



Water Crisis

If there is no water, there can be no life. The moon is a sterile desert because there is no water in it. It is to be remembered that it was water that determined the location of human settlements. Water shortage has become a matter of concern throughout the world. It is estimated that in a few years from now more than one third of the people in the world could face water scarcity. In some places, even today there is an acute shortage of water.

Water is essential for the human body to function. The human body is largely made of water. Body water content declines with age, from about 75% in babies to 60% in adults. Although we can live for up to 50 days without food, without water we will survive only a few days, even in a cool climate. Large quantities of water are used for growing crops and in industry.

SOURCES OF WATER

Ocean(sea water)	97%
Glaciers & Poles as ice (fresh water)	2%
Ground Water, Water in Ponds, Lakes and Rivers (available fresh water)	1%

Fresh water has been the constant and essential companion of human beings throughout history. Water is used in great quantities in agriculture and industries. But our planet is poorly

FACT

Although most lakes are fresh water, some such as the Great Salt Lake and the Dead Sea are salt water lakes.

endowed with fresh water. Most water is rendered useless to humans by dilution with salt in the oceans. Only 2.5 percent is available as fresh water of which 2/3 is locked up in ice and snow. Nature is unkind in depositing almost eighty percent of rain over the sea. The rain that falls over the land has a great potential value.

Out of the total water resources, 97% remains in oceans as salty water, 2% remains as ice and rest 0.6 to 1.0% remains as fresh water either on surface or as ground water. That is, the available fresh water resources are very limited.

The fresh water is used by living organisms for their different activities. Of the three major uses of water that is domestic, agricultural and industrial, agriculture is the greatest consumer of water (65% of total consumption). Industries consume 25% and 5% is used for domestic purpose.

Most of the times we do not think of the importance of fresh water. We are not aware of the need to preserve fresh water.

FORMS OF WATER

Pure water is tasteless, odourless, and colourless. Water can occur in three states : solid (ice), liquid (water), or gas (vapour).



Solid



Liquid

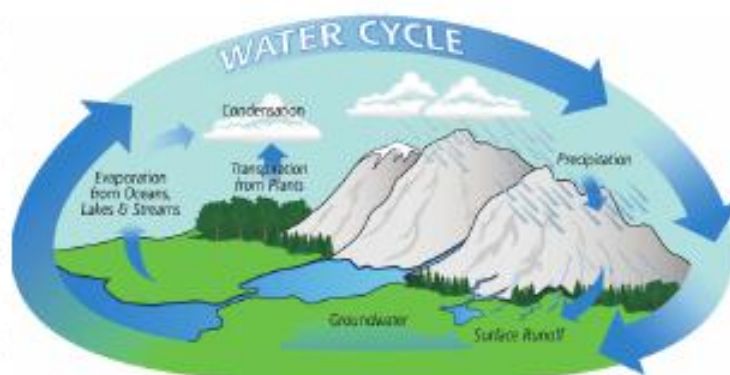


Gas

Solid water (ice) is frozen water. When water freezes, its molecules move farther apart, making ice less dense than water. This means that ice will be lighter than the same volume of water, and so ice will float in water. Water freezes at 0° Celsius, 32° Fahrenheit.

Liquid water is wet and fluid. This is the form of water with which we are most familiar. We use liquid water in many ways, including washing and drinking.

Water as a gas vapour is always present in the air around us. You cannot see it. When you boil water, the water changes from a liquid to a gas or water vapour. As some of the water vapour cools, we see it as a small cloud called steam. This cloud of steam is a mini version of the clouds we see in the sky. At sea level, steam is formed at 100° Celsius, 212° Fahrenheit.



Water Cycle

The water vapour attaches to small bits of dust in the air. It forms raindrops in warm temperatures. In cold temperatures, it freezes and forms snow or hail. The continuous cycling of water among its three forms keeps the total amount of water on the earth constant even when the whole world is using it.

Such a continuous cycling of water from the earth to the atmosphere and from the atmosphere to the earth is called **water cycle**.

Groundwater as an Important Source of Water

Ground water is the water that is found beneath the earth's surface. Have you ever driven on a road cut along the side of a hill and seen what looked like layers inside the earth? If we could see the ground beneath us, it would look very similar. The top layer of earth is dirt, but as you get deeper, the dirt changes into layers of solid rock. Believe it or not, each of these layers has many small spaces and cracks filled with water. The upper limit of this layer is called the **water table**.

The groundwater provides soil moisture for plant growth. It is drawn in the form of well water and pump water and is used for our daily activities. The groundwater contains many dissolved salts. It may also contain some bacteria and other organisms.

Ground water is replenished by the rain water and the water in rivers, lakes and ponds.

Infiltration is the process by which water on the ground surface enters the soil. The rainwater and water from rivers, ponds, lakes etc. seeps through the soil and fills the empty spaces and cracks deep below the ground. Infiltration rate in soil science is a measure of the rate at which soil is able to absorb rainfall or irrigation. The groundwater thus gets recharged by this process.

USES OF WATER

For drinking and for life processes

Water is essential for human beings and other animals. It regulates the body temperature and do other functions.

In agriculture

Without water, plants cannot conduct the process of photosynthesis. Plants absorb nutrients from soil with help of water.

Agricultural food crops grow under proper irrigation facilities, where regular water supply is available.

In industries

Industries are the greatest consumers of water. It is used for purposes such as processing, cleaning, transportation, dilution, and cooling in manufacturing facilities. It is also used for hydroelectric production.

For recreational purposes

Water is used for swimming, fishing, sailing, and other water sports.

For domestic purposes

Water is used for bathing, clothes and utensils, cleaning the house, and sanitation. Our household tasks will not be complete in the absence of water.

Depletion of Water

Water table at a place is affected by the following factors:

Climate

Greater precipitation, ice melt in the spring and summer and high humidity that prevents water from evaporating can raise the water table.

Soil Characteristics

Soil characteristics that affect the water table include permeability and composition. Soils with a high clay content are relatively impermeable, while sandy soils allow water to pass through and raise the water table.

Human Activities

Human activities such as clear cutting, farming, pumping, irrigation and construction can affect the water table.

One of the another important causes for the lowering of water table are poor rainfall (the water that seeps in) and excessive use of the groundwater. Water drawn from under the ground gets replenished by seepage of rainwater. The water table does not get affected as long as we draw as much water as is replenished by natural processes. However water table may go down if the water is not sufficiently replenished. Thus, there arises three situations:

- (i) When groundwater pumped out is equal to that charged (seeped) in by the natural processes, the water table remains unchanged.
- (ii) When the groundwater pumped out is more than the water that seeps in, the water table gets lowered.
- (iii) When the groundwater pumped out is less than the water that seeps in, the water table rises up.

The gradual and continuous lowering of water table is called depletion of water table or depletion of groundwater.



Handpump

Causes of Depletion of Water Table

The factors responsible for the depletion of water table are as follows:

Increasing population

As a result of increasing population, all the facilities such as houses, shops, roads, offices, pavements, etc. increase to fulfil the increasing demands. This, in turn, decreases the open area for seepage of water into the ground.

Increasing industries

Most of the stages of manufacturing processes in industries require water. If the

number of industries increases, the water required by them will also increase. Therefore, increase of industries contributes to the depletion of water table.

Agricultural activities

India is an agricultural country and agriculture is impossible without water. The water for agriculture is mainly utilised from ground water, rain water and canal water. As there is no rainfall in many places, agriculture cannot be entirely depended on rain water. Also, canal water is available in a few places only. Therefore, ground water is the main source of water for agricultural activities and this causes depletion of water table.

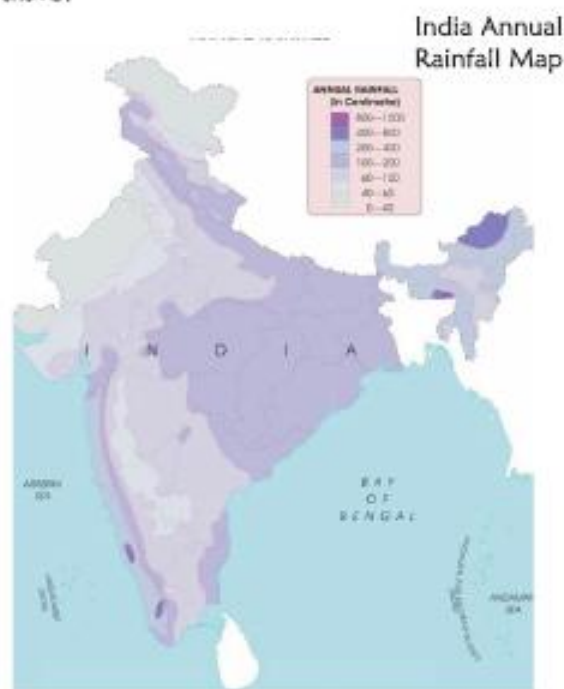
NATURAL DISTRIBUTION OF WATER

The distribution of water over the globe is uneven. Some places have good amount of rain and are water-rich and have thick forest cover. Some areas have very little rainfall and are deserts.

India is a vast country and the rainfall is not the same everywhere. Some regions have excessive rains which result in floods. Other regions have very little rainfall and suffer from droughts.

Based on the annual rainfall, a distinction can be made between 6 major climatic zones as is shown in Table.

Human activities have affected the uneven distribution of water on the earth very much.



Rainfall Map of India

Management of water resources :

In order to avoid water stress and water scarcity, the following steps should be taken for the better management of water resources.

- Treating and recycling of municipal waste water, effluents of industries and sewage for agricultural use.
- Effective rain water harvesting in urban and rural areas.
- Maintaining river systems effectively without disturbing their original paths.
- Constructing small water reservoirs instead of big dams.
- Protection of wet lands.

Climatic Zone	Annual Rainfall (mm)	Wet period (months)
Desert	less than 100	0-1
Arid	100-400	1-3
Semi-arid	400-600	3-4
Sub-humid	600-1200	4-6
Moist sub-humid	1200-1500	6-9
Humid	more than 1500	9-12

- (vi) Preventing over exploitation of ground water.
- (vii) Checking overgrazing and deforestation.

On individual level, we should take following steps :

- (i) Leaking taps should be repaired immediately.
- (ii) Turn off taps while brushing. Better use a mug for brushing and a bucket for bathing.
- (iii) Mop the floor instead of washing. Clean your car with a wet cloth/sponge instead of washing it with a water jet.
- (iv) Use drip irrigation to water plants in the fields. Drip irrigation is a technique of watering plants by making use of narrow tubings which deliver water directly at the base of the plant.



Agriculture

RECHARGING OF GROUNDWATER

Recharging of groundwater by making the rainwater to seep/percolate into the soil is called **rainwater harvesting**. Collection of rainwater for future use is called rainwater harvesting. India has a long history of different rainwater harvesting structures; especially in rain deficient areas. For example: tankas and bawris had been in use in the northwestern part of India; especially in Rajasthan. Tanka is an underground tank to collect rainwater. Bawri is an open tank in which water used to be collected. Steps were made around the bawri so that one can easily access the water.

Rainwater harvesting can also be done in modern homes. The runoff rainwater from the rooftop should be collected in an underground reservoir. Such reservoirs are usually filled with sand and gravel to filter out impurities from water. The water can either be used directly or can be channelized to recharge the underground water.



Rainwater Harvesting

◀ REWIND

- ▶ Water is essential for all living beings. There can be no life without it.
- ▶ Water exists in three forms: solid, liquid and vapour.
- ▶ Though water is maintained by the water cycle, yet there is an acute scarcity of water in many parts of the globe.
- ▶ There is an uneven distribution of water. Much of it has resulted from human activities.
- ▶ Rapid growth of industries, increasing population, growing irrigation requirements and mismanagement are some of the causes for water shortage.
- ▶ We need to be worried about the wastage during the supply of water through pipes, the leaking taps in buildings and other places. Unnecessary use of water and overdrawing from groundwater should be avoided. Recharge of water to the ground should be increased.
- ▶ The need of the hour is that every individual uses water economically.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- World Water Day is celebrated on
 (i) 22nd March (ii) 24th March (iii) 28th February (iv) 1st March
- Water exists in
 (i) Three forms (ii) Two forms (iii) Five forms (iv) No form
- The solid form, snow and ice, is present _____ of the earth.
 (i) At the poles (ii) At the equator
 (iii) Present in the atmosphere of the earth (iv) In the oceans
- The process of seeping of water into the ground is called
 (i) Infiltration (ii) Aquifer (iii) Water table (iv) Bawri
- The process of ground water getting stored between the layers of hard rock below the water table is known as
 (i) Infiltration (ii) Aquifer (iii) Water table (iv) Bawri
- Depletion of water table is due to
 (i) Increase in population (ii) Industrial and Agricultural activities
 (iii) Scanty rainfall (iv) All of these

2 Fill in the blanks :

- Three forms of water are ice, _____, and water vapour.
- Water bearing layer of the earth is _____.
- Distribution of water on globe is _____.
- Rivers and lakes are the main source of _____ water.
- _____ increase the water table.
- Plants need _____ to get nutrients from the soil to prepare their food.

3 Write T for true and F for false :

- Human body contains about 70% of water.
- Ocean water can be used for washing and cooking.
- Deforestation is responsible for decreasing water table.
- Recycling of waste water is not possible.
- Water is a non-renewable natural resource.
- There is an even distribution of water.

4 Match the following :

- | | |
|--------------------|----------------------------|
| a. ground water | (i) solid form of water |
| b. bawri | (ii) gaseous form of water |
| c. snow | (iii) wastage of water |
| d. drip irrigation | (iv) water management |
| e. leaking taps | (v) small water |
| f. vapour | (vi) water table |

5 Answer these questions in short :

- What is fresh water?
- Describe interchangeability of the states of water.
- Why we cannot drink sea water?
- How does groundwater help plants?
- What is difference between water level and water table?
- What is infiltration?

6 Give answers in detail :

- Explain the factors responsible for the depletion of water table.
- Explain how groundwater is recharged.
- What do you mean by water table?
- What is drip irrigation? What is its importance?
- What are main reasons for decreasing of water table in cities?
- Write difference between ocean water and river water.

NEP QUERY



PROJECT WORK

- In the laboratory : Collect information with help of internet and make a pie-graph on 'How much water is available for human consumption'.

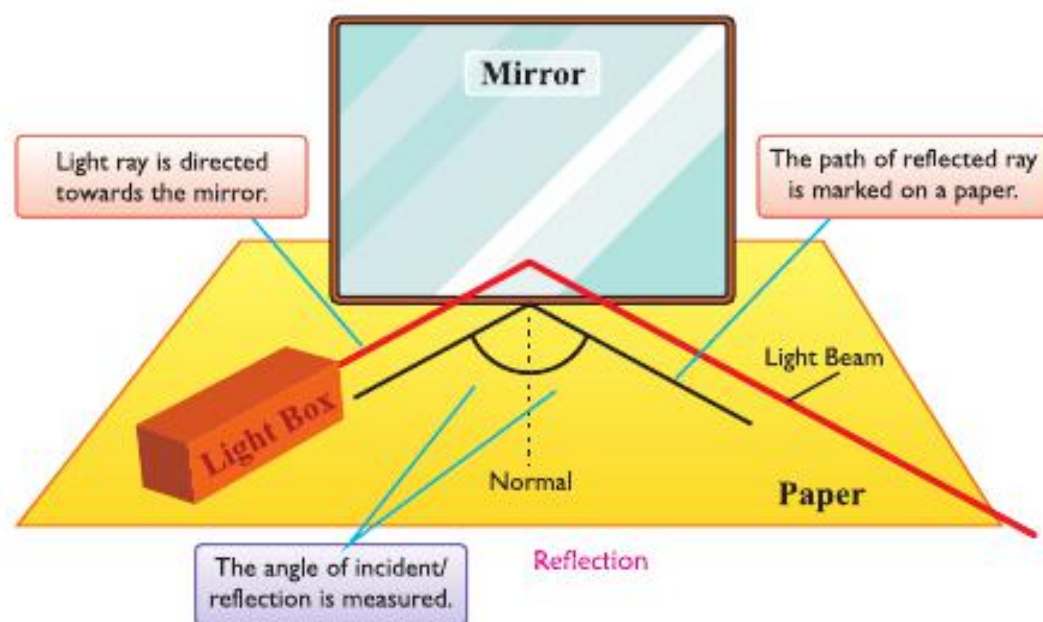
(Learning by Experiment)

COMMUNICATION SKILLS

- Hold a group discussion in your class on the topic- "Useful and adverse effects of constructing dams on rivers".

(Discussion)

emit light are not seen in the dark. An object is only seen if light is incident on it and reflected back to our eyes.



When a ray of light strikes a plane mirror, the light ray reflects off the mirror. Reflection involves a change in direction of the light ray. When light falls on a smooth or a polished surface, the surface shines. It changes the direction of light falling on it. For example: a shining stainless steel plate or a shining stainless steel spoon. The plate has a plane surface and the spoon has a curved surface. First, we shall study about the plane polished surfaces.

As the name suggests the shape of a plane mirror is a plane area. It is made from a clear plane glass sheet, usually thin and pasted with suitable reflecting abrasive (for example, mercury) on one side. Once this pasting is done, the glass becomes opaque but due to the reflecting property of the abrasive, the plane glass sheet becomes a plane glass reflector or a plane glass mirror.

Image Formation in Plane Mirrors

When we see in a plane mirror, it appears as if a similar face is there behind the mirror. The face which appears behind the mirror is the image of our face formed by the mirror. Our face itself is the object.

Let us do more activities to study some other properties of the image formed by a plane mirror.



Activity-2

- To find the location of virtual image formed by a plane mirror.

Materials required : Centimetre graph sheet, eraser.

Method : 1. Place a strip of plane mirror on a line on the graph sheet.

2. The image of the graph sheet is seen inside the mirror.

3. Place an eraser or sharpener at the boundary of the second line.



4. Note the position of the image inside the mirror.

5. Repeat by placing the eraser at different positions and observe the position of the image every time.

Observation : We see that in each case the image is at the same distance behind the mirror as the object is in front of it.

Have you ever noticed strange letters in front of an ambulance? Actually it is nothing but the word AMBULANCE written such that drivers in vehicles ahead can read the word.

We thus see from these activities that the image formed by a plane mirror has these characteristics :

- (i) The distance from the object to the mirror will be the same as the distance from the image to the mirror.
- (ii) The image is of the same size as the object.
- (iii) The image is upright as compared to the object.
- (iv) The image is laterally inverted as compared to the object.



Reflective Image

SPHERICAL MIRROR

A spherical mirror is a mirror which has the shape of a piece cut out of a spherical surface. The curved shining surface of a spoon acts as a mirror. You can see your image in a new spoon of stainless steel or any curved shining surface.

Types of Spherical Mirrors

The two types of spherical mirrors are concave and convex. Each type reflects images in a different fashion. This is due to the curve of the mirror. A good example of how a curved mirror alters an image can be seen in a fun house mirror. The image reflected back can make a person look tall and skinny or short and fat.

Concave Mirror

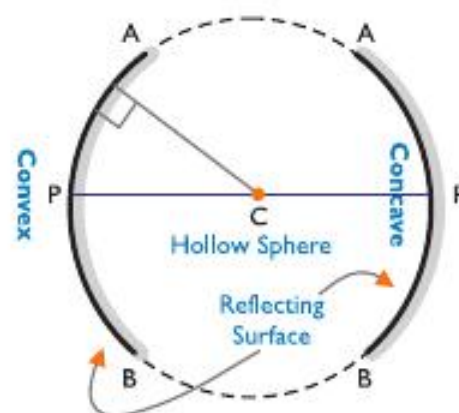
A concave mirror is one that curves in like a bowl. The image is seen inverted. The image can be smaller, the same size or magnified depending on the distance from the concave mirror.

Convex Mirror

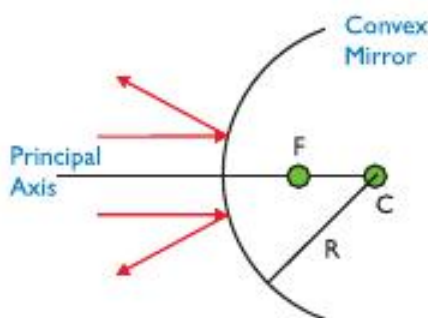
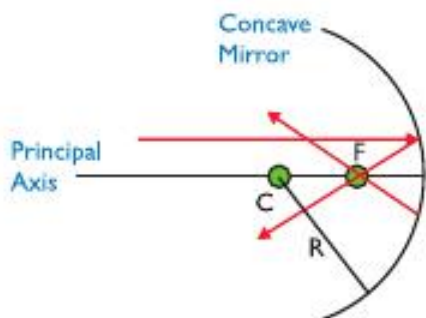
A convex mirror is shaped like the outside of a bowl. The middle of the convex mirror is raised from the edges. Images will always appear erect and virtual. The size of the image will be reduced.



Curved Mirrors



Curved Mirrors



The centre of a mirror is called its **pole**. The line passing through the centre of curvature of the mirror and the pole is called the **principal axis** of the mirror. It is perpendicular to the

mirror at the pole. Its centre of curvature, C, is the centre of the sphere it was cut from. R, the mirror's radius of curvature is the radius of the sphere. The focal point F (the point where parallel rays are focussed) is located half the distance from the mirror to the centre of curvature. The focal length, f, is :

Focal length of a spherical mirror : $f = R / 2$

This bright spot obtained on the paper sheet in the above activity is the image of the Sun. Notice that this image is formed on a screen. An image formed on a screen is called a **real image**. The image formed by a plane mirror could not be obtained on a screen. Such an image is called a **virtual image**.

Now let us try to obtain on the screen the image of a candle flame formed by a concave mirror.

Uses of Concave Mirrors

1. The reflectors of torches are concave in shape to reflect the light of the bulb to form a powerful beam of light.
2. Very big concave mirrors are used in telescopes to view stars and planets.
3. Shaving mirrors are another example that makes use of concave mirrors because of their curved and reflective surface.
4. Dentist and ENT doctors also use concave mirrors in their examinations procedure to obtain a larger image than the original of teeth, ear or skin etc.
5. Concave mirrors are also used in torches on the same principle as they are used in vehicles headlights.
6. They are also used in solar powered gadgets. The parallel rays of the sun are focussed at the focal point of the mirror and then the reflected rays are used for heating purposes like cooking, heating water, recharging power backups etc.
7. They are used in electron microscopes and magnifying glasses to get a larger view of smaller objects under study.
8. Another use of concave mirror is in visual bomb detectors.
9. The flash light mirror of camera also makes use of concave mirrors.



Concave Mirror

Uses of Convex Mirror

1. Convex mirrors are used as rear view mirrors in vehicles. Since a convex mirror forms smaller images of objects, it can be used to view a much larger area than would be possible with a plane mirror.

2. Used in shops for safety.



Convex Mirror

LENSES

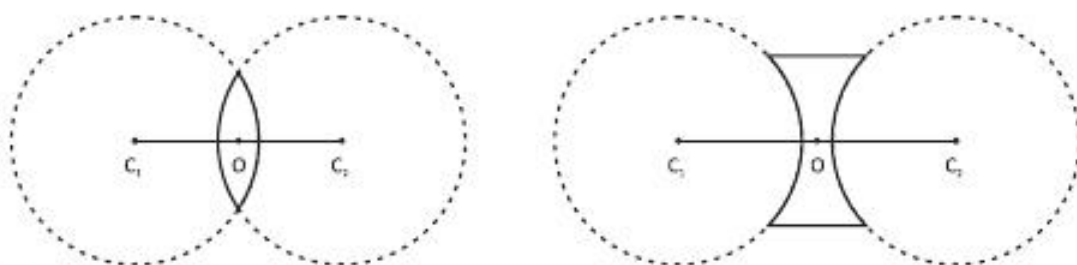
We all are familiar with lenses as they are widely used in spectacles. Lenses are transparent and light can pass through them.

A lens is made of two transparent surfaces in which at least one surface is curved. Convex and concave lenses in which both surfaces are either convex or concave are the most common types.

A concave lens is thinner at its centre than at its edges. A convex lens makes the object magnified, when viewed through it.

A convex or concave lens has two curved surfaces. Each curved surface is a part of a sphere. The line joining the centres of the two spheres called C_1 and C_2 of which the lens forms a part is called the principal axis of the lens.

Centre of the lens is called the optical centre (O).



Converging lens and Diverging lens

A converging lens is a lens that converges rays of light that are travelling parallel to its principal axis. Converging lenses can be identified by their shape; they are relatively thick across their middle and thin at their upper and lower edges. A diverging lens is a lens that diverges rays of light that are travelling parallel to its principal axis. Diverging lenses can also be identified by their shape; they are relatively thin across their middle and thick at their upper and lower edges.

Converging Lenses



thicker across the middle
thinner at its edges
serves to converge light

Diverging Lenses



thinner across the middle
thicker at its edges
serves to diverge light

Formation of Image in a Concave Lens

For all distances of the object from the lens, a concave lens forms a diminished, erect, virtual image of the object on the same side of the lens as the object.

SUN LIGHT : WHITE OR COLOURED

Have you seen the rainbow in the sky? The rainbow is seen as a large arc in the sky with many colours, in the opposite direction of the Sun, when it rains. The rainbow is a spectacular demonstration of white light as a combination of many colours. Rainbows occur when sunlight from behind the observer falls on water droplets. So, we infer that sunlight consists of many colours. When observed carefully, there are seven colours in a rainbow: violet, indigo, blue, green, yellow, orange, red. To remember them easily, the first letters of the colours make the order VIBGYOR. The pattern formed by these colours is called a **spectrum**.

All these colours come from the sunlight. When the Sun shines through water droplets in the air after it rains, or near a waterfall, the sunlight gets split into its different colours. This is how a rainbow is formed. Soap bubbles appear colourful because their thin surface splits light into different colours.

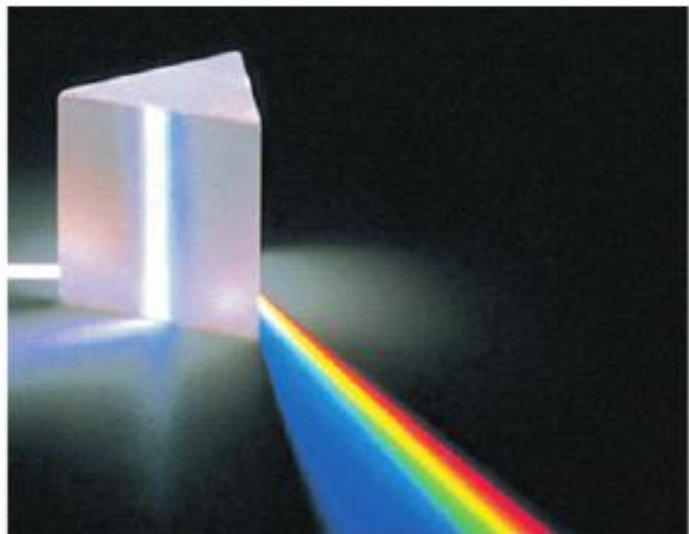
The famous scientist Sir Isaac Newton in the seventeenth century was the first to show that white light consists of seven colours. Newton set up a prism near his window, and projected a beautiful spectrum 22 feet onto the far wall. Further, to prove that the prism was not coloring the light, he refracted the light back together.

Newton's disc is a circular disc with segments painted in the seven colours of the spectrum. The disc is supported on a stand. It is provided with a handle to rotate the disc. When the disc is rotated fast, the colours disappear and the disc appears almost white.

Conversely, we can make white light from the seven colours of the rainbow.



Newton's Disc



Prism

- Rectilinear Propagation of Light
- Reflection of Light

- Spherical Mirror
- Lenses

- Sunlight : White or Coloured

We see beams of light coming from torches, headlamps of scooters, cars, buses and engines of train. Observing these beams of light, we find that the light travels in straight lines.

RECTILINEAR PROPAGATION OF LIGHT

Light always travels in a straight line. The following experiment proves that light always travels in a straight line. When the candle is viewed through a straight tube, its flame can be seen. In case of a bent tube, the flame cannot be seen.



Activity-1

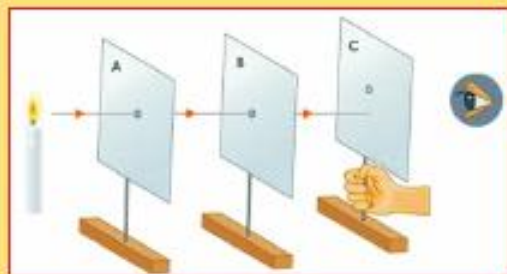
- ▶ To show that light travels along a straight line.

Materials required : Three rectangular equal pieces of cardboard with wooden supports, candle.

Method : Take three cardboards A, B and C and make a pinhole at their centers. Place a burning candle on one side of A and arrange the cardboards in such a way that the three pinholes and the candle flame are in a straight line. The candle flame will be visible through the pinhole of the cardboard C. Now slightly displace any one of the cardboards and try to see the flame through the pinhole of the cardboard C. The flame will not be visible. From this it is clear that light travels in a straight line.

Observation : When we slightly tilt any of the cardboard to left or right, we cannot see the candle flame.

Conclusion : On tilting, the rectilinear position of three holes is disturbed and we cannot see the candle flame. Thus, the experiment shows that light travels along a straight line.



REFLECTION OF LIGHT

We find our images in proper proportions in plane mirrors due to regular reflection.

A plane mirror is the simplest form of mirrors and the most commonly used. We look ourselves in front of a mirror at least a few times a day. In addition to the cosmetic purposes, plane mirrors have also important applications in medical, industrial and scientific fields.

The characteristic of reflection of light enables us to see objects. Objects that do not

◀ REWIND

- ▶ Light travels along straight lines.
- ▶ Any polished or a shining surface acts as a mirror.
- ▶ An image which can be obtained on a screen is called a real image.
- ▶ An image which cannot be obtained on a screen is called a virtual image.
- ▶ The image formed by a plane mirror is erect. It is virtual and is of the same size as the object. The image is at the same distance behind the mirror as the object is in front of it.
- ▶ In an image formed by a mirror, the left side of the object is seen on the right side in the image, and right side of the object appears to be on the left side in the image.
- ▶ A concave mirror can form a real and inverted image. When the object is placed very close to the mirror, the image formed is virtual, erect and magnified.
- ▶ Image formed by a convex mirror is erect, virtual and smaller in size than the object.
- ▶ A convex lens can form real and inverted image. When the object is placed very close to the lens, the image formed is virtual, erect and magnified. When used to see objects magnified, the convex lens is called a magnifying glass.
- ▶ A concave lens always forms erect, virtual and smaller image than the object.
- ▶ White light is composed of seven colours.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. Splitting of white light into seven colours is called

(i) Reflection of light	(ii) Rectilinear propagation of light
(iii) Lateral inversion	(iv) Dispersion
- b. When white light is passed into a prism it splits into

(i) Seven colours	(ii) Eight colours
(iii) Five colours	(iv) Three colours
- c. Sunlight is

(i) Red in colour	(ii) Green in colour
(iii) Yellow in colour	(iv) White in colour
- d. The light that is reflected from a compact disc (CD) consists of

(i) Seven colours	(ii) Eight colours
(iii) Five colours	(iv) Three colours
- e. Image formed by a convex mirror is a

(i) Virtual image	(ii) Erect
(iii) Diminished	(iv) All of these

- f. When an object is placed between the focus and the centre of curvature of concave mirror, the image formed is
- | | |
|----------------|-------------------|
| (i) Real | (ii) Inverted |
| (iii) Enlarged | (iv) All of these |

2 Fill in the blanks :

- Convex mirror is used as _____ mirror in automobiles.
- A _____ refers to a collection of rays.
- _____ is perpendicular to reflecting surface.
- Incident angle is always equal to _____ angle.
- Shadows are formed when light falls on an _____ object.
- Another use of _____ mirror is in visual bomb detectors.

3 Write T for true and F for false :

- Light is a form of energy which cannot be seen.
- Light travels slower than sound.
- Plane mirror is used in periscope.
- Bending of rays of light due to change in medium is called refraction.
- Plane mirror has fixed focal length.
- Newton's disc is a circular disc with segments painted in the seven colours of the spectrum.

4 Match the following :

- | | |
|------------------|---------------------|
| a. The moon | (i) Concave mirror |
| b. The sun | (ii) Non-luminous |
| c. The rainbow | (iii) Reflector |
| d. Mirror | (iv) Seven Colours |
| e. The rear-view | (v) Luminous |
| f. Searchlight | (vi) Concave mirror |

5 Answer these questions in short :

- What kind of the image formed by a plane mirror?
- Which mirror can form a real image? Write about the position of object.
- Which alphabets of English remained same when their images were seen in plain mirror?
- Write two uses of a concave mirror in daily life.

- e. Name the three kinds of mirrors. State the nature of the image formed in them, when held close to the face.
- f. Why does a paper held in sunshine at focus of a convex lens burns?

6

Give answers in detail :

- a. Write about the uses of concave and convex mirrors.
- b. How can be obtain the image of the Sun on a screen with the help of a convex lens?
- c. White light consists of seven colours, explain it by an experiment.
- d. What is rectilinear propagation of light?
- e. Write difference between concave and convex mirror.
- f. Why convex mirror is used as rear view mirror? Write difference between reflection and refraction of light.

NEP QUERY



PROJECT WORK

- ♦ In the laboratory : Hold a concave mirror towards a distant object. Adjust the position of the concave mirror till a clear and well defined image is obtained on the wall or on a screen. Measure the distance from the concave mirror and the wall or screen. Repeat for different objects and record your observations.

(Learning by Experiment)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- "Use of lenses in making spectacles."

(Discussion)

Electric Current and Its Effects

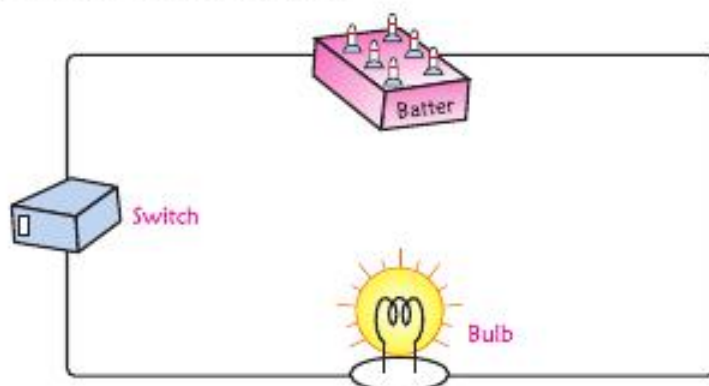
14

- Electric Components and Their Symbols
- Heating Effect of Electric Current
- Magnetic Effect of Current
- Electromagnet

Amit and Sumit are studying at night and the electricity goes off. Amit searches the table desk for the torch and the batteries. Sumit tries to insert the batteries in the torch, he tries for a few minutes and the torch lights up. We all are familiar with such a situation.

This is possible because we make a circuit for electricity.

The path of flow of electricity through various electric components connected to a source of electricity is called an electric circuit.



ELECTRIC COMPONENTS AND THEIR SYMBOLS

An electric circuit can be represented by circuit diagram. The various electrical components are shown by using standard symbols in circuit diagrams.

Electric components of some basic circuits are cells, bulbs, wires, switches, etc.

An electric cell has a positive terminal and a negative terminal. The symbol of an electric cell has a longer line and a shorter line. The longer line is the positive terminal and the shorter line is the negative terminal. In a battery, the positive terminal of one cell is connected to the negative terminal of another cell. Devices like torches, transistors, toys, TV remote controls use batteries. To connect cells together, one after the other position, the positive terminal of one cell is connected to the negative terminal of the next cell.



Bulb



Electric Lamp

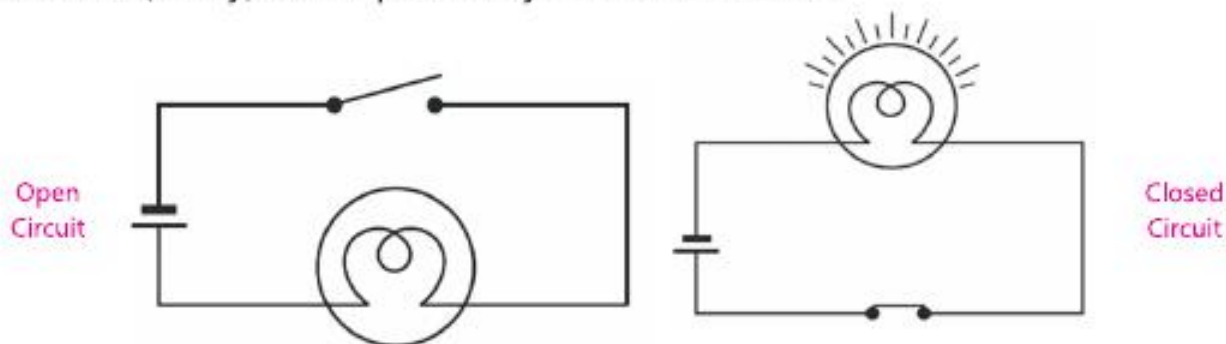


Fan

An electric bulb is represented by a circle with a curved wire inside it. The curved wire represents the filament of the bulb. Connecting wires are represented by straight lines. Each switch has two points. When the two points are not connected, the circuit remains open (OFF). When the two points in the switch are connected, the circuit gets closed. (ON).



A switch (or key) can be placed anywhere in the circuit.



HEATING EFFECT OF ELECTRIC CURRENT

The bulb becomes hot when we put it on for some time. It is the filament of the bulb that heats up due to current flowing through it. We have seen an electric iron, electric cooker and electric heater. All these contain a coil of wire made up of nichrome. This coil is called filament of the appliance.

When an electric current flows through a wire, the wire gets hot. This is known as the heating effect of electric current. It is a matter of common experience that a wire gets heated up when electric current flows through it. The factors that affect the production of heat in a wire through which an electric current flows are the length and thickness of the wire, the duration of flow of current, and the material of the wire. Nichrome is an alloy made by melting iron, nickel, chromium and magnesium together. It does not melt even at higher temperature. The filaments of electric bulbs are generally made of the metal tungsten.

The change of electric energy to heat energy when electricity passes through a resistor (or conductor) is called heating effect of electric current.

Application of the Heating Effect of Electric Current

Some appliances that work on the property of the heating effect of electric current are electric room heater, electric roti maker, electric iron, toaster, hair dryer, electric stove, immersion water heater, food warmer, electric coffee maker, electric rice cooker and geyser. These appliances have coils of wire known as heating elements which produce heat. The amount of heat produced in a wire depends on the nature of its material, length and thickness. The heat produced also depends on the amount of the current passed and duration for which the current is passed. As current flows through these electrical appliances,

the heating elements turn bright orange or red in colour. This is because a huge amount of heat is produced. For different requirements, the wires of different materials and different lengths and thickness are used. An electric bulb also works on the heating effect of electric current. The filament of an electric bulb gets heated to such a high temperature that it starts glowing. Different appliances have different types of heating elements. These are called an element. The type of heating element depends on the function of the appliance. Some appliances are required to produce more heat than others.



Electric Geyser



Electric Appliances

Electric Bulb

Incandescent bulb is an example of a device which works because of the heating effect of electric current. The filament of bulb is made of tungsten. Tungsten has very high melting point and that is the reason it is used in electric bulb. Moreover, argon gas is filled inside the bulb. Argon prevents the filament from catching fire. Bulb glows and gives out light. We use electric bulbs to obtain light but it also gives heat. This is not desirable. This results in the wastage of energy. CFL's do not depend on the heating effect of electricity to produce light, since they do not use filaments. Using CFL's instead of ordinary bulbs minimises wastage of electricity. In CFL's, light is generated using two electrodes. The fluorescent coating inside each tube makes the light brighter.



Bulb



CFL

Electric Fuse



Fuse

When excessive electric current flows through a circuit, the wires or the appliances may get heated and can catch fire. To avoid fire, a safety device known as a fuse is connected in series in the circuit. Electric fuse has a body made of ceramic and two points for attaching the fuse wire. An electrical fuse is a weakest part of an electrical circuit which breaks when more than predetermined current flows through it. The materials used for fuse wires are mainly tin, lead, zinc, silver, antimony, copper,

aluminum etc. The fuse wire melts whenever there is overload in the wiring. This breaks the circuit and helps in preventing damage to costly appliances and to the wiring. In electrical devices a glass fuse is often used. This is a small glass tube, in which lies the fuse wire. Thus further damage to the electric appliances and wires is prevented.

A fuse is a small piece of wire as shown in figure. It is made of a special alloy that gets heated quickly and melts. If the current in the circuit is too high, the fuse wire gets hot and melts. This leaves a gap in the circuit. Automatically the circuit is broken and flow of electricity is stopped. This protects appliances from getting burnt due to the passage of too large a current through them.

These days Miniature Circuit Breaker (MCB) is increasingly being used in place of fuses. These are switches which automatically turn off when current in a circuit exceeds the safe limit. It has a switch which goes OFF automatically if there is overheating. This breaks the circuit. If we turn them on, the circuit is once again complete. The advantage of miniature circuit breakers have over fuses is that they can be reset (manually or automatically) to restore normal operation, whereas fuses need to be replaced after every single operation. The MCB can be reset by hand and the circuit becomes complete once again.



MCB

Reasons for Excessive Current

When all the appliances are connected to the same socket, they draw more current, and so the load increases. When the insulation on the wires is torn, two wires carrying current may touch each other directly. This causes a spark, which leads to fire. This is termed as a **short circuit**. If a fuse is not used, overloading and short circuits occur which results in fire.

Short Circuit and Overload : When live wire and earth wire touch each other by mistake or removal of insulation on the wires due to wear and tear, huge current flows with big sparks and fire. This is called short circuit. When many devices are connected to a single socket it may also cause large (excessive) current. This situation is called **overload**. Large currents may come from the supply for various reasons.



Different Types of Electric Fuse

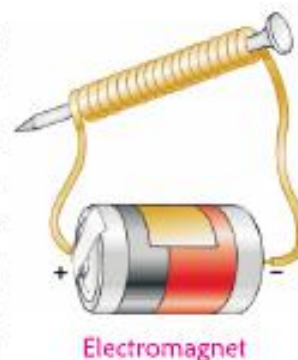
MAGNETIC EFFECT OF ELECTRIC CURRENT

In 1800s scientist Hans Oersted observe that, current flowing in a circuit effects the direction of needles of the compass.

ELECTROMAGNET

Magnetism is defined as the force exerted by a magnetic field. A magnetic field is defined

as the energy exerted by the magnet. Magnetic effect of electric current is used to make very powerful magnets. These electromagnets work as long as current is passed in them. So they are temporary magnets. Permanent magnets have much smaller power than these electromagnets. However, electromagnets can be made of small power also. An electromagnet is made by winding a large number of turns of an insulated copper wire on a soft iron core. The strength of the electromagnet depends on the number of turns in the coil, the strength of the current flowing through the coil and nature of the core material. Soft iron core produces very strong electromagnets.



Electromagnets, like permanent magnets, have poles and magnetic fields. Actually, whenever an electrical current passes through a wire, it produces a magnetic field. Every electrical appliance or cord has a weak magnetic field around it. The difference between an electromagnet and a permanent magnet is that you can reverse direction of the current in an electromagnet. The greatest electromagnet that is known today is the Earth.

Uses of Electromagnets

Electromagnets have several uses and practical applications. They can be found in various everyday electronics equipments and are also used for industrial purposes.

- (i) The electromagnets can be made very strong and when attached to a crane can lift very heavy loads.
- (ii) Scrap yards also use large electromagnets to separate iron and other ferrous metals mixed with non-ferrous materials.
- (iii) Doctors use tiny electromagnets to take out small pieces of magnetic material that have accidentally fallen in the eye.
- (iv) Many toys operated with cells have electromagnets inside them.
- (v) Many appliances such as telephones, microphones, speakers, etc. have electromagnets in them.
- (vi) An electromagnetic lock can be used to lock a door by creating a strong field in an electromagnet that is in contact with a magnetic plate.

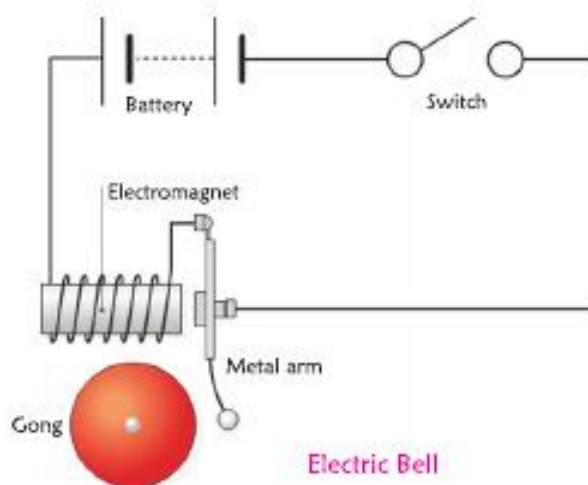
Electric Bell

We are quite familiar with electric bell. These are used in houses. It works on the principle of magnetic effect of electric current. An electric bell consists of a coil of wire on an iron piece (U shaped). The coil acts as an electromagnet when connected to a battery (DC supply) through a push button type switch.

An iron strip with a hammer at one end is kept close to the electromagnet. There is a contact screw near the iron strip. When the iron strip is in contact with the screw, the current flows through the coil.

Electric bells like the ones used in most schools also contain an electromagnet. When the current flows through the circuit, the electromagnet makes a magnetic field.

- › The electromagnet attracts the springy metal arm.
- › The arm hits the gong, which makes a sound.
- › The circuit is broken now the arm is out of position.
- › The electromagnet is turned off and the springy metal arm moves back.
- › The circuit is complete again.
- › The cycle repeats in quick succession as long as the switch is closed.



Chemical Effect of Electric Current

When electric current is passed through some solutions such as copper sulphate or silver nitrate, a chemical reaction takes place and the solutions break up into their constituent elements in the ionic form. The disassociation of solution into the constituent element is called **electrolysis**. This effect is called **chemical effect of electric current**. One of the main application of this effect is electroplating. The process of depositing a thin layer of metal with the help of electric current is called **electroplating**.

Why is Electricity Dangerous?

Electricity is a part of our everyday life. Sometimes it can be dangerous. An electric shock is a lot painful and dangerous. If we follow these safety rules, we would be safe and sound:

- Don't touch an electric wire which has fallen from power lines.
- Never touch electrical appliances with wet hands.
- Don't enter any metal object into electric sockets.
- Don't overload power sockets. Overloaded sockets can cause fire .
- If a person has been electrocuted, don't touch the body of that person. Use a non

◀ REWIND

- ▶ It is convenient to represent electric components by symbols. Using these, an electric circuit can be represented by a circuit diagram.
- ▶ When an electric current flows through a wire, the wire gets heated. It is the heating effect of current. This effect has many applications.

- ▶ Wires made from some special materials melt quickly and break when large electric currents are passed through them. These materials are used for making electric fuses which prevent fires and damage to electric appliances.
- ▶ An electric bulb glows when electric current passes through it.
- ▶ In a closed electric circuit, the electric current passes from one terminal of the electric cell to the other terminal.
- ▶ Electromagnets have several uses and practical applications.
- ▶ Switch is a single device that is used to either break the electric circuit or to complete it.
- ▶ Wastage of electricity can be reduced by using fluorescent tube lights in place of bulbs.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- | | | |
|--|---|--|
| a. Fuse wire is made up of alloys of | (i) High melting point | (ii) Low melting point |
| | (iii) High boiling point | (iv) Low boiling point |
| b. Coils of heating devices are made up of | (i) Tungsten | (ii) Copper |
| | (iii) Mercury | (iv) Nichrome |
| c. Wire carrying current produce | (i) Magnetic field | (ii) Chemical effect |
| | (iii) Heating effect | (iv) All of these |
| d. MCB in electric circuit stand for | (i) Minimum Current Breaker | (ii) Maximum Current Breaker |
| | (iii) Miniature Circuit Breaker | (iv) Molar Current Breaker |
| e. Electromagnet is based on | (i) Heating effect of electric current | (ii) Magnetic effect of electric current |
| | (iii) Chemical effect of electric current | (iv) Electro-chemical effect of electric current |
| f. Filament of electric bulb is made up of | (i) Nichrome | (ii) Silver |
| | (iii) Copper | (iv) Tungsten |

2 Fill in the blanks :

- An electric circuit can be represented by _____ diagram.
- Wire of _____ metal is used as connective wires.
- Beside light, electric bulb also produces _____.
- _____ iron core produces very strong electromagnets.
- The full form of MCB is _____.
- To prevent fire in circuits, _____ is used.

3 Write T for true and F for false :

- Electromagnet is used in electric fan.

- b. When current flows through the conductor magnetic field is produced.
- c. Fuse is used to measure voltage.
- d. Chemical effect of electric current is used for electroplating.
- e. Coil of heating devices are made up of good conductor.
- f. Tungsten is a bad conductor of electricity.

4 Answer these questions in short :

- a. What do you understand by electric circuit?
- b. What is the meaning of closed electric circuit?
- c. Which materials produce heat energy when electric current is passed through them and why?
- d. What is the role of fuse in a circuit?
- e. Why do we use nichrome in an electric heater?
- f. Write difference between electromagnet and permanent magnet.

5 Give answers in detail :

- a. How does a fuse wire protect an electric circuit?
- b. Show experimentally about the heating effect of current.
- c. Describe a simple experiment to show that an electric current produces a magnetic effect.
- d. Write the process of making an electromagnet practically.
- e. What is an electric bell? Write the steps of its working.
- f. What is a fuse? Write its function.

NEP QUERY



PROJECT WORK

- ♦ In the laboratory : Make four electromagnets with 20, 40, 60 and 80 turns. Connect them one by one to a battery of 2 cells. Bring the electromagnet near a box of pins. Count the number of pins attracted by it. Compare the strengths of the electromagnets.

(Learning by Experiment)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic–“Fuse in an electric circuit and its advantages and disadvantages”.

(Discussion)

Movement of Substances

Transportation and Excretion

15

- Circulatory System Heart
- Blood
- Blood Transfusion
- Excretion
- Transport of Substances in Plants
- Excretion in Plants
- Transpiration

All plants need water. The wilted leaves recover when water is added to the soil, which means that water has been conducted upward into the leaves. You have also learnt that the leaves for photosynthesis need water. Likewise, the food produced in the leaves has to be transported to other parts of the plant including the stem, the roots flowers and fruits etc. All this transportation is the function of conducting tissues. Similarly, in animals, the food absorbed by the gut has to be carried to all the body parts, oxygen absorbed in the lungs has to be transported to every cell of the body, the carbon dioxide produced in the cells has to be carried to the lungs for elimination, and the poisonous body wastes like urea has to be transported to the kidneys for elimination in urine, and so on. All such functions are the outcome of a transport system.

CIRCULATORY SYSTEM

The circulatory system in humans (and many animals) consists of the following :

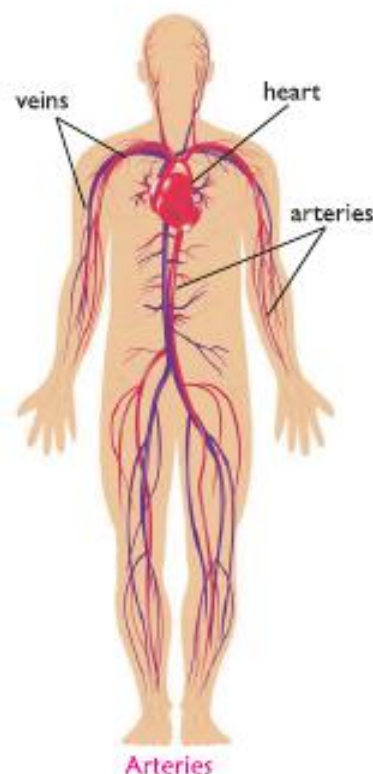
- a main pumping station, heart.
- network of blood vessels, arteries and veins
- blood, a fluid medium for transportation

Blood Vessels

The blood travels throughout the body through blood vessels. The three types of blood vessels are arteries, capillaries and veins.

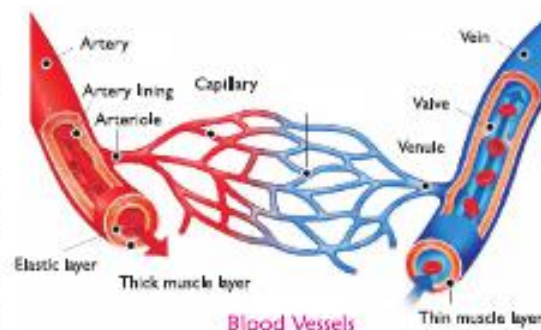
Arteries

Arteries are the blood vessels that carry blood away from the heart. Arteries have thick and flexible walls. Most of the arteries carry oxygenated blood, but pulmonary arteries carry deoxygenated blood to the lungs. Arteries divide many times to smaller tubes, called capillaries.



Pulse Rate

The number of pulse beats (jerks) per minute is called the pulse rate. A resting person, usually has a pulse rate between 72 and 80 beats per minute. On an average it is about 74 times per minute in a healthy adult person. Pulse beat in children is more. After running the pulse beat increases. In fever and illness the pulse-rate varies.



Capillaries

Capillaries are the smallest blood vessels in the body. They are so small that red blood cells flow through them one cell at a time. Food and oxygen from the blood of capillaries diffuse into the cells. Waste materials and carbon dioxide from the cells diffuse into the blood of capillaries. Capillaries again join to form the larger blood vessels called veins.

Veins

Veins are the blood vessels that bring blood back to the heart. Most of the veins bring deoxygenated blood back to the heart, but pulmonary veins bring oxygenated blood from the lungs to the heart.

FACT Human red blood cells have a life span of approximately 120 days.



Activity

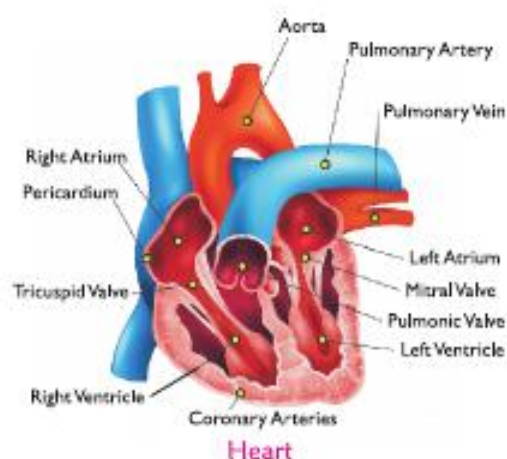
- ▶ To Place the index and middle fingers of your one hand on the wrist of other hand where your wrist joins the base of your thumb. Move the two fingers slightly until you locate your pulse.
Now count the number of beats that you feel in one minute.
Record the result.
Now run for two minutes. Again, measure your pulse and record the result.
What do you conclude from this activity?



Differences among Arteries, Veins and Capillaries

Arteries	Capillaries	Veins
Away from heart	Wall only one cell thick really thin - diffusion	Blood into heart
Mainly carry oxygenated blood	Smallest blood vessel	Most carries de-oxygenated blood
High pressure	—	Lower pressure

No valves	—	Valves
Thick outer wall (pumps a lot more blood)	—	Thin outer walls
Thick inner layer of muscle and elastic fibres	—	Thin inner layer
Narrow central tube- small lumen (where the blood flows through)	—	Wide central tube - large lumen



HEART

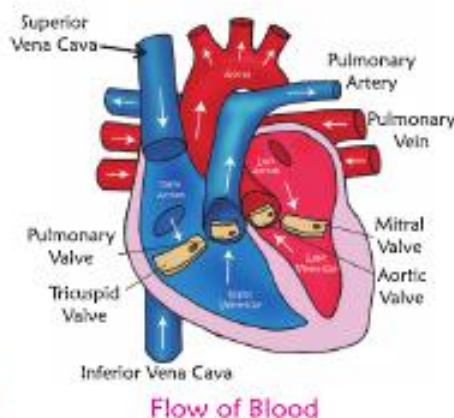
Our heart is a muscular organ about the size of our fist. It appears that the heart is on the left side of the chest. Actually the heart lies in the middle of the chest between the two lungs. It pumps oxygen-poor blood (deoxygenated blood) to the lungs and oxygen-rich blood (oxygenated blood) to the body. There are four chambers in our heart; two upper chambers called **atria** (singular atrium) and two lower chambers called **ventricles**. The ventricles of our heart are larger than the

atria. Both atria contract at the same time, and so the ventricles. The blood passes from the atria into the ventricles. There is a valve between each atrium and ventricle on the both sides of the heart. These valves keep the blood flowing in one direction. Deoxygenated blood from the body enters the right atrium and oxygenated blood from the lungs enters the left atrium of our heart. The right ventricle pushes the blood to the lungs and the left ventricle pushes the blood to the body.

How does our Heart Work?

Human heart acts as a double pump. Blood from lungs and other body parts enters the atria. The two atria contract at the same time and push the blood to ventricles. It is one pump. Now both ventricles contract at the same time and pump the blood to lungs and other body parts. It is the second pump. Our heart beats about 70 times a minute. We can feel our heart beat (pulse) while placing our fingers below the base of the thumb on the underside of our wrist.

The heart contracts rhythmically throughout life. Each heart beat consists of a systole and a diastole. The total duration is about 0.8 seconds. The total heart action involving all the three phases men-tioned above is called the **cardiac cycle**.



The heart takes in blood low in oxygen from the body. It pumps it through the right side of the heart and on to the lungs. In the lungs, the blood passes through very small blood vessels and absorbs oxygen. The freshly oxygenated blood is pumped back through the heart and on to the rest of the body where the oxygen provides fuel for muscles and organs.

The heart muscle contracts in two stages to squeeze blood out of the heart. This is known as **systole**.

In the first stage, the upper chambers (atria) contract at the same time, pushing blood down into the lower chambers (ventricles).

Blood is pumped from the right atrium down into the right ventricle and from the left atrium down into the left ventricle.

In the second stage, the lower chambers contract to push this blood out of the heart to either the body via your main artery (aorta) or to the lungs to pick up oxygen.

The heart then relaxes known as **diastole**. Blood fills up the heart again, and the whole process, which takes a fraction of a second, is repeated.

Heart Beat

The heart beats non-stop all the time. The heart beat is due to the rhythmic contraction and relaxation of the heart muscles which make up the atria and the ventricles. Please note that the two atria (left atrium and right atrium) contract together and relax together. Similarly, the two ventricles (left ventricle and right ventricle) contract together and relax together. The contraction of two atria is immediately followed by the contraction of the two ventricles. The heart beats (or beating of heart) circulates the blood in the human body.

When a person runs, his/her heart beat increases.

Bloodless Animals : The bloodless animals includes mollusk, sponge, hydra etc. The water in which they live, brings food and oxygen as it enters their bodies. The water carries away waste materials and carbon dioxide as it moves out. Thus, these animals do not need a circulatory fluid like the blood.

Blood Pressure

Human body contains 4 to 6 litres of blood. The pressure at which blood is pumped around the body by the heart is called **blood pressure**. The blood pressure of a person is always expressed in the form of two values called '**systolic pressure**' and '**diastolic pressure**'. In order to understand this, we should first know the meaning of 'systole' and 'diastole'.

The phase of the heart beat when the heart contracts and pumps the blood into arteries is called '**systole**'. The phase of heart beat when the heart relaxes (or expands) and allows the chambers to fill with blood is called '**diastole**'.

The blood pressure of a person is expressed in terms of millimetres of mercury (which is written as mm Hg). The normal blood pressure values are : Systolic pressure : 120 mm Hg, Diastolic pressure: 80 mm Hg. This is usually written as 120/80.

If the blood pressure of a person remains above the normal value (120/80), the condition is called **high blood pressure** or **hypertension**. The increased blood pressure can damage blood vessels and the result may be the failure of kidneys and heart. People with

high blood pressure may have the symptoms such as, headache, fatigue, blurred vision, nosebleeds, etc. Foods high in fat, high salt intake, smoking, obesity, diabetes, lack of exercise, etc. raise the risk of high blood pressure. The decreased pressure of the blood in the ventricles when they are filling blood from the auricles is called **diastolic pressure**.

BLOOD

Everybody is familiar with the sight of blood - the red fluid that oozes out of your body when you've sustained a cut or a deep injury. The composition of blood is actually quite complex. Blood is crucial to our survival.

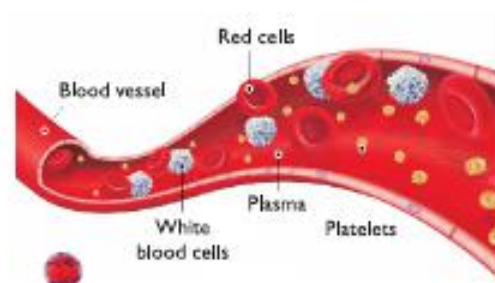
Blood is the medium in which dissolved gases, nutrients, hormones and waste products are transported. Blood along with the heart and the blood vessels (e.g. veins and arteries) comprises the circulatory system of the body. The circulatory system helps in maintaining balanced conditions within the body (i.e. homeostasis). For example: oxygen is picked up by blood as it passes through the lungs. This blood in turn flows through successively narrower blood vessels: from arteries to arterioles and finally the capillaries, where the oxygen rich blood delivers its oxygen to the cells.

Blood is composed of a straw-coloured liquid called **plasma** which contains suspended cells. The different specialized cells found in blood are :

- Red blood cells
- White blood cells
- Platelets

Red Blood Cells (RBCs)

These are the familiar discotic shaped cells which make up 99% of the cells in the blood. They are the principal carriers of the red coloured **haemoglobin** molecules. Haemoglobin is an iron containing protein and binds about 97% of all oxygen in the body. The unique surface shape of red blood cells is nature's design to maximize surface area in order to facilitate absorption and release of oxygen.



Cells in Blood

White Blood Cells (WBCs)

The role of blood is not confined to the transportation of materials within the body. The white blood cells are a vital source of defense against external organisms. White blood cells also serve as 'sanitary engineers' cleaning up dead cells and tissue debris that would otherwise accumulate to and lead to problems. Thus they protect us from the infection.

Platelets

Platelets are very minute cells. They help in clotting of blood.

FACT

Platelets counts go down during dengue fever.

Functions of Blood

- (i) Blood carries oxygen from the lungs to different parts of the body.
- (ii) Blood carries carbon dioxide from the body cells to the lungs for breathing out.
- (iii) Blood carries digested food from the small intestine to all the parts of the body.
- (iv) Blood carries hormones from the endocrine glands to different organs of the body (where they are needed).
- (v) Blood carries a waste product called urea from the liver to the kidneys for excretion in urine.
- (vi) Blood protects the body from diseases. This is because white blood cells kill the bacteria and other germs which cause diseases.
- (vii) Blood regulates the body temperature. This is because the blood capillaries in our skin help to keep our body temperature constant at about 37°C.

BLOOD TRANSFUSION

A blood transfusion involves taking blood from one person (the donor) and giving it to someone else.

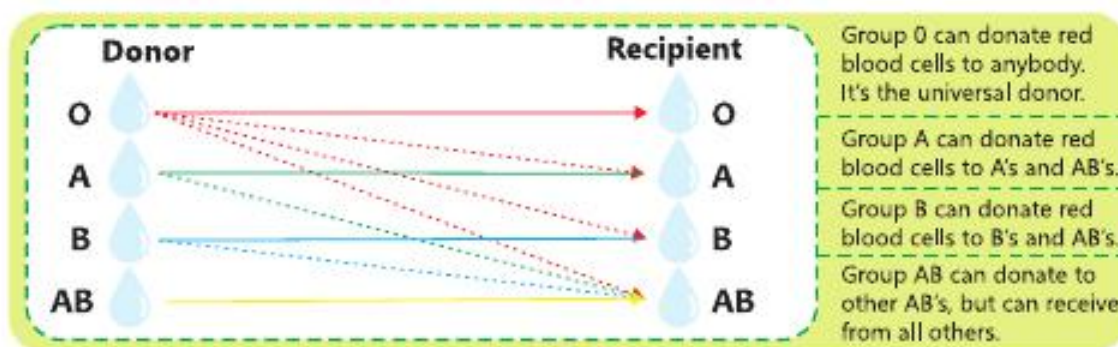
We may need a blood transfusion for a number of reasons, including :

- (i) To replace blood lost during major surgery, childbirth or a severe accident.
- (ii) To treat anaemia that has failed to respond to other treatments; anaemia is a condition where a person has low levels of red blood cells.
- (iii) To treat inherited blood disorders, such as thalassaemia or sickle cell anaemia.

There are four major blood groups determined by the presence or absence of two antigens A and B on the surface of red blood cells:

- › **Group A** : has only the A antigen on red cells (and B antibody in the plasma)
- › **Group B** : has only the B antigen on red cells (and A antibody in the plasma)
- › **Group AB** : has both A and B antigens on red cells (but neither A nor B antibody in the plasma)
- › **Group O** : has neither A nor B antigens on red cells (but both A and B antibody are in the plasma)

There are very specific ways in which blood types must be matched for a safe transfusion. See the chart below:



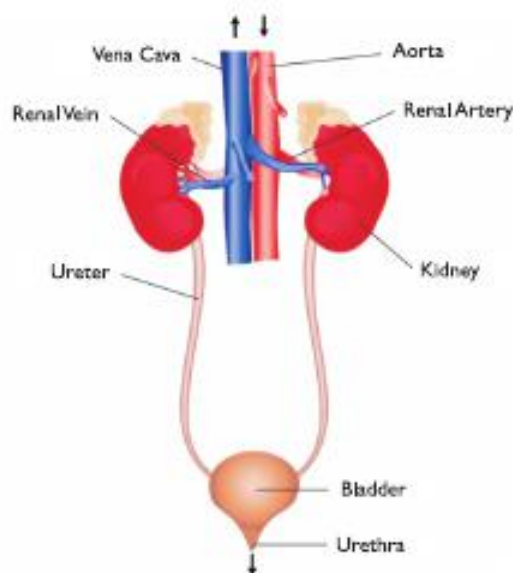
EXCRETION

Several chemical reactions take place inside a living body during life. This is known as metabolism.

Due to chemical reactions a large number of waste products, like carbon dioxide, water, urea, uric acid, etc., are being constantly produced in the body. At the same time not only toxic products but some gases also produced. These waste products should be eliminated from the body. If these waste products are allowed to accumulate in the body, they may cause serious damage. The removal of waste products from the body is called excretion.

Excretory System in Human

The human body has different organs for the removal of wastes from the body. These are our **lungs** and **kidneys**. Our **lungs** excrete carbon dioxide. Our kidneys excrete urea. The kidneys are the main excretory organs of the human body. The kidneys are bean-shaped organs located at the small of the back near the spinal column. The left kidney sits slightly higher than the right one. The size of an adult kidney is approximately 4 inches (10 centimeters) long and 2 inches (5 centimeters) wide. To maintain human life, it is necessary for at least one of the kidneys to function properly. Blood carries waste products to the kidneys via the renal artery. Inside each kidney, blood is transported to 1.2 million filtering units called nephrons. The cells in nephrons take in the liquid portion of the blood and filter out impurities (urea, mineral salts, and other toxins). Necessary substances such as certain salts, water, glucose (sugar), and other nutrients are returned to the blood stream via the renal vein. The waste is filtered from the blood and the purified blood is sent to the normal circulation. The waste along with water is transferred to the urinary bladder through the **ureters**. The content of the ureter is called **urine**. Urine contains 95% water, 2.5% urea and 2.5% other wastes. Urine is expelled out from time to time. An adult human being normally passes about 1 to 2 litre of urine in 24 hours.



Excretory System

How Kidneys work?

The kidney is not one large filter. Each kidney is made up of about a million filtering units called **nephrons**. Each nephron filters a small amount of blood. The nephron includes a filter, called the **glomerulus**, and a **tubule**. The nephrons work through a two-step process. The glomerulus lets fluid and waste products pass through it; however, it prevents blood cells and large molecules, mostly proteins, from passing. The filtered fluid then passes through the tubule, which sends needed minerals back to the bloodstream and removes wastes. The final product becomes urine.

Artificial Kidney

A normal human being has two kidneys which work with same efficiency. Sometimes a person's kidneys stop working due to infection/injury. If one kidney is damaged; then the other kidney is sufficient for excretory needs. But failure of both the kidneys would lead to death because of the accumulation of waste products in the blood.

The haemodialyzer or artificial kidney is a machine used to filter the blood of a person whose kidneys are damaged. The process is called **haemodialysis**. Haemodialysis is defined as the separation of small molecules from large molecules in a solution by interposing a semipermeable membrane between the solution and water. It works on the principle of dialysis, which is the diffusion of small solute molecules through a semipermeable membrane.

Sweating

Sweat is produced by sweat glands in the body, which produce a watery fluid and fluid containing substances such as proteins and lipids. We sweat from up to 5 million sweat glands located all over the skin on our body. These glands produce a watery fluid called sweat. These glands are called sweat glands and their function as sweating. Sweat glands excrete waste from the blood circulated in them. This sweat contains urea, salt, excess water and other wastes. However, sweating that is triggered by emotions or stress only happen in certain areas of the body, such as the armpits, palms and head.



Sweating

Sweating is rapid in summer and slow in the cold season. When the weather is hot and has high humidity (amount of water vapour in the air), the sweat glands produce a lot of sweat.

We must know that sweating helps our body

- Maintain proper temperature and keep you from overheating.
- Expel toxins, which supports proper immune function and helps prevent diseases related to toxic overload.
- Kill viruses and bacteria that cannot survive in temperatures above 98.6 degrees Fahrenheit.
- Clean the pores, which will help eliminate blackheads and acne.

Loss of dissolved salts and water of the blood in sweating causes muscular pains(cramps). Oral Dehydrating Solution or ORS (water containing salt and glucose with a little lemon or citric acid) should be taken at regular intervals.

Excretion in Some Other Animals

In simple animals like amoeba, paramecium, sponges and hydra most of the metabolic wastes are removed from the body by direct diffusion through the cell membrane. There is no special organs of excretion.

Animals such as earthworm and leech have tubular structures called **nephridia** to get rid of body waste. Insects excrete through **malpighian tubules**. Aquatic animals like fishes excrete metabolic wastes in gaseous form which directly dissolve in water.

Some land animals like birds, lizards and snakes excrete a semi-solid white coloured compound called **uric acid**. Kidney is the main excretory organ in vertebrate animals.

TRANSPORT OF MATERIALS IN PLANTS

Plants are living organisms like animals. They need water, minerals and food for their survival. The transport of materials in plants is carried out by the **vascular tissues**. These tissues extend from the tip of the root to the tip of the leaves.

Vascular Tissues

There are two kinds of vascular tissues found in plants. These are **xylem** and **phloem**.

The **xylem** transports water and nutrients up wards from the roots to the leaves. Evaporation of water from the leaves during transpiration produces a pulling force causing water to move up.

Phloem carries the food prepared by the leaves downwards to all parts of the plant including roots. However, in some plants the material made in the roots is transported by phloem upwards to the leaves.

Thus, xylem and phloem transport substances in plants.

Transport of Water, Minerals and Food

Plants absorb water and minerals through the roots. The root hairs increase the surface area of the root for the absorption of water and minerals dissolved in water. The root hairs come in contact with the water present between the soil particles.

The water moves from the root hair to the xylem in the root. The absorbed water then moves up the stem through the xylem by the force developed in the leaves by the process of transpiration. This is called **transpiration pull**.

The fluid containing water and dissolved nutrients is called **sap**. The movement of sap through the xylem is called **ascent of sap**.

The food prepared in the leaves is transported from the leaves to other parts of the plant. This process is called **translocation** and it mainly takes place through phloem.

EXCRETION IN PLANTS

Plants have no special organs for excretion. The waste product of respiration i.e. carbon dioxide is partly used during photosynthesis. The excess amount of carbon dioxide escapes through the **stomata** and **lenticels**. Excess water evaporates mostly from the stomata and also from the outer surface of stem, leaves, fruits, etc. throughout the day. This process is called **transpiration**.

Some of the waste products of photosynthesis are collected in vacuoles in the leaves and barks of trees. Plants get rid of them by shedding the leaves and barks.

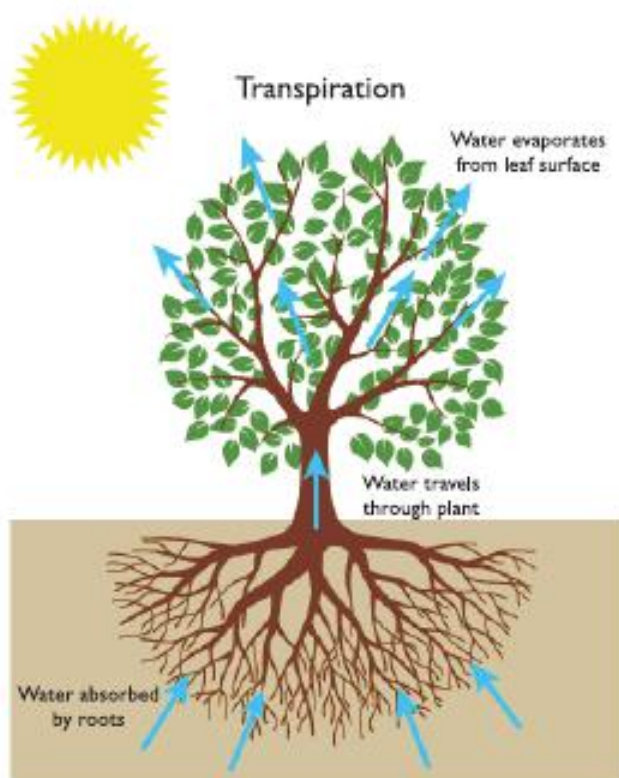
Some waste products are transformed into harmless products and stored inside the plant body. **Rubber** and **raphides** are examples of such products. Plants also secrete waste in the form of gum, latex, resin and oil which are useful for us.

TRANSPIRATION

A large quantity of water is absorbed by the roots from the soil. But only a small quantity of the total absorbed water is used by the plant in making of food. the remaining water is lost into the atmosphere in the form of water vapour.

This loss of water in the form of water vapour from the aerial parts of a plant is called **transpiration**.

Most of the loss of water takes place from leaves. As water diffuses out of the leaf through stomata, osmosis to replace it. Water is constantly being taken from the top of the vessels to supply the cells in the leaves. This reduces the pressure at the top of the xylem vessels, so that water flows up them. This process is known as the **transpiration stream**.



◀ REWIND

- ▶ Animals (including humans) and plants need food, water and oxygen for survival.
- ▶ The organ system which removes these substances is called excretory system.
- ▶ Blood is the medium in which dissolved gases, nutrients, hormones and waste products are transported.
- ▶ The human heart beats about 70-80 times per minute in an adult person. This is called heart rate.
- ▶ Veins carry blood from all parts of the body back to the heart.
- ▶ Excretory system of humans consists of two kidneys, two ureters, a urinary bladder, and urethra.
- ▶ Salts and urea are removed along with water as sweat.
- ▶ Water and mineral nutrients are absorbed by roots from the soil.
- ▶ Nutrients are transported along with water to the entire plant via the vascular tissue called xylem.
- ▶ A lot of water is lost by plants in the form of vapour through stomata during transpiration.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. In plants, food is transported through
 (i) Xylem (ii) Phloem (iii) Stomata (iv) Guard cells
- b. Human heart consists of
 (i) Striated muscle (ii) Smooth muscle
 (iii) Voluntary muscle (iv) Cardiac muscle
- c. Which blood group is called as universal donor?
 (i) AB (ii) A (iii) B (iv) O
- d. Which chamber of heart receives oxygenated blood from heart?
 (i) Left atrium (ii) Left ventricle (iii) Right atrium (iv) Right ventricle
- e. In plants xylem and phloem comprise the
 (i) Food storing tissue (ii) Vascular tissue
 (iii) Protective tissue (iv) Reproductive tissue

2 Fill in the blanks :

- a. The liquid part of blood is called _____.
- b. Kidney is made of tiny structures called _____.
- c. _____ carry impure blood from organs to heart.
- d. Main excretory product of human being is _____.
- e. Sweat contain water and _____.
- f. Haemoglobin is present in _____.

3 Write T for true and F for false :

- a. Throbbing of heart is called pulse.
- b. Human heart consists of two chambers.
- c. Removal of waste from the body is called excretion.
- d. Dialysis is the process of removing nitrogenous waste from blood using machine.
- e. Transport of water and minerals takes place via phloem.
- f. In man, carbon dioxide is excreted through kidneys.

4 Answer these questions in short :

- a. Why is transport of materials necessary in an animal?
- b. What is the role of platelets in the blood?
- c. What do you understand by stomata? Give two functions of stomata.
- d. Name all the components of blood.
- e. Why is it necessary to excrete waste products?
- f. What is the function of red blood cells?

5 Give answers in detail :

- a. What are the function of the major components of blood?
- b. Write a note on the function of the heart.
- c. Explain the human heart with a neat diagram.
- d. How dialysis technique is useful?
- e. How does a kidney work?
- f. What is transpiration? Write its function?

NEP QUERY



PROJECT WORK

- ♦ In the laboratory : To demonstrate transpiration, place a potted plant on a plate. Enclose the pot in a polythene bag and tie the mouth of the polythene bag around the stem of the plant with a thread. Now cover the plant with a glass jar. Leave it undisturbed for sometime. We see tiny droplets of water on the inner surface of the glass jar. The water comes from no where else but the leaves, since the pot (soil) has been covered completely. This shows that plants give out water vapour through transpiration.

(Learning by Experiment)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- "Transpiration is a necessary evil".

(Discussion)

Rest and Motion

16

- Time
- Simple Pendulum
- Motion
- Speed
- Distance-Time graph

TIME

We use the word time very often in our daily life. In some situations, without using the word 'time', we express the duration of time like 'so late', 'so early', etc.

Now-a-days, we use different instruments like electronic clocks, digital clocks, quartz clocks, etc. to measure time. We can find stop clocks almost in all cell phones. In our daily life stop clocks help measure pulse rate, etc. Apart from this, they also help us to accurately measure the time in running races, swimming competitions etc.

A few decades ago people used pendulum clocks that have now become rare. People in old times noticed the natural events carefully. They found that many events in nature repeat themselves regularly after definite intervals of time. For example, the unit of time used by them was

- The time interval between sunrise to sunset or to the next sunrise,
- One full moon to another full moon,
- From one season to another, etc.

There are certain events in nature that can help us track time. For example, the phases of the moon indicate the time of the month. There are man-made structures that were constructed to measure time. For example, the Jantar Mantar in Jaipur, which is basically a sun clock, is used to measure time. Water clocks and sand clocks were also used hundreds of years ago to measure time.



Sun Dial

A sundial is a device that tells the time of day by the apparent position of the Sun in the sky. In common designs such as the horizontal sundial, the sun casts a shadow from its style onto a surface marked with lines indicating the hours of the day. The hourglass is sometimes referred to as a sand clock or a sandglass. Mechanical clocks use oscillatory motion to measure time. The time period of a pendulum is the time taken for one complete oscillation. The working of a pendulum clock is based on the time period of its pendulum.



Sand clock

Winding clocks and wrist watches were developed from modifications to the pendulum clock. A quartz clock is more accurate than other clocks because its oscillations are regulated by a quartz crystal.

SIMPLE PENDULUM

A simple pendulum consists of a relatively massive object hung by a string from a fixed support. Given figure-1 shows the pendulum in the rest (mean) position. The massive object is affectionately referred to as the pendulum bob. When the bob is displaced from equilibrium and then released, it begins its back and forth vibration about its fixed equilibrium position. The motion is regular and repeating, an example of periodic motion.

The position A where the simple pendulum rests is called the **mean position** of the simple pendulum. If the simple pendulum is displaced from its mean position (A) to position (C) and then released, we observe that the simple pendulum begins to move towards the mean position (A) and then to extreme position (B) and back to (B) through (A). Thus the bob of the simple pendulum keeps moving to and fro between the two extreme positions B and C about the mean position (A). Such a to and fro motion which repeats itself at regular intervals of time about the mean position is called an oscillatory motion or harmonic motion. When the bob of the pendulum moves from one extreme position say B, to the other extreme position C and then back to the extreme position B, then we say that the simple pendulum has executed one oscillation or one vibration.

The time taken by the simple pendulum to execute one complete oscillation is called its **Time period** (T).

The maximum displacement of the bob from its mean position (AB or AC) is called **amplitude**.

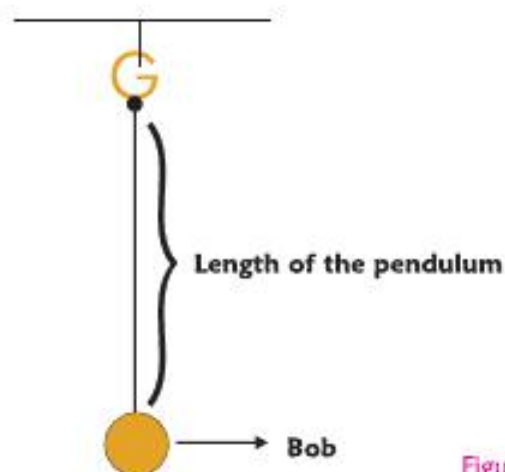


Figure (1)

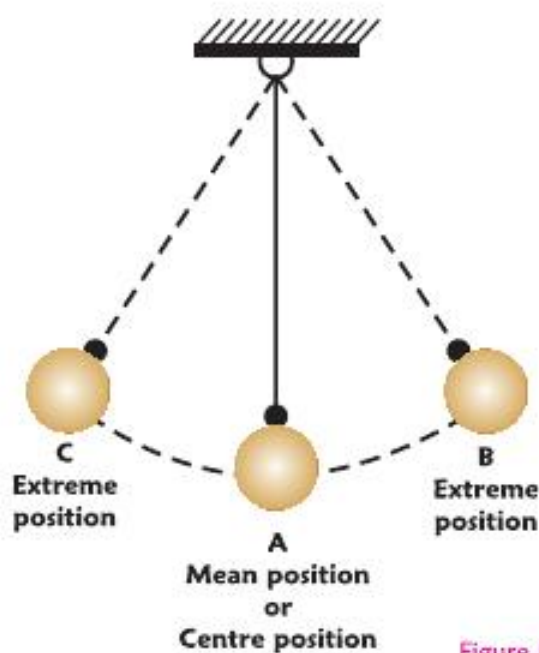


Figure (2)

Numerical Example 1 : A simple pendulum takes 120 seconds to complete 50 oscillations. What is the time period of the pendulum?

Solution : Time period = Total time taken $\div$ Number of oscillations
= 120 seconds $\div$ 50 = 2.4 s

Numerical Example 2 : The distance covered by the bob of a simple pendulum in one complete oscillation is 18 cm. What is the amplitude of the oscillation?

Solution : Amplitude = $\frac{1}{4}$ of the distance covered in one oscillation
= $\frac{1}{4} \times 18 \text{ cm} = 4.5 \text{ cm}$

Units of Time

The standard unit of time is second. Bigger units of time are minute, hour, day, year, etc.

60 seconds	1 minute
60 minutes	1 hour
24 hours	1 day
365 days	1 year
10 years	1 decade
10 decades	1 century
10 centuries	1 millennium

MOTION

In our daily life, we come across a variety of objects. Some of them like buildings, tables, chairs, plants, etc are stationary i.e. they do not change their position from one place to another. Also, there are other objects like moving cars, running trains, etc. which change their positions from one place to another.

When a body does not change its position from one place to another with respect to another body present in its surroundings i.e., the reference body, the body is said to be at rest.

When a body changes its position from one place to another with respect to another body in its surroundings i.e., the reference body, the body is said to be in motion. However all the objects on the Earth's surface, at rest or in motion are moving with the moving Earth in space. Thus their rest and



motion states are relative to the Earth only.

An object is said to be in motion if it changes its position with respect to its surroundings in a given time.

An object is said to be at rest if there is no change in its position with respect to its surroundings in a given time.

It should be known that rest and motion is always a relative term. They depend upon the position of an observer. For example: to a person sitting inside a moving train, everything inside the train appears to be at rest, while the trees, buildings, fields etc. outside the train appear to be running past the train in opposite direction i.e., they appear to be in motion.

Again, to a person standing outside the train, the trees, buildings, fields etc. outside the train appear to be at rest, while everything inside the train appears to be in motion. So, rest and motion are always relative terms.

Types of Motion

An object may move in a straight line or in a curved line like a circle. It may move at random or oscillate on a swing. It may roll like a rolling ball on a plane. It may vibrate like the string of a guitar. The motion may be periodic exactly repeating itself like the motion of a needle in a clock. Some common types of motion are:

Linear motion is the most basic of all motions. Linear motion is the type of motion in which all parts of an object move in the same direction and each part moves an equal distance.

Rotatory motion is motion in a circle. This type of motion is the starting point of many mechanisms. Example: a spinning wheel.

Reciprocating motion is back and forth motion. Example: the up and down motion of a yo-yo.



Activity-1

- Take a string and tie a stone at one of its ends. Now, tie the other end of the string to a rigid surface so that it can swing freely without touching any object. Release the string attached to the stone from an extreme position and let it oscillate freely. You have to count ten oscillations, that is the stone must return ten times to the same position from where you had released it. Use a stopwatch to note the time taken by the pendulum for ten oscillations. Now, divide the recorded time by ten. You will get the time period of oscillations. Repeat the activity with different number of oscillations and again calculate the time period. You will observe that the time period is constant. You will observe that the time period always comes out to be the same. This shows constancy of time period of the pendulum.

Uniform and Non-Uniform Motion

A body is said to be in motion if its position keeps on changing with time (with respect to the observer). But in our daily life we experience certain motions in which, the change in position of objects remains the same for a time interval, in some other motions, the change in position of object will not be the same for a given time interval.

Imagine the movement of hands in a wall clock and the movement of a butterfly in a garden. In these two cases hands of wall clock and the butterfly are in motion. They change their positions with time.

SPEED

Many objects in the world around us are moving. To compare how fast they are moving, we need to know their speed. We may have observed the speedometer in motor cars which tells us how fast the car is travelling. Usually, the speedometer in a vehicle shows the distance travelled in kilometers and the needle shows the speed of the journey in kilometres per hour.



Average Speed : We experience in our everyday life that most of the objects in motion do not have the same speed throughout the journey. To express the speed of the object, we consider its average speed. How to calculate average speed.

Average speed = Total distance travelled / Total time taken to travel the distance

If a car is moving with a speed of 60 kilometres per hour, it means that it covers a distance of 60 kilometres in one hour. However, a car seldom moves with a constant speed for one hour. It starts moving slowly and then picks up speed. So we usually consider only the total distance covered by it in one hour. We do not bother whether the car has been moving with a constant speed or not during that hour. The speed calculated here is actually the average speed of the car.

Average speed = 60 kilometres / 1 hour = 60 kilometres per hour

Units of Speed

Depending on the need and context, speed is measured in different units.

Speed = Distance ÷ Time

Unit of speed = Unit of distance ÷ Unit of time

= m ÷ s = metre per second (or m/s)

Unit of speed in S.I. system is metre per second (m/s)

Another unit commonly used for speed is kilometre per hour (Km/h).

Range of Speeds

We see the animals moving at different speeds in everyday life and on TV screens.

Maximum speed that some animals can attain			
Animal	Speed (kmph)	Animal	Speed (kmph)
Peregrine falcon	320	Giraffe	50
Cheetah	112	Cat (domestic)	48
Lion	80	Human	45
Hyena	65	Elephant	42
Zebra	65	Squirrel	20
Rabbit (domestic)	60	House mouse	13
Garden snail	0.05	Giant Tortoise	0.25

Rockets, launching satellites into orbits around the Earth, often attain speeds up to 28,800 km/h. Light travels at a speed of 300,000 km/s.

Numerical Example 3 : A bus travels 120 km in 1.5 h. Calculate its speed.

Solution : Speed of the bus = Distance $\div$ Time
= 120 km $\div$ 1.5 h = 80 km/h

Measuring Speed

To measure speed, we measure distance and time first.

DISTANCE-TIME GRAPH

Information about movement can be presented in a number of ways. Data for the total distance moved at different times during movement can be recorded in a table. Alternatively, the same information can be presented in a graph.

The motion of a car

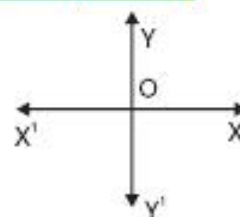
You can make the graph by following the steps given below.

To learn how to plot a graph, we shall consider a simple case of motion of a car moving along a straight road. We shall assume that the car is moving on a level road along a straight-line path. The distance moved by it in every twelve minutes is shown in table.

Time	0 min	12 min	24 min	36 min	48 min	60 min
Distance moved from the starting point	0 km	10 km	20 km	30 km	40 km	50 km

Steps :

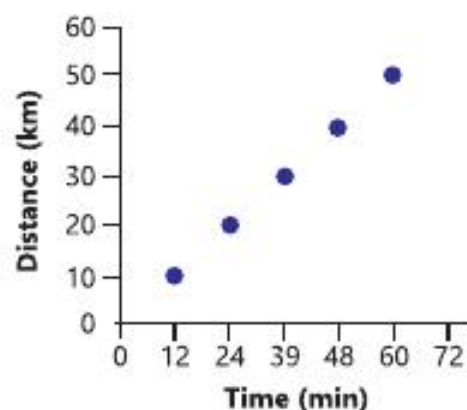
1. To draw a line graph we first draw two perpendicular lines. The horizontal line XOX' is known as the x-axis and the vertical line YOY' as y-axis. The point of intersection of XOX' and YOY' is



called the origin. It has been shown by point o in figure. Let x-axis represent the time and y-axis the distance. The positive values of the quantities on the x-axis are shown along ox while those on the y-axis are shown along OY.

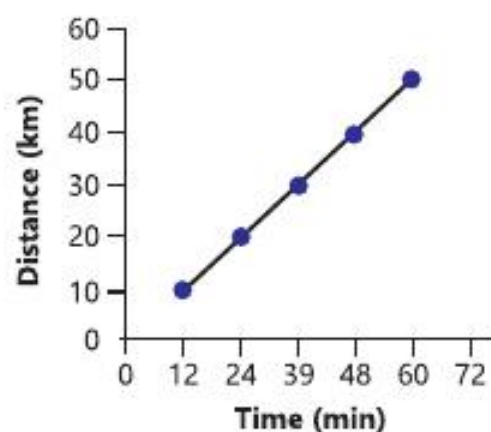
- Now, we have to decide upon suitable scales to represent the two quantities on the graph. It is always advantageous to select a scale by carefully studying: (i) the total range of data i.e., the difference between the lowest and the highest value for each quantity and their intermediate values, and (ii) the size of the graph paper available. In the example we have taken, the distance varies in a range of 50 km in steps of 10 km while time has a range of 60 minutes in steps of 12 minutes. Suppose we have a graph paper having a length 25 cm and width 15 cm.

- One of the appropriate scales to accommodate the given data could be distance 5 km = 1 cm time 6 min = 1 cm One can, however, choose any other convenient scale to draw the graph. Once we have decided the scale, the next step is to plot the given data on the graph. We make a decision about the quantity to be shown along the x-axis and that along the y-axis. We begin with marking the points corresponding to values given for the two quantities one on the x axis and the other on y-axis. In the case of the data given in table-1, for example, points marked at every 2 cm on the x-axis would correspond to a time interval of 12 min. Similarly, points marked at every 2 cm on the y-axis would correspond to a distance of 10 km.



- Next, we have to mark the point on the graph paper corresponding to each set of the two values, distance and time. Let us consider a particular value of time and corresponding value of distance. We note that at time 12 min, the distance moved is 10 km. we draw a line parallel to the distance axis (or perpendicular to x-axis) at the point that marks 12 min on the x-axis. Similarly, we draw a line parallel to the time axis at the point that marks 10 km on y-axis. The point where the two lines intersect each other gives the location of the point corresponding to the set of values under consideration. The points corresponding to other set of values are similarly marked on the graph. Once all the points for a given data are marked on the graph paper, these are joined together by a smooth curve. In the present case, this curve is a straight line.

If the distance-time graph is a straight line, it indicates that the object is moving with a constant speed (uniform motion). However, if the speed of the object keeps changing the graph can be of any other shape.



◀ REWIND

- ▶ Motion is a common experience in our life.
- ▶ An object is said to be in motion if it changes its position.
- ▶ We measure time by using different instruments like electronic clocks, digital clocks, etc.
- ▶ Speedometer helps us to know the speed of a vehicle.
- ▶ Unit of speed in SI system is metres per second.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. Change in position of a body is called
(i) Motion (ii) Speed (iii) Rest (iv) Acceleration
- b. A car is moving with 72 km/hrs. The speed of car in m/s is
(i) 20 m/s (ii) 25 m/s (iii) 30 m/s (iv) 40 m/s
- c. Stop watch is used to measure
(i) Correct time (ii) Total distance travelled
(iii) Exact time during the event (iv) All of these
- d. Pendulum clock is based on
(i) Moon clock (ii) Simple Pendulum
(iii) Sand clock (iv) Sun dial
- e. S.I unit of time is
(i) Second (ii) Minute (iii) Hour (iv) Day
- f. Distance-time graph is straight line for
(i) Non-uniform motion (ii) Uniform motion
(iii) Accelerated motion (iv) Constant motion

2 Fill in the blanks :

- a. Distance travelled by a body in unit time is called _____.
- b. _____ of object helps us to decide which one is moving faster than the other.
- c. Distance covered = _____ x time.
- d. A mineral used in the crystal form in watch is _____.
- e. A body is said to be in _____ if it covers equal distance in equal interval of time.
- f. All clocks are based on _____ events.

3 Write T for true and F for false :

- a. Speed is equal to distance per unit time.
- b. Unit of speed is km/hr.
- c. Speedometer is used to measure speed of vehicle at particular time.

- d. Car always travels with uniform motion.
- e. S.I unit of distance is kilometre.
- f. We measure time by using different instruments like electronic clocks, digital clocks, etc.

4 Answer these questions in short :

- a. What is a pendulum? What for a pendulum is used?
- b. Write differences between motion and rest.
- c. What do you mean by periodic motion?
- d. Show the shape of the distance-time graph for the motion in the following cases by a rough diagram : (i) a car moving with constant speed (ii) a car parked on a side road.
- e. A simple pendulum takes 6 seconds to complete 20 oscillations. Find the time period of the pendulum.
- f. A truck travels 540 km in 4.5 hrs. Find the speed of truck.

5 Give answers in detail :

- a. Give an experiment to find the time period of a simple pendulum of a certain length.
- b. Distinguish between uniform and non-uniform motion giving one example of each.
- c. How is time measured in ancient time?
- d. Define average speed.
- e. Arrange the following slowest to fastest speed. (i) Speed of a car (ii) Speed of an aeroplane (iii) Speed of a bicycle (iv) Speed of walking (v) Speed of auto-rickshaw
- f. Differentiate between linear motion and rotatory motion.

NEP QUERY



PROJECT WORK

- ♦ In the laboratory : Make a model of a sand clock which can measure a time interval of two minutes. Use a plastic bottle with narrow mouth inverted on another bottle of narrow mouth using your creative idea.

(Learning by Experiment)



TEST PAPER

1



1 Tick (✓) the correct one :

- a. Gills are the respiratory organs in
(i) fish ☐ (ii) cockroach ☐ (iii) amoeba ☐ (iv) earthworm ☐
- b. Water can drain easily through
(i) sandy soil ☐ (ii) loamy soil ☐ (iii) clayey soil ☐ (iv) alluvial soil ☐
- c. Binary fission do not take place in
(i) amoeba ☐ (ii) paramecium ☐ (iii) yeast ☐ (iv) bacteria ☐
- d. Single-celled organisms among the following is
(i) human being ☐ (ii) yeast ☐ (iii) cow ☐ (iv) amoeba ☐
- e. Sheep are reared for getting
(i) cotton ☐ (ii) jute ☐ (iii) wool ☐ (iv) rayon ☐

2 Fill in the blanks :

- a. _____ rate of water is different in different types of soil.
- b. Muscle cell produce _____ during sudden activity.
- c. _____ is found in centre of our tooth.
- d. Cotton and jute are _____ fibres.
- e. Yeasts reproduce by _____.

3 Write T for true and F for false :

- a. Frogs breathe through their skin as well as their lungs. ☐
- b. Soil is affected by climate factors. ☐
- c. Ruminants are plant-eating animals. ☐
- d. Camel is a wool-giving animal. ☐
- e. Bryophyllum reproduce by leaf. ☐

4 Answer the following questions :

- a. What is the uppermost layer of soil?
- b. Why do we breathe?
- c. Describe the nutrition in a amoeba.
- d. What is sericulture?
- e. Why is reproduction necessary for living organisms?
- f. What is the role of stomata?
- g. Write a note on galvanisation.
- h. What is climate? How is it differ from weather?



TEST PAPER

2

1 Tick (✓) the correct one :

- a. All acids contain
(i) oxygen ☐ (ii) nitrogen ☐ (iii) carbon ☐ (iv) hydrogen ☐
- b. Heat is a form of
(i) temperature ☐ (ii) energy ☐ (iii) power ☐ (iv) work ☐
- c. The braver part of a tree above the stem is known as
(i) canopy ☐ (ii) crown ☐ (iii) herb ☐ (iv) shrub ☐
- d. Filament of electric bulb is made up of
(i) nichrome ☐ (ii) silver ☐ (iii) copper ☐ (iv) tungsten ☐
- e. Which blood group is called a universal donor?
(i) AB ☐ (ii) A ☐ (iii) B ☐ (iv) O ☐

2 Fill in the blanks :

- a. Litmus is extracted from _____.
- b. The SI unit of heat is _____.
- c. _____ are lungs of our earth.
- d. Distribution of water on globe is _____.
- e. A _____ is refers to a collection of rays.

3 Write T for true and F for false :

- a. Water soluble bases are called alkalies.
- b. 100°C is equal to 373°K .
- c. Plants do not depend on animals.
- d. Water is a non-renewable natural resource.
- e. Light travels slower than sound.

4 Answer the following questions :

- a. What is neutralisation?
- b. What is temperature?
- c. Explain a food chain.
- d. What is infiltration?
- e. Write two uses of a concave mirror in daily life.
- f. What is a fuse? Write its function.
- g. How does a kidney work?
- h. How is time measured in ancient time?